Permanent Labor Income Inequality and Stabilization Policies

Eustache Elina* Raphaël Huleux†

October 2026

Click here for the latest version

Abstract

We study how the distribution of permanent labor income (PLI) affects the transmission of fiscal policy. Standard heterogeneous-agent models have two sharp implications: (i) the distribution of PLI is neutral for aggregate variables, and (ii) the share of hand-to-mouth households does not vary with PLI. Using US household survey data, we show instead that the hand-to-mouth share falls from 64% in the bottom PLI quintile to 22% in the top one. We then build a heterogeneous-agent New Keynesian (HANK) model with a non-homothetic bequest motive that can reproduce this gradient and breaks the neutrality. In standard HANK models, the multiplier of a fiscal package depends mostly on how long the debt lasts. In ours, it depends at least as much on who receives the payout and who finances it, because low-PLI households spend extra income quickly while high-PLI households save most of it. In an application to a regressive fiscal package, the One Big Beautiful Bill Act, the model gives a cumulative multiplier of -0.06, against 0.57 in a standard HANK model with the same aggregate marginal propensity to consume. For the same reason, the effect of a rate cut depends on who receives the interest the government saves on its debt. In the model, the rise in PLI inequality since 1970 has made redistributions between PLI groups move output more.

Keywords: Inequality, permanent income, monetary policy, fiscal policy, interest rates, heterogeneous agent model.

JEL codes: D63, E21, E24, E52, E62

*Banque de France, eustache.elina@banque-france.fr

†CEBI, Department of Economics, University of Copenhagen, rph@econ.ku.dk

This work would not have been possible without the fruitful discussions with Leonard Bocquet who shares the starting idea of the paper. Our work also largely benefited from the helpful supervision and guidance of Axelle Ferrière, Tobias Broer, David Jinkins, Jeppe Druedahl,Birthe Larsen, and Karl Harmenberg. We also thank Edouard Challe, Xavier Ragot, Riccardo Cioffi, Morten Olsen, Jean-Baptiste Michau, Julien Matheron, Alex Clymo, Jeppe Druedahl, Søren Hove Ravn, Jacob Sundram, Charles Labrousse, John Kramer, and Yann Perdereau for their helpful comments. Center for Economic Behavior and Inequality (CEBI) is a center of excellence at the University of Copenhagen, founded in September 2017, financed by a grant from the Danish National Research Foundation (Grant DNRF134). All remaining errors are our own.

1 Introduction

Labor-income inequality in the United States has risen sharply since 1980, with the share of pre-tax labor income earned by the top one percent rising from 8% to 14% (Piketty et al. 2018, Saez & Zucman 2020). Most of this rise comes from the persistent component of earnings, as gaps in lifetime earnings have widened (DeBacker et al. 2011, Guvenen et al. 2022), while transitory earnings volatility has fallen (Bloom et al. 2017, Braxton et al. 2021). Our thus paper asks if the distribution of permanent labor income (PLI) affect the transmission of fiscal policy?

In standard heterogeneous-agent models, the answer is no. With homothetic preferences, consumption is linear in permanent income (Straub 2019), and its distribution has no impact on aggregate variables.[1] In these models, four decades of rising permanent inequality have no effect on stabilization policy.

However, the data reject one implication of these models: that hand-to-mouth status is unrelated to permanent labor income. In the PSID, there exists a strong negative relationship between the share of hand-to-mouth households and their PLI. The share of hand-to-mouth households falls from 64% in the bottom PLI quintile to 22% in the top one. Moreover, PLI predicts hand-to-mouth status better than any other household characteristic we observe. These facts extend the saving-rate evidence of Dynan et al. (2004), Straub (2019). Households with higher permanent income save more of it, and households with low permanent income live hand to mouth.

We develop a two-asset HANK model with liquid and illiquid assets (Kaplan & Violante 2014, Kaplan et al. 2018), in which these facts arise from one departure from the benchmark. In our model, households have a non-homothetic bequest motive, so that bequests are a luxury good (Carroll 1998, De Nardi 2004, Kumhof et al. 2015, Michau 2018, Straub 2019, Mian et al. 2021). High-PLI households save a large share of their income, because they value the bequest they leave on top of the buffer that wealth provides against shocks. Their saving pushes down the equilibrium interest rate. At the bottom of the distribution, consumption is low and its marginal utility high, so the bequest motive matters little. Facing a low interest rate, these households dissave

[1] We state this result as Proposition 4 in Appendix E.3.

toward the borrowing constraint. The share of hand-to-mouth households then falls steeply with PLI, as in the data.

Calibrated to the United States in 2019, the model reproduces the HtM–PLI gradient of the PSID, for households with little net worth and for those with little liquid wealth, along with the top 1% wealth share and the liquid share of wealth. We compare it with a homothetic benchmark in which discount factors differ across households, as in Carroll et al. (2017) and Auclert et al. (2025). This heterogeneous beta model has the same hand-to-mouth shares and MPC out of a transfer, but its MPCs are uncorrelated with permanent labor income by construction, so it misses the HtM–PLI gradient.

Matching the HtM–PLI gradient has two implications for fiscal policy. First, the multiplier of a fiscal package depends much more on who receives the payout and who finances it. Low-PLI households are more likely to be hand-to-mouth, and hence on average have a higher marginal propensity to consume: they spend an income shock within a few quarters, while high-PLI households save most of it. A pure redistribution from high-PLI to low-PLI households thus has a cumulative multiplier of 1.2, against 0.1 in the heterogeneous beta model. Second, how persistent the deficit is matters less, because the income an expansion creates goes mostly to high-PLI households, who spend it slowly.

Together, these effects are large enough to flip the sign of a stimulus that redistributes away from low-PLI households. In an application to the One Big Beautiful Bill Act, the cumulative multiplier is -0.06 in the bequest-motive model and 0.57 in the heterogeneous beta model. This gap comes from the fact that the law is regressive: it cuts taxes in proportion to income and finances part of the cut with transfer cuts and tax increases that rise much less with income. This has a large negative effect in the bequest-motive model, and only a modest one in the heterogeneous beta model.

The same logic applies to monetary policy, because a rate cut also affects the government budget constraint. It lowers the interest bill on public debt and raises tax receipts, and the fiscal rule pays this windfall back to households as a rebate. Moving the rebate from low to high earners sharply lowers the present value of the output response in the bequest-motive model, but barely changes it in the heterogeneous beta model.

Monetary transmission thus depends more on the fiscal rule than in a standard HANK model, which argues for closer coordination between the central bank and the fiscal authority.

Finally, we study rising PLI inequality by re-solving the model under the 1970 PLI distribution. When inequality increases, high-PLI households receive a larger share of labor income and save most of it, which lowers the real rate from 2.5% to 2.0%. Households below the top move toward the borrowing constraint. Both changes widen the gap between the MPC out of a transfer and the MPC out of labor income, so incidence matters more. Redistributions between PLI groups move output more, and the effect of a rate cut depends more on who receives the rebate. At a given debt duration, deficits move output less and finance themselves less. Under an unchanged fiscal rule, the lower rate lengthens the debt duration and offsets this loss.

Related Literature. Our paper makes three contributions. First, we document that hand-to-mouth status is concentrated among households with low permanent labor income, and that this concentration has increased since 1999. Second, we extend the intertemporal Keynesian cross (IKC) to make the incidence of fiscal and monetary policy explicit, and we extend the neutrality result of Straub (2019) from aggregates such as consumption and the interest rate to every sequence-space Jacobian. Third, we quantify the consequences for stabilization policy, both under the 2019 distribution and across the rise in PLI inequality since 1970.

Monetary policy transmission with heterogeneous households. Micro data show that a large share of households hold little liquid wealth (Kaplan et al. 2014). With such households, monetary policy works mostly through indirect general-equilibrium effects (Kaplan et al. 2018), which arrive slowly (Holm et al. 2021) and depend on the fiscal response (Alves et al. 2020). Monetary shocks also move inequality in income and consumption (Coibion et al. 2017), and this redistribution is a transmission channel (Auclert 2019). This literature studies transmission along wealth and liquidity. We add permanent labor income. Patterson (2023) studies the closest mechanism and shows that high-MPC workers are more exposed to business-cycle earnings fluctuations, a covariance that amplifies recessions. We study the same covariance along the permanent component of earnings, where it is negative and its size depends on the PLI distribution. On

the empirical side, the closest paper is Aguiar et al. (2025), who characterize who the hand-to-mouth households are in the PSID. We add that hand-to-mouth status lines up with permanent labor income, and that this relationship has tightened as earnings inequality rose.

The intertemporal Keynesian cross. Auclert et al. (2024) show that the output response to a fiscal shock is priced by the intertemporal MPCs (iMPCs) of the households who receive and repay it, and Auclert et al. (2025) survey the framework's applications to fiscal and monetary policy. We generalize the cross in two directions. First, the benefit and financing incidence of every instrument enters explicitly, so that the same equation prices deficit-financed packages, balanced-budget redistributions, and any combination of payout and financing instruments. Second, through the government budget, the cross extends to monetary policy, since the interest savings from a rate cut are a package whose incidence the fiscal rule sets. Wolf (2025) shows that uniform transfers can reproduce the path of output and inflation of any rate cut, and in his rate policy the interest savings are rebated uniformly. We let the incidence of this rebate vary, and find that it changes the effect of the cut. Angeletos et al. (2024) show that deficits partly finance themselves through the tax receipts of the expansion they create, more so when the fiscal adjustment is delayed, provided households spend new income quickly. This new income goes largely to high-PLI households. When MPCs fall with PLI, these households spend it slowly, so a longer debt duration raises output less and deficits finance themselves less.

Fiscal policy and inequality. Brinca et al. (2016) show across countries that fiscal multipliers rise with wealth inequality, and Brinca et al. (2021) that fiscal consolidations are more contractionary where income inequality is higher. In both, inequality matters through the share of constrained households. Ferriere & Navarro (2025) show that government spending has a larger multiplier when it is financed by taxes on high earners, mostly because their labor supply responds less to taxes, while MPCs vary little with income in their model. We add the correlation between MPCs and permanent labor income, which makes the multiplier depend on who receives a package and who finances it, and which rising inequality strengthens.

Non-homothetic bequest motive. Our saving motive comes from Straub (2019), who

documents that saving rates rise with permanent income, proposes a non-homothetic bequest motive to match this, and traces its consequences for wealth inequality and the long-run interest rate. A broader literature uses similar preferences: De Nardi (2004) for the concentration of wealth at the top, Carroll (1998) for the saving rates of the rich, Kumhof et al. (2015) for the build-up of household debt before the Great Recession, Michau (2018) and Mian et al. (2021) for the over-accumulation of savings, and Gaillard et al. (2023) for the relative tails of the income and consumption distributions.[2] This literature uses non-homotheticity to explain long-run trends in saving and wealth, and we use it to study stabilization policy. Melcangi & Sterk (2020) find that rising stock-market participation has strengthened U.S. monetary policy since 1980. We study how the rise in labor-income inequality changed transmission.

Outline. Section 2 documents the three PSID facts. Section 3 presents the model, and Section 4 its calibration. Section 5 derives the incidence-explicit IKC and compares the bequest-motive model with the heterogeneous beta model under the 2019 distribution. Section 6 measures the effect of rising PLI inequality on fiscal and monetary transmission.

2 Permanent Labor Income and Hand-to-Mouth Status

This section documents three empirical facts using the Panel Study of Income Dynamics (PSID, 1999–2019). First, hand-to-mouth (HtM) status is strongly concentrated among households with low permanent labor income. Second, permanent labor income predicts hand-to-mouth status better than any other household characteristic we observe, and it accounts for part of the household-specific component of hand-to-mouth status. Third, the concentration has increased over the sample period.

Data and definitions. We use the PSID's biennial waves from 1999 to 2019, covering 32,378 household-year observations from 4,945 households. Each wave records income for the preceding calendar year, so statistics dated by income year run from 1998 to 2018.[3] We follow Aguiar et al. (2025) and Kaplan et al. (2014) and classify a household as

[2] De Nardi et al. (2016) survey the related evidence on bequest motives.

[3] Sample selection follows Aguiar et al. (2025): heads aged 25–64, real labor earnings and disposable income above $2,000, and at least three waves observed per household (required to measure permanent

hand-to-mouth in net worth (HtM_{NW}) if its net worth is below two months of household earnings. We classify a household as a *wealthy hand-to-mouth* (HtM_{LIQ}) if its liquid assets fall below one week of earnings, but its net worth is above the hand-to-mouth in net worth threshold. Liquid assets are composed of checking and savings accounts, money market funds, certificates of deposit, government bonds and directly held stocks, net of unsecured debt. Appendix A reports summary statistics by permanent income quintile.

Our baseline measure of permanent labor income (PLI) follows Straub (2019) and is built in two steps. First, we regress the log household labor earnings y_{it} of household i in income year t on year fixed effects d_t, an age cubic $f(\cdot)$, and controls X_{it} for family size and a dual-earner indicator,

$$\log y_{it} = d_t + f(\text{age}_{it}) + \delta' X_{it} + \hat{y}_{it}, \tag{1}$$

where $\hat{y}_{it}$ is the residual. Second, we average the residuals within each household across its T_i observed waves,

$$\bar{y}_i = \frac{1}{T_i} \sum_t \hat{y}_{it}, \tag{2}$$

and use $\bar{y}_i$ as the household's PLI.[4] Appendix A.2 shows that the facts below are robust to two other measures of PLI: a household fixed-effects estimator, and an empirical Bayes estimator that shrinks $\bar{y}_i$ toward the mean for households observed in few waves.

Fact 1: HtM status declines steeply with permanent income. The total hand-to-mouth share falls from 64% in Q1 (lowest PLI) to 22% in Q5 (highest PLI), as panel (a) of Figure 1 shows. Both definitions of hand-to-mouth follow this pattern: the HtM_{NW} share falls monotonically from 38% to 12%, and the HtM_{LIQ} share from 26% to 10%. A household in the bottom quintile of permanent income is more than three times as likely to be hand-to-mouth in net worth as one in the top quintile.

Panel (a) shows a stock: the share of households that are hand-to-mouth at a point in

income). The household earnings measure sums head and spouse labor income and government transfers, net of the employee payroll tax; the same measure is the denominator of both hand-to-mouth thresholds below. All dollar amounts are deflated with the PCE price index (2009 base). Statistics are weighted by PSID family weights.

[4] This measure includes richer first-stage controls than the PLI measure of Aguiar et al. (2025) (which conditions only on an age cubic and year), though the two are highly correlated.

time. A stock alone cannot tell whether the same households remain hand-to-mouth over time. Panel (b) therefore shows the flows: four-year transition probabilities into and out of hand-to-mouth status by PLI quintile, summing over the two definitions of hand-to-mouth. Among households that are hand-to-mouth in wave t, the probability of remaining hand-to-mouth four years later (wave $t+2$) falls from 78% in Q1 to 51% in Q5. Among households that are *not* hand-to-mouth in wave t, the probability of becoming hand-to-mouth falls from 35% in Q1 to 10% in Q5. The flows show the same gradient as the stock.

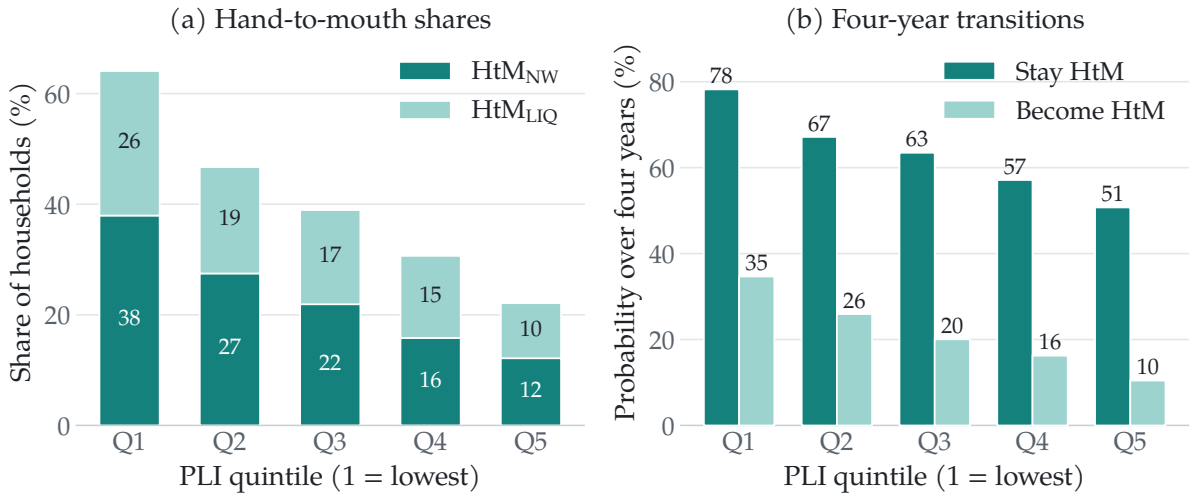


Figure 1: Hand-to-mouth status by PLI quintile: stocks and flows (PSID, 1999–2019)

Notes: Panel (a): HtM_{NW} = net-worth hand-to-mouth: net worth below two months of earnings (Aguiar et al. 2025). HtM_{LIQ} = wealthy hand-to-mouth: net worth above the threshold but liquid assets below one week of earnings (Kaplan et al. 2014). Shares are computed among households with non-missing hand-to-mouth status. Panel (b): "Stay HtM" = probability of being hand-to-mouth (HtM_{NW} or HtM_{LIQ}) in wave $t+2$ (four years later) conditional on being hand-to-mouth in wave t; "Become HtM" = probability of becoming hand-to-mouth conditional on not being hand-to-mouth in wave t. PLI quintiles computed from the averaged residuals of Straub (2019). Weighted by PSID family weights.

Fact 2: PLI is the main observable predictor of HtM status. Among the household characteristics we observe, PLI is the one that best predicts hand-to-mouth status. Many of these characteristics are correlated with PLI, so we regress hand-to-mouth status on year effects and five groups of characteristics: PLI, deviations of current earnings from PLI, age, family structure and race. A Shapley decomposition then splits the explained variance among the groups (Appendix A.3). PLI accounts for 44% of the explained variance, about twice as much as age or family structure (panel (a) of Figure 2). Current earnings do not predict hand-to-mouth status better than PLI. Holding age and family structure fixed, PLI raises the explained variance by 7.2 points and current earnings by 6.6, and current earnings add only 0.5 points once PLI is included.

Aguiar et al. (2025) find that household fixed effects account for an important share of the variation in hand-to-mouth status, and they interpret these persistent differences as heterogeneity in preferences. In our sample, household fixed effects explain 44% of the variance of hand-to-mouth status (panel (b)). The five groups of characteristics together explain 18%, so most of this household component remains unexplained by the characteristics we observe. PLI is the largest part of what we can explain. On its own, it explains 9% of the variance, half of what the five groups explain together and a fifth of the household component.

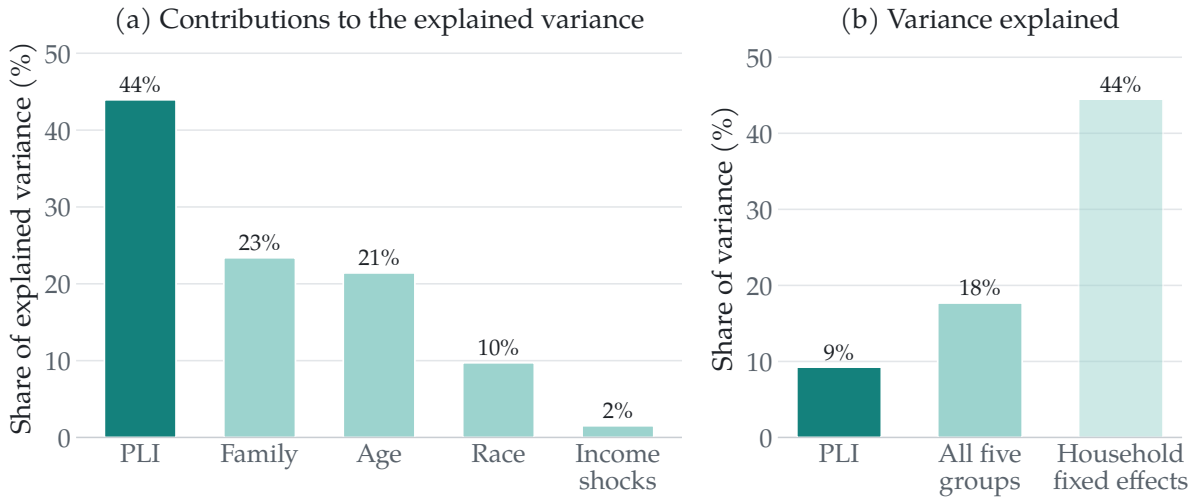


Figure 2: What explains hand-to-mouth status (PSID, 1999–2019)

Notes: Panel (a): share of the explained variance of total hand-to-mouth status (HtM_{NW} or HtM_{LIQ}) attributed to each group of household characteristics by a Shapley decomposition, net of year effects. A group's Shapley value is its average gain in R^2 over all orders in which the five groups can be added to the regression, so the values sum to the R^2 of the five groups, 17.7%. Panel (b): R^2 of PLI alone, of the five groups together, and of household fixed effects (adjusted for the number of households), net of year effects. Groups and standard errors in Table 7. Weighted by PSID family weights.

Fact 3: The HtM–PLI gradient has steepened since 1999. Between 1999 and 2019, the gap in HtM_{NW} shares between low-PLI (Q1–Q2) and high-PLI (Q4–Q5) households widened from 14 to 22 percentage points (panel (a) of Figure 3). The aggregate HtM_{NW} share rose from 19% in the 1999 wave to 28% in the 2013 wave before declining modestly, but low- and high-PLI households diverged. The share among low-PLI households rose from 26% to 33%, while the share among high-PLI households ended the sample at 11%, slightly below where it started, after a temporary rise around the Great Recession.

Panel (b) shows the same steepening without binning households into quintiles. For

each wave t we estimate the weighted linear probability model

$$\mathrm{HtM}_{i,t} = a_t + b_t \widehat{\mathrm{PLI}}_i + u_{i,t}, \tag{3}$$

where $\mathrm{HtM}_{i,t}$ is an indicator for $\mathrm{HtM}_{\mathrm{NW}}$ status and $\widehat{\mathrm{PLI}}_i$ is the household's averaged log-earnings residual, which is fixed over time. The slope b_t measures the change in the probability of being hand-to-mouth per log point of permanent labor income; because $\widehat{\mathrm{PLI}}_i$ does not vary over time, changes in b_t come only from changes in who is hand-to-mouth. The gradient nearly doubles over the sample, from -0.10 to -0.18. It steepens both before 2008 and after 2010, so the divergence in Panel (a) does not come from the 2008–2009 recession and housing bust alone. The steepening also lines up with inequality. Interacting PLI with the wave-level dispersion of residual earnings yields a negative and strongly significant coefficient (Table 8 in Appendix A).

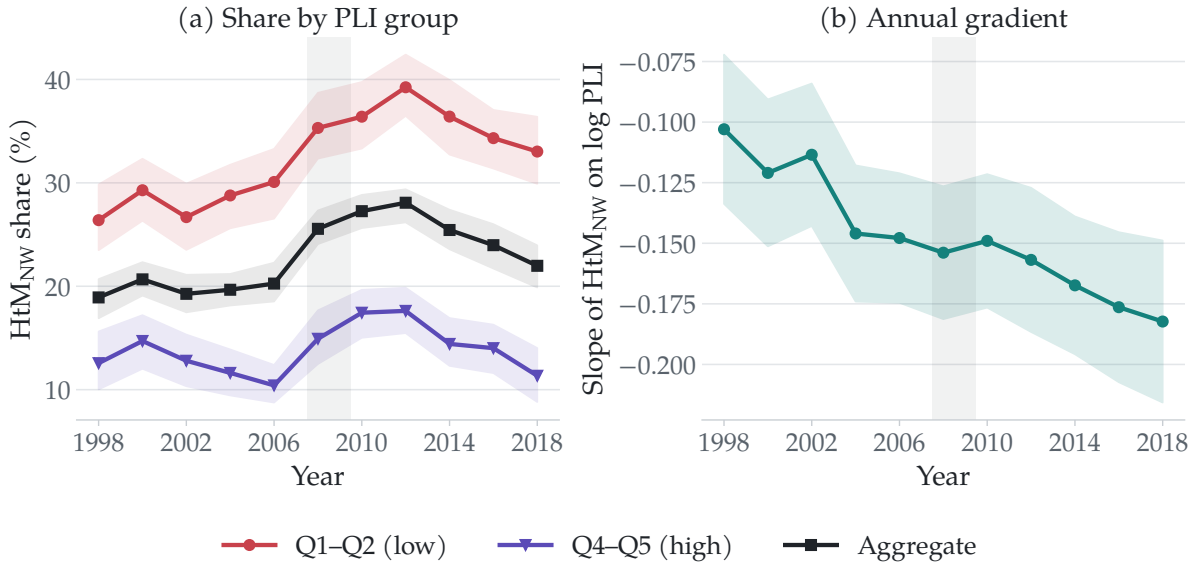


Figure 3: The steepening of the HtM–PLI gradient (PSID, 1999–2019)

Notes: Panel (a): weighted $\mathrm{HtM}_{\mathrm{NW}}$ shares by PLI group; shaded bands are 95% bootstrap confidence intervals. Panel (b): slope coefficients of weighted linear probability regressions of $\mathrm{HtM}_{\mathrm{NW}}$ status on log PLI, estimated separately by wave; the shaded band is the 95% confidence interval (robust standard errors). PLI groups (Panel a) and the continuous PLI measure (Panel b) are based on the averaged residuals of Straub (2019). The shaded bar marks the Great Financial Crisis (2008–2009).

3 A HANK Model with Permanent Labor Income Inequality

The previous section established a strong HtM–PLI gradient in the data. We now introduce a model that reproduces it. The model is a standard two-asset HANK model with one main addition, a non-homothetic bequest motive. With a bequest motive that is a luxury good, high-PLI households save more of their income than low-PLI households, and MPCs fall with PLI. Figure 4 gives a graphical overview of the different building blocks of the model.

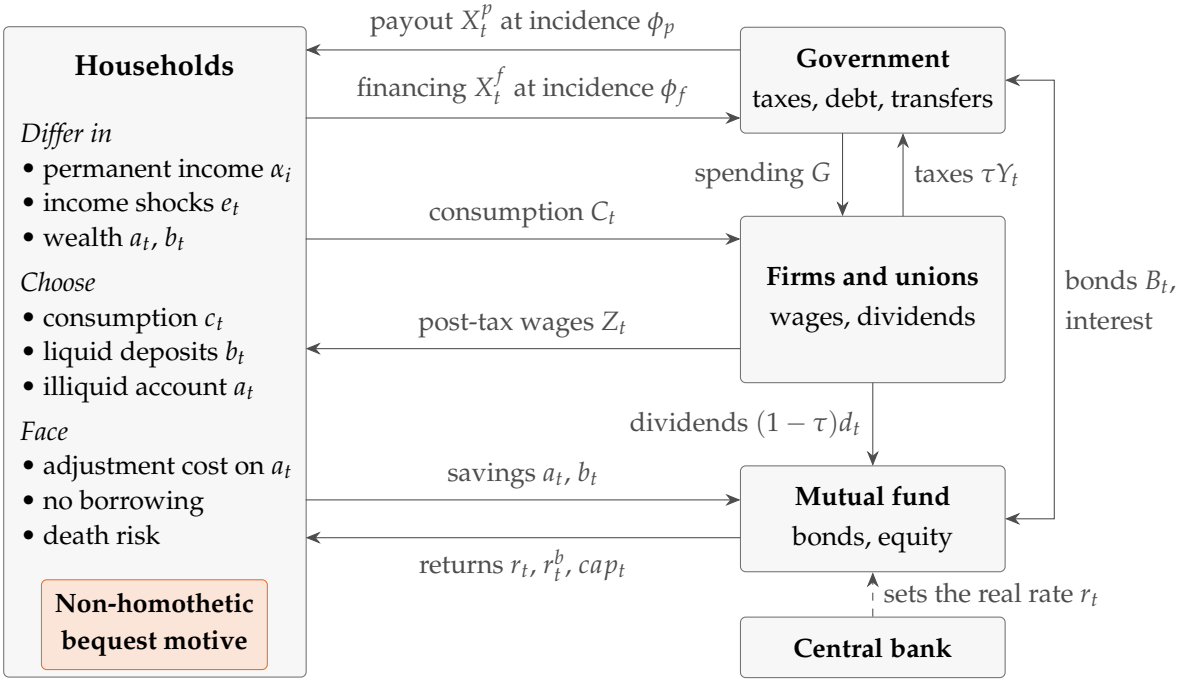


Figure 4: The blocks of the model

Note: Solid arrows show payments, and the dashed arrow shows a policy choice. The goods, labor and asset markets clear in every period. All the heterogeneity sits in the household block. The shaded box is the main departure from a standard two-asset HANK model.

Throughout the paper, we study how the economy responds to a one-time shock, starting from a deterministic steady state. Time is discrete and denoted t, there is no aggregate risk, and monetary and fiscal shocks are modelled as MIT shocks.[5]

[5]An MIT shock is a one-time unexpected shock, after which the transition to the steady state is deterministic and perfectly foreseen.

3.1 Households

The economy has a unit mass of households. Each household belongs to a PLI group i, with permanent productivity α_i, and faces idiosyncratic productivity risk. It holds its wealth in two accounts, liquid deposits b_t and illiquid wealth a_t, and can borrow in neither. Moving funds into or out of the illiquid account is costly. Every period, a household dies with probability ξ, leaves its wealth as a bequest, and is replaced by a newborn of the same PLI group.

Preferences. Households derive utility from consumption and, when they die, from bequests. Period utility functions are

$$u(c_t) = \log(c_t), \qquad v(a_t + b_t) = \gamma \frac{(a_t + b_t + \zeta)^{1-\eta}}{1-\eta},$$

where γ is the intensity of the bequest motive, ζ a Stone-Geary shifter, and η the curvature on wealth.

The bequest motive is the main departure from a standard HANK model, and it makes saving a luxury good. Without it, the household problem would scale linearly with permanent income: a household with twice the PLI would consume and save twice as much in every state, so the savings rate would not depend on PLI (Straub 2019). Two features of the bequest motive break this linearity. First, we assume that the curvature on wealth is below that of log consumption, $\eta < 1$. As a household gets richer, its marginal utility of consumption thus falls faster than its marginal value of bequests, and it saves a larger share of its income. Second, the Stone-Geary shifter ζ lowers the marginal value of bequests at low wealth, so the bequest motive is essentially weaker for poorer households. Together, these features generate a savings rate that increases with PLI and, ultimately, the HtM–PLI gradient we observe in the data. Section 4.3 discusses this mechanism in more detail and explains how the strong saving of wealthy households lowers the equilibrium return and keeps low-PLI households near the borrowing constraint.

Labor productivity. A household's labor productivity $\alpha_i e_t$ is the product of its PLI α_i and an idiosyncratic component e_t. Each PLI group i has mass ω_i, with $\sum_i \omega_i = 1$, and the total endowment of labor productivity is normalized to one, $\sum_i \omega_i \alpha_i = 1$. An increase

in PLI inequality is thus a mean-preserving spread in the distribution of $\{\alpha_i\}$. The idiosyncratic component $e_t = z_t \nu_t$ is the product of a persistent component, with $\log z_t$ an AR(1) with persistence ρ_e and unconditional standard deviation σ_e, and a transitory component, with $\log \nu_t$ iid normal with standard deviation σ^{tr}. Both have mean one. The persistent component z_t predicts future productivity, while ν_t only moves current income.

Income. Each household receives the share $\alpha_i e_t$ of aggregate post-tax labor income $Z_t \equiv (1-\tau)\, w N_t$, where w is the real wage and N_t hours. The real return r_t between $t-1$ and t is the ex-ante real rate set at $t-1$, so the date-0 return is predetermined when a shock arrives. The illiquid account earns r_t plus the capital gain cap_t of Section 3.4, and deposits earn $r_t^b = r_t - \kappa(1+r_t)$, where κ is an intermediation wedge. Finally, the government pays a payout X_t^p and raises its financing X_t^f, distributed across households with the incidence weights ι_{ϕ_p} and ι_{ϕ_f} of Section 3.5.

Adjustment costs. Adjusting the illiquid asset position is subject to the quadratic cost

$$\Psi(a_t, a_{t-1}) = \frac{\psi_1}{2} \frac{\left(a_t - (1 + r_t + cap_t)\, a_{t-1}\right)^2}{(1 + r_t + cap_t)\, a_{t-1} + \psi_0}, \tag{4}$$

which follows the convex cost of Kaplan et al. (2018) and Auclert et al. (2021). This feature of the model allows us to match the two definitions of hand-to-mouth we use in our empirical section. Poor hand-to-mouth households have a lower total wealth $a_t + b_t$, and "liquid" hand-to-mouth have low liquid wealth b_t but potentially high levels of illiquid wealth a_t.

Household problem. A household with deposits b_{t-1}, illiquid wealth a_{t-1}, productivity (z_t, ν_t) and PLI α_i chooses consumption c_t and its two account balances b_t and a_t to

solve

$$V_t(b_{t-1}, a_{t-1}, z_t, \nu_t, \alpha_i) = \max_{c_t, b_t, a_t} \left\{ u(c_t) + \xi\, v(a_t + b_t) + (1-\xi)\beta\, \mathbb{E}\left[V_{t+1}(b_t, a_t, z_{t+1}, \nu_{t+1}, \alpha_i) \,\middle|\, z_t \right] \right\},$$

(5)

$$\text{s.t. } c_t + b_t + a_t + \underbrace{\Psi(a_t, a_{t-1})}_{\text{adjustment cost}} = \underbrace{(1 + r_t^b) b_{t-1} + (1 + r_t + cap_t) a_{t-1}}_{\text{financial income}} + \underbrace{\alpha_i e_t Z_t}_{\text{post-tax labor income}}$$
$$+ \underbrace{\iota_{\phi_p}(\alpha_i e_t)\, X_t^p - \iota_{\phi_f}(\alpha_i e_t)\, X_t^f}_{\text{government payout and financing}}, \quad b_t \geq 0, \quad a_t \geq 0. \tag{6}$$

Relative to a standard two-asset HANK model, the only new terms are the value of bequests $\xi\, v(a_t + b_t)$ and the PLI α_i, which scales labor income and enters the incidence of transfers.

Bequests. Newborns start with no illiquid wealth, and estates reach them as deposits, in proportion to their permanent productivity: a newborn of group i receives $\alpha_i \mathcal{E}_t / \xi$, where $\mathcal{E}_t$ is the aggregate flow of bequests

$$\mathcal{E}_t = \xi \Big(A_{t-1}(1 + r_t) + cap_t\, \bar{S} \Big), \tag{7}$$

A_{t-1} is total wealth, and $\bar{S} = \int a\, d\Phi_{ss}$ is steady-state illiquid wealth, with Φ_t the joint distribution over the two accounts, idiosyncratic productivity, and permanent productivity. Mortality keeps the wealth distribution non-degenerate despite the bequest motive and pins down its Pareto tail, as we discuss in Section 4.

3.2 Unions

A union sets a common level of hours N_t for all households. Nominal wages are rigid, so the union supplies the hours that firms demand, and output is determined by aggregate demand.

As a result, transfers have no effect on labor supply in our model. Hours are common to all households and set by demand, so neither a payout nor its financing changes how much a household works, through income effects or through incentives. However, in reality, a transfer can lower the labor supply of its beneficiaries, in particular low-PLI

households, for whom it is large relative to earnings. As in standard in the HANK literature, we abstract from this margin and focus exclusively on the demand effect of fiscal policy.

3.3 Firms

A competitive final good producer combines a continuum of imperfectly substitutable intermediate inputs j,

$$Y_t = \left(\int_j y_{jt}^{\frac{\varepsilon-1}{\varepsilon}} dj \right)^{\frac{\varepsilon}{\varepsilon-1}},$$

while intermediate firms produce using labor, $y_{jt} = n_{jt}$, and set flexible prices. All firms are identical, so $y_{jt} = Y_t$ and $n_{jt} = N_t$ in every period. Each intermediate firm charges the constant markup $\mu = \frac{\varepsilon}{\varepsilon-1}$ over its marginal cost, so the real wage is constant at $w = \frac{1}{\mu}$ and dividends are a constant share of output:

$$d_t = Y_t - wN_t = Y_t \frac{\mu - 1}{\mu}.$$

The value of the firm p_t is given by the no-arbitrage condition between equity and bonds, where dividends are paid net of the tax τ:

$$\frac{p_{t+1} + (1-\tau)d_{t+1}}{p_t} = 1 + r_{t+1}. \tag{8}$$

3.4 Mutual fund

A mutual fund receives all household savings, liquid and illiquid, and invests them in the two assets available in the economy, government bonds B_t and firm equity p_t, so that $A_t \equiv A_t^b + A_t^a = p_t + B_t$, with A_t^b aggregate deposits and A_t^a aggregate illiquid wealth. It pays r_t on illiquid wealth and r_t^b on deposits, and the spread $\kappa(1 + r_t)$ on deposits pays for the real resource cost of providing liquidity. In steady state, the no-arbitrage condition (8) ensures that the return on equity equals the return on bonds, and $cap_t = 0$.

The capital gain term arises only at date 0, on impact after a monetary or fiscal shock, and is paid to the illiquid account. At $t = 0$, the jump in p_0 and d_0 was not priced into

$p_{-1} = p_{ss}$. The unexpected excess return (or underperformance) on equity is paid to illiquid accounts in proportion to their size, and the balance sheet of the mutual fund at $t = 0$ is

$$A_{ss}(1 + r_{ss}) + cap_0 \bar{S} = p_0 + (1 - \tau)d_0 + (1 + r_{ss})B_{ss}, \tag{9}$$

which implies that capital gains in period 0 are equal to

$$cap_0 = \frac{p_0 - p_{ss} + (1 - \tau)(d_0 - d_{ss})}{\bar{S}}.$$

3.5 Government

Budget. The fiscal authority taxes labor income and dividends at the flat rate τ, issues real one-period bonds B_t, and spends public spendings G. It also makes two transfers to households, both zero in steady state. We call X_t^p, the positive transfer that the government pays to households, the *payout*, and the households who receive it the *beneficiaries*. We call X_t^f, the negative transfer that households pay to the government to finance the payout, the *financing*, and the households who pay it the *payers*. In practice, a payout can be a transfer or a tax cut, and the financing a tax increase or a transfer cut. The budget constraint is

$$B_t + \tau Y_t + X_t^f = (1 + r_t)B_{t-1} + G + X_t^p. \tag{10}$$

Incidence of transfers. An important mechanism in this paper is the relative incidence of the payout X_t^p and the financing X_t^f. Each has its own incidence: ϕ_p for the payout, which we call the *payout incidence*, and ϕ_f for the financing, the *financing incidence*. More precisely, a transfer of aggregate size X_t gives a household with earnings $\alpha_i e_t$ the amount $\iota_\phi(\alpha_i e_t)\, X_t$, with

$$\iota_\phi(\alpha_i e_t) = \frac{(\alpha_i e_t)^\phi}{\mathbb{E}\left[(\alpha e)^\phi\right]}, \tag{11}$$

so that the amounts add up to X_t. At $\phi = 0$, every household receives the same amount, a lump-sum transfer. At $\phi = 1$, the transfer is proportional to earnings, the incidence of labor income itself. At $\phi = 2$, a household earning twice the average receives four times what an average earner receives. Negative values target low earners. Both transfers go

through the liquid account.[6]

Fiscal shocks. A fiscal shock is composed of three elements:

1. a path for public debt B_t, which determines the path of deficits;
2. a path of payout X_t^p to beneficiaries at incidence ϕ_p;
3. a path of financing X_t^f by payers at incidence ϕ_f. Given B_t and X_t^p, X_t^f is the residual of the budget constraint (10).

In practice, both paths are AR(1). The payout decays at the rate ρ_X, $X_t^p = \rho_X^t X_0^p$, and debt rises with the payout and falls back with persistence ρ_B,

$$B_t - B_{ss} = \rho_B \left(B_{t-1} - B_{ss}\right) + X_t^p. \tag{12}$$

Debt duration. It is convenient to define D, a measure of the persistence of a deficit, to derive some analytical formulas for the cumulative multiplier in Section 5.3. Formally, D is the present value of the extra debt per unit of payout in present value,

$$D \equiv \frac{\sum_{t\geq 0} Q_t \left(B_t - B_{ss}\right)}{\sum_{t\geq 0} Q_t \, X_t^p}, \qquad Q_t \equiv \prod_{s=1}^{t} \frac{1}{1+r_s}, \tag{13}$$

where Q_t discounts date-t amounts to date 0 at the path of the real rate, with $Q_0 = 1$. It is the average number of quarters that a unit of payout stays in public debt before payers finance it, weighted by present value (Appendix B.1). For example, a government that borrows a payout and repays it in one go, with interest, after eight years has a debt duration of eight years. A purely redistributive shock, financed within the quarter, has $D = 0$. Under Equation (12), the government repays gradually, and D averages over these repayments.

After a fiscal shock, the central bank keeps the real rate at r_{ss} (Section 3.6), so $Q_t =$

[6] We restrict the incidence of the financing to $\phi_f \geq 0$. A more regressive financing would exceed the income of the poorest households, those with a low transitory shock, for a payout of 1% of quarterly output. A payout can take any incidence.

$(1 + r_{ss})^{-t}$. The debt path (12) then gives, for any path of payouts,

$$D = \frac{1 + r_{ss}}{1 + r_{ss} - \rho_B}. \tag{14}$$

The debt duration thus depends only on debt persistence and on the steady-state real rate.

3.6 Central bank and monetary policy

The central bank follows a real rate rule, keeping the *ex ante* real interest rate at its steady-state level absent shocks,

$$r_{t+1} = r_{ss} + \varepsilon_t^r, \tag{15}$$

where ε_t^r is an exogenous policy shock. After a fiscal shock, the real rate thus stays constant at r_{ss}.

Note that this real rate rule, along with the absence of nominal bonds, make inflation irrelevant for all real variables in our model, as in Auclert et al. (2024). Our model thus only require the presence of nominal rigidities to make output demand-constrained, but the actual variations in inflation are irrelevant for output, and we thus abstract from studying them in this paper.

A monetary policy shock also affects the government budget, by lowering the interest bill and, as output rises, raising tax receipts τY_t. We make two assumptions about what the government does with this windfall. First, it raises no financing, $X_t^f = 0$. Second, it returns the windfall to households through the payout X_t^p, following the fiscal rule

$$X_t^p = (1 + r_{ss} - \rho_B)(B_{ss} - B_{t-1}). \tag{16}$$

After a rate cut, debt falls below B_{ss}, so households receive a transfer at incidence ϕ_p, which we call the rebate. Each quarter, the rule pays out the interest on the gap between debt and its steady-state level, plus a share $1 - \rho_B$ of this gap, so debt returns to B_{ss} with the same persistence ρ_B as after a fiscal shock. The rebate thus reaches households

with an average delay of D quarters.[7]

3.7 Market clearing and equilibrium

Aggregate demand from households is their spending plus the goods used up by adjustment and intermediation, and aggregate wealth integrates the two accounts:

$$C_t = \int \left(c_t + \Psi_t\right) d\Phi_t + (1-\xi)\,\kappa(1+r_t)A^b_{t-1}, \qquad A^a_t = \int a_t\, d\Phi_t, \qquad A^b_t = \int b_t\, d\Phi_t.$$

Definition 1 (Equilibrium). *A competitive equilibrium is a path of household decisions* $\{b_t, a_t, c_t\}_{t\geq 0}$*, prices* $\{r_t, p_t\}_{t\geq 0}$*, the date-0 capital gain* cap_0*, fiscal policy* $\{X^p_t, X^f_t, B_t\}_{t\geq 0}$*, aggregates* $\{Y_t, N_t, d_t, \mathcal{E}_t\}_{t\geq 0}$ *and distributions* $\{\Phi_t\}_{t\geq 0}$ *such that:*

1. *Households solve their problem* (5) *given prices and hours;*
2. *The value of the firm satisfies the no-arbitrage condition* (8)*;*
3. *The union supplies the hours that firms demand;*
4. *The mutual fund balance sheet* (9) *holds;*
5. *The government budget constraint* (10) *holds, with debt following* (12) *after a fiscal shock and the payout following the fiscal rule* (16) *after a monetary shock;*
6. *The central bank follows the rule* (15)*;*
7. *The sequence of distributions* $\{\Phi_t\}$ *is consistent with the policy functions;*
8. *Markets clear in every period:*
   (a) *the asset market,* $A^a_t + A^b_t = p_t + B_t$*;*
   (b) *the goods market,* $Y_t = C_t + G$*;*
   (c) *the labor market,* $\int_j n_{jt}\, dj = N_t$*.*

[7] Appendix B.1 derives the timing of the rebate. The real rate moves after a monetary shock, but this changes the present value of the extra debt only at second order, so the delay is still given by Equation (14).

4 Calibration and Empirical Fit

We calibrate the model at the quarterly frequency to the United States in 2019.[8] The calibrated model reproduces the HtM–PLI gradient in both net-worth and liquid hand-to-mouth status, the wealth-to-output ratio and most of the top-1% wealth share, and it delivers an aggregate 1-year MPC of 0.32 that is not targeted. Standard parameters, the income process, the liquid return, and the fiscal block are set outside the model. Two restrictions on the Pareto tails of wealth and consumption pin down the curvature of the bequest motive and the death probability. The remaining household parameters match hand-to-mouth shares by PLI quintile and the wealth distribution, and the discount factor clears the asset market. Table 1 reports all parameter values.

4.1 Calibration strategy

Bequest motive and death probability. Following Gaillard et al. (2023), we derive an analytical approximation of the Pareto tails of wealth and consumption in the model, which yields

$$\theta^a = -\frac{\log(1-\xi)}{\log(1+r)}, \qquad \theta^c = \frac{\theta^a}{\eta}.$$

Appendix C.1 provides the derivation. Targeting a wealth tail of $\theta^a = 1.48$ and a consumption tail of $\theta^c = 3.11$, both from Gaillard et al. (2023), at the quarterly real rate of 0.5% (2% annualized), gives $\eta = 0.48$ and $\xi = 0.0074$.[9]

Remaining household parameters. Given the curvature on wealth η and the death probability ξ, we choose the intensity of the bequest motive γ, the Stone-Geary shifter ζ, and the adjustment cost parameters ψ_0 and ψ_1 by least squares on the following moments:

- The total and the liquid hand-to-mouth shares in each PLI quintile in the 2019 PSID wave;
- The top-1% wealth share in the 2018 DINA series (34%);
- The liquid share of net worth in the 2019 PSID wave (21.8%), with directly held stocks counted as liquid, as in Aguiar et al. (2025), and negative net liquid wealth

[8] We label the calibration 2019, the year of the latest PSID wave we use. The 2019 wave records income year 2018, and we use the 2018 DINA distributions accordingly.

[9] This implies an annual death probability of roughly 2.9%, and thus an expected (working) lifetime of about 34 years.

set to zero.

The model classifies households as in Section 2: hand-to-mouth in net worth when total wealth is below two months of labor income, and liquid hand-to-mouth when deposits are below one week of labor income and net worth is above that threshold. For these targets, PLI is measured from pre-tax labor income, the income concept of the model. The bequest parameters γ and ζ mostly move total hand-to-mouth shares and the top-1% share, and the adjustment-cost parameters ψ_0 and ψ_1 mostly move the liquid share and liquid hand-to-mouth shares. The fit yields $\gamma = 11.0$ for the intensity of the bequest motive, $\zeta = 19.8$ for the Stone-Geary shifter (about the size of aggregate wealth), and $\psi_0 = 54.6$ and $\psi_1 = 18.7$. The discount factor $\beta = 0.983$ clears the asset market at the 2% annual rate.

Liquid return. The intermediation wedge κ is set so that deposits pay no real interest at the 2019 rate, $r^b = 0$, a wedge of 2% a year. In 1970 and along transitions, the liquid return moves one for one with r.

Labor income. Permanent labor income levels $\{\alpha_i\}$ match the labor income shares of seven groups (the four bottom quintiles, P80–90, P90–95, and the top 5%) computed from the DINA microdata of Piketty et al. (2018) for 2018. The idiosyncratic process is estimated by minimum distance on the autocovariances of residual log pre-tax labor income in our PSID sample, following Straub (2019, Appendix D.4) (Appendix A.5). The persistent component has annual persistence $\rho_e = 0.94$ and unconditional standard deviation $\sigma_e = 0.38$, and the transitory component an annual standard deviation $\sigma^{tr} = 0.20$. We discretize the persistent component with the Rouwenhorst method.

Government and firms. The rest of our calibration follows Auclert et al. (2025) closely. We normalize steady-state quarterly output to $Y_{ss} = 1$ and set public debt to $B_{ss} = 4$, an annual debt-to-output ratio of 100%. Government spending is $G_{ss} = 0.20$, and the tax rate $\tau_{ss} = 0.22$ is the residual that closes the government budget constraint. Debt persistence is $\rho_B = 0.975$ in both the fiscal and the monetary experiments, a debt duration of 8.4 years. The markup $\mu = 1.11$ is chosen so that total wealth, $p_{ss} + B_{ss}$, matches an annual wealth-to-output ratio of 5.[10]

[10]The markup determines steady-state profits and therefore the equity value, $p_{ss} = (1 - \tau)d_{ss}/r_{ss}$.

Table 1: Calibration of the two-asset HANK model

Parameter	Description	Value	Source
Households			
β	Discount factor	0.983	Asset market clearing at annual r of 2%.
η	Curvature on wealth	0.48	Pareto coeff. on wealth/cons. tails (1.48/3.11).
γ	Strength of bequest motive	11.0	Joint fit: total and liquid HtM by PLI quintile (PSID), top-1% share (DINA), liquid share (PSID).
ζ	Stone-Geary shifter	19.81	Joint fit, same moments.
ψ_0	Adjustment cost, offset	54.6	Joint fit, same moments.
ψ_1	Adjustment cost, scale	18.7	Joint fit, same moments.
κ	Deposit wedge (annual)	0.020	Liquid return of zero at the 2019 real rate.
σ^{tr}	Std of transitory income shocks (annual)	0.20	PSID income process.
ξ	Death probability (quarterly)	0.0074	Wealth Pareto tail 1.48.
ρ_e	Persistence of income shocks (annual)	0.94	PSID, minimum-distance estimation.
σ_e	Uncond. std of persistent component	0.38	PSID, minimum-distance estimation.
Government and Firms			
B_{ss}	Public debt	4.00	Annual B/Y of 100%.
G_{ss}	Gov. spending	0.20	National accounts.
τ_{ss}	Tax rate	0.22	Residual (gov. budget).
ρ_B	Persistence of public debt	0.975	Fiscal rule.
μ	Markup	1.11	Annual W/Y of 5.0.

4.2 Model fit

Figure 5 compares hand-to-mouth shares by PLI quintile in the model and in the data. The model reproduces the HtM–PLI gradient of Section 2. The total hand-to-mouth share falls from 66% in the bottom quintile to 15% in the top quintile, against 62% and 18% in the 2019 wave, and the aggregate share is 38.5% in both. Liquid hand-to-mouth status also declines with PLI, from 29% to 13%, against 25% and 9%. The main miss is net-worth hand-to-mouth status at the top, 2% in the top quintile against 9% in the data. The PSID count includes households with negative net worth, which the model rules out.

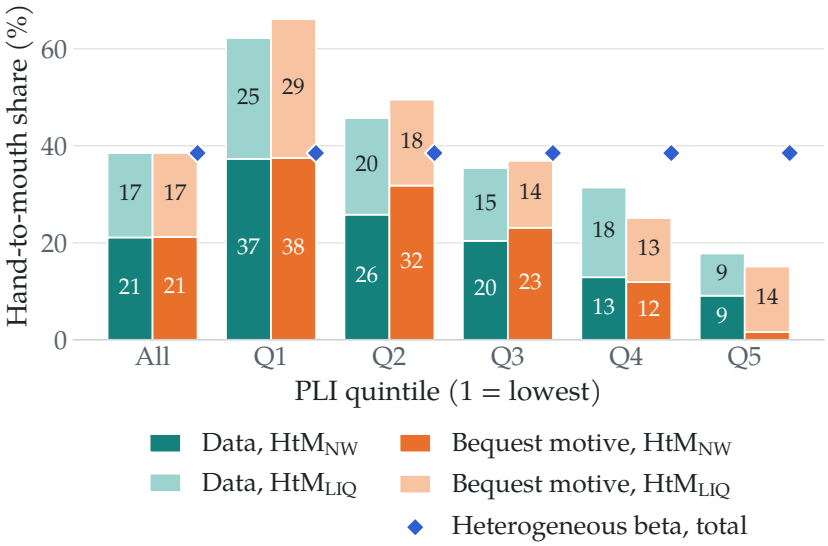


Figure 5: Hand-to-mouth shares by PLI quintile

Note: Hand-to-mouth shares in the 2019 PSID wave, the calibration targets, and in the bequest-motive model, in the aggregate and by quintile of PLI, measured from pre-tax labor income, with net-worth and liquid hand-to-mouth stacked. Diamonds: total hand-to-mouth share in the heterogeneous beta model of Section 4.4.

The model also reproduces the fall of hand-to-mouth status with current labor income, from 66% to 12% across quintiles against 59% and 18% in the PSID, which determines who receives the payouts and who finances them in Section 5 (Appendix C.2). This match follows largely from the PLI targets, because PLI drives most differences in current labor income in the model.

Table 2 compares the model wealth distribution with the 2018 DINA series of Piketty et al. (2018). The top-1% wealth share is 33%, against 34% in the data. The model understates the middle 40% (19% against 30%) and overstates the top 10% (80% against

69%). The liquid share of wealth is 21.5%, close to its PSID target of 21.8%. The model has the empirical Pareto coefficient $\theta^a = 1.48$ only asymptotically, and its tail is thinner between \$10 million and \$1 billion, so the top-0.1% share is 8%, against 18% in the data (Appendix C.1).

Table 2: Wealth distribution and hand-to-mouth shares, 2019

	Bottom 50%	Middle 40%	Top 10%	Top 1%	Top 0.1%	HtM (NW)	HtM (LIQ)
Data	1.4	29.5	69.1	34.4	18	21.1	17.4
Model	0.9	19.4	79.7	32.7	8.3	21.2	17.4

Note: Wealth shares (%) of total wealth, liquid plus illiquid, in the DINA data for 2018 and in the model. HtM (NW) and HtM (LIQ): net-worth and liquid hand-to-mouth shares (%), PSID 2019 wave.

4.3 How a non-homothetic bequest motive matches both the wealth distribution and the HtM–PLI gradient

Standard HANK models with only idiosyncratic shocks generate too little wealth inequality and too few hand-to-mouth households when calibrated to the observed wealth-to-output ratio (Kaplan et al. 2018, Auclert et al. 2024), and hence an aggregate MPC well below empirical estimates. Households accumulate buffers up to a target and then dissave, so large fortunes never emerge, and matching the wealth-to-output ratio assigns too much wealth to the bottom and middle of the distribution, which lowers their MPCs.

A bequest motive fixes both problems, because it adds a saving motive that strengthens with wealth. Households value the wealth they bequeath, and this motive is non-homothetic, so bequests are a luxury good: once their consumption needs are met, the richest households devote almost all marginal income to additional saving. The model thus matches the high saving rates documented for the top of the distribution (Dynan et al. 2004, Kumhof et al. 2015, Straub 2019). As wealth grows, these households hold it in the illiquid account and their wealth approaches the accumulation rule $a_t \approx (1+r)a_{t-1}$ (Appendix C.1).

Households at the bottom of the PLI distribution behave differently. Their consumption is low, so its marginal utility is high and the bequest motive matters little next to it, and they behave like buffer-stock savers. The strong saving of wealthy households

pushes the equilibrium return down, so that $(1-\xi)\beta(1+r) = 0.981$, about 7% of impatience per year, which weakens the incentive of these households to save and keeps them near the borrowing constraint. High-PLI households value bequests more relative to their higher consumption and hold wealth far above the hand-to-mouth threshold. The model therefore concentrates hand-to-mouth status at the bottom of the PLI distribution, as in Figure 5. Under homothetic preferences, buffers would scale with permanent income, and the hand-to-mouth share would be flat across PLI quintiles.

The liquid account adds a second margin. Savers hold most of their wealth in the illiquid account, which pays 2% a year against nothing on deposits. Since moving funds is costly, a bad transitory shock can leave them short of liquidity with positive net worth, so liquid hand-to-mouth households appear in every PLI quintile.

4.4 Homothetic benchmark

To isolate the role of the non-homothetic bequest motive, we build a homothetic benchmark with $\gamma = 0$ and heterogeneous discount factors (Appendix C.3). In this heterogeneous beta model, the discount factor follows a two-state Markov chain independent of permanent income (Krusell & Smith 1998, Carroll et al. 2017, Auclert et al. 2025), and the adjustment-cost scale ψ_0 is proportional to PLI, so preferences and frictions are homothetic. Its patient share, spread and adjustment-cost curvature ψ_1 reproduce the bequest-motive model's aggregate 1-year MPC, net-worth and liquid hand-to-mouth shares exactly. The two models have the same aggregate 1-year MPC, 0.32, but hand-to-mouth status in the heterogeneous beta model depends on the discount-factor type, which is independent of PLI, so the hand-to-mouth share is 38.5% in every PLI quintile (diamonds in Figure 5).

The empirical literature finds that MPCs out of transfers fall with income when nothing else is held fixed, and that this relation weakens or reverses once wealth is held fixed. The bequest-motive model reproduces both patterns, while the heterogeneous beta model reproduces only the second (Appendix C.4).

5 Fiscal Incidence and the Cumulative Multiplier

This section studies how matching the HtM–PLI gradient changes the transmission of fiscal and monetary policy. We do so by comparing the bequest-motive model with the heterogeneous beta model of Section 4.4.

We find three results. First, the multiplier depends less on how long the debt lasts, and deficits finance themselves less, because the income an expansion creates goes largely to high-PLI households, who spend it slowly. Second, the multiplier depends much more on who receives the payout and who finances it. This effect is large: a package that redistributes toward high earners can have a negative multiplier even when deficits finance part of it. Third, the effect of a rate cut depends much more on who receives the interest savings, which the fiscal rule pays back to households.

Appendix G shows that these results also hold, mostly more strongly, in a model without a bequest motive in which the discount factor rises with PLI, fitted to the same gradient.[11] Our results are thus not an artefact of the specific preferences we use, but come from matching the HtM–PLI gradient observed in the data.

5.1 Aggregate consumption and the intertemporal Keynesian cross

We start our analysis by deriving the intertemporal Keynesian cross of Auclert et al. (2024) in our setting. We then show that fiscal policy moves output through two channels: the standard debt-financing effect and a redistribution effect.

Aggregate consumption responds to five aggregate paths, each through an iMPC matrix. A household with idiosyncratic earnings e in PLI group i receives post-tax labor income $Z_t \equiv (1-\tau)wN_t$ in proportion to $\alpha_i e_t$, the payout X_t^p and its financing X_t^f in proportion to $\iota_{\phi_p}(\alpha_i e_t)$ and $\iota_{\phi_f}(\alpha_i e_t)$, and the date-0 capital gain, of aggregate size $dcap \equiv \bar{S}\, cap_0$, in

[11] In this model, $\gamma = 0$ and each PLI group has its own discount factor, rising with its PLI rank, with the schedule and ψ_1 fitted to the same twelve moments as the bequest-motive model (Appendix G.2). It reproduces the HtM–PLI gradient but misses the liquid share of wealth, 13% against 21.8% in the data, and its lump-sum deficit multiplier at $\rho_B = 0.975$ is 0.19, against 0.41.

proportion to its illiquid wealth, and it faces the real rate r_t.[12] To first order,[13]

$$d\mathbf{C} = \mathbf{M}^{Z}\, d\mathbf{Z} + \mathbf{M}^{\phi_p} d\mathbf{X}^{p} - \mathbf{M}^{\phi_f} d\mathbf{X}^{f} + \mathbf{M}^{r} d\mathbf{r} + \mathbf{m}^{cap}\, dcap, \tag{17}$$

where the entry (t, s) of the iMPC matrix $\mathbf{M}^{x}$ is the change in aggregate spending at date t after a unit change in x at date s (Auclert et al. 2024). Note that this decomposition holds in both the bequest-motive model and the heterogeneous beta model, so every difference between them comes from their iMPC matrices.

With the real rate held fixed, substituting the financing from the government budget constraint (10) and the debt rule (12), and imposing goods market clearing $d\mathbf{Y} = d\mathbf{C}$, gives an intertemporal Keynesian cross for fiscal policy (Appendix D.1),

$$(\mathbf{I} - \mathbf{K})\, d\mathbf{Y} = \underbrace{\mathbf{M}^{\phi_f}\left[\mathbf{I} - \sum_{k \geq 1} \pi_k \,(1+r)^k\, \mathbf{L}^k\right] d\mathbf{X}^{p}}_{\text{Debt-financing effect}} + \underbrace{\left(\mathbf{M}^{\phi_p} - \mathbf{M}^{\phi_f}\right) d\mathbf{X}^{p}}_{\text{Redistribution effect}}, \qquad \mathbf{K} \equiv (1-\tau)\overline{\mathbf{M}}^{Z} + \tau \mathbf{M}^{\phi_f}, \tag{18}$$

where $\pi_k \equiv \frac{1}{D}\left(1 - \frac{1}{D}\right)^{k-1}$ depends only on the debt duration D, and $\overline{\mathbf{M}}^{Z} \equiv \frac{1}{\mu}\mathbf{M}^{Z} + \frac{\mu - 1}{\mu}\, \mathbf{m}^{cap}\mathbf{q}'$ is the iMPC matrix of post-tax income, paid as wages and dividends. The right side is the first impulse to demand, and the matrix $\mathbf{K}$, the dynamic counterpart of the Keynesian multiplier, amplifies it. Extra spending raises output, and hence wages, dividends and tax receipts, which return to households and are spent in turn.

The debt-financing effect depends on the debt duration D and the financing incidence ϕ_f. Borrowing lets the government pay the payout today and raise the financing only later: payers repay the payout with interest k quarters later, with weight π_k. Each quarter, households repay a share $1/D$ of the outstanding debt and its interest, so the weights π_k sum to one and the average delay of repayment, $\sum_k k\, \pi_k$, is exactly the debt duration D.[14] The package has zero present value, since payers eventually finance the

[12] We count the spending of newborns, who inherit the estates of dying households, with that of the household whose estate they inherit (Appendix D.1). With this convention, Equation (18) reproduces the full model to numerical precision.

[13] Bold letters denote sequences and the matrices operating on them; $\mathbf{q} \equiv \left(1, (1+r)^{-1}, (1+r)^{-2}, \dots\right)'$ is the steady-state discount vector, $\mathbf{L}$ the lag matrix and $\mathbf{I}$ the identity. As in Auclert et al. (2024) and Auclert et al. (2025), the iMPC matrices are the sequence-space Jacobians of consumption. Appendix D contains all derivations and proofs.

[14] A pure redistribution, financed within the quarter, has $D = 0$: households repay the transfer in the same quarter, and the debt-financing effect is zero. Appendix D.1 gives the cross for any debt path.

whole payout, but it raises demand today, because Ricardian equivalence does not hold in our setting: households spend more out of current income than they cut in response to the future financing.

The redistribution effect depends only on who receives the payout and who pays the financing, ϕ_p and ϕ_f. It arises because the payout and the financing reach households with different iMPCs. Demand rises when beneficiaries raise their spending faster than payers cut theirs, even in a pure redistribution without debt, $D = 0$.

In the rest of the paper, we measure the output response to a package by its cumulative multiplier $\mathcal{M} \equiv \mathbf{q}' d\mathbf{Y} / \mathbf{q}' d\mathbf{X}^p$, the present value of output per present value of the payout. Because the intertemporal Keynesian cross is linear, we can also split the multiplier into a debt-financing effect and a redistribution effect,[15]

$$\mathcal{M}(\phi_p, \phi_f, D) = \underbrace{\mathcal{M}(\phi_f, \phi_f, D)}_{\text{Debt-financing effect}} + \underbrace{\mathcal{M}(\phi_p, \phi_f, 0)}_{\text{Redistribution effect}} . \tag{19}$$

5.2 Incidence and iMPCs

The HtM–PLI gradient changes both effects through the iMPC matrix of a transfer at each incidence. To understand this, note that a household spends extra income the same way whether it comes from wages, a transfer or interest on deposits. Households in different states, however, spend the same income differently. Thus, in the aggregate, the iMPC matrix of an income flow therefore depends only on how the income is distributed across households. This lets us split the iMPC matrix of any transfer into three parts: the response to an income shock equally distributed between households (a lump-sum transfer), a PLI term, from how the transfer is distributed across PLI groups, and an idiosyncratic term, from how it is distributed within them. Proposition 1 states this result formally.

Proposition 1 (Incidence and iMPCs)**.** *Let $m_{t,s}$ be the iMPC of a household and $\bar{m}_{i,t,s} \equiv \mathbb{E}_i[m_{t,s}]$ the average iMPC of PLI group i, with population share ω_i. Since the distribution*

[15] Counting heirs, households spend all their income in present value, so $\mathbf{q}'\mathbf{K} = \mathbf{q}'$ and $\mathbf{I} - \mathbf{K}$ is not invertible, as in Auclert et al. (2024). Both impulses have zero present value, so the cross has solutions, and asset market clearing selects one (Appendix D.1).

of e is the same in every PLI group, the incidence weight (11) *factors as $\iota_\phi = \tilde{\alpha}_i^\phi \tilde{e}^\phi$, where $\tilde{\alpha}_i^\phi \equiv \alpha_i^\phi / \sum_j \omega_j \alpha_j^\phi$ is what a household of group i receives relative to the average, so that $\sum_i \omega_i \tilde{\alpha}_i^\phi = 1$, and $\tilde{e}^\phi \equiv e^\phi / \mathbb{E}[e^\phi]$. Then, for every incidence ϕ, the iMPC matrix of a transfer is that of a lump-sum transfer plus two covariances between what households receive and their iMPCs,*

$$\mathbf{M}_{t,s}^\phi = \mathbf{M}_{t,s}^0 + \underbrace{\mathrm{Cov}_\omega\left(\tilde{\alpha}_i^\phi,\, \bar{m}_{i,t,s}\right)}_{\textit{PLI term}} + \underbrace{\sum_i \omega_i \tilde{\alpha}_i^\phi \,\mathrm{Cov}_i\left(\tilde{e}^\phi,\, m_{t,s}\right)}_{\textit{idiosyncratic term}}, \tag{20}$$

where Cov_ω is the covariance across PLI groups, weighted by population shares, and Cov_i the covariance within group i.

Proof. Appendix D.2. □

The PLI term is zero in the heterogeneous beta model and large in the bequest-motive model. The idiosyncratic term arises in any model with uninsurable earnings risk, since households with high current earnings save for worse times and have low iMPCs. The PLI term arises only when average iMPCs differ across PLI groups. Under homothetic preferences, with discount factors independent of PLI and adjustment costs proportional to PLI, as in the heterogeneous beta model, every PLI group spends extra income the same way, and the PLI term is zero at every incidence (Lemma 3). In the bequest-motive model, average iMPCs fall steeply with PLI. Many low-PLI households are hand to mouth, while high-PLI households value bequests as a luxury good and save extra income, much of which is spent only by their heirs.

The PSID points to a large PLI term. Hand-to-mouth status is the usual empirical marker of a high MPC (Aguiar et al. 2025), and the left panel of Figure 6 shows the share of a transfer that reaches hand-to-mouth households at each incidence. In the 2019 PSID wave, as in the bequest-motive model, about half of a transfer targeted to low earners ($\phi = -1$) reaches hand-to-mouth households, against about 15% of a transfer concentrated on high earners ($\phi = 2$). Most of this difference arises across PLI groups, in the PLI term of Proposition 1. In the heterogeneous beta model, hand-to-mouth households receive the same share at every incidence.

Since hand-to-mouth households have high MPCs, the 1-year MPCs of the two models

follow the same pattern as their hand-to-mouth shares (right panel of Figure 6).[16] With the HtM–PLI gradient, a transfer to high earners raises consumption less in the first year, and a transfer to low earners raises it more. A transfer concentrated on high earners ($\phi = 2$) has a 1-year MPC of 0.09 in the bequest-motive model, against 0.25 in the heterogeneous beta model, and most of its gap with a lump-sum transfer comes from the PLI term, the shaded area.

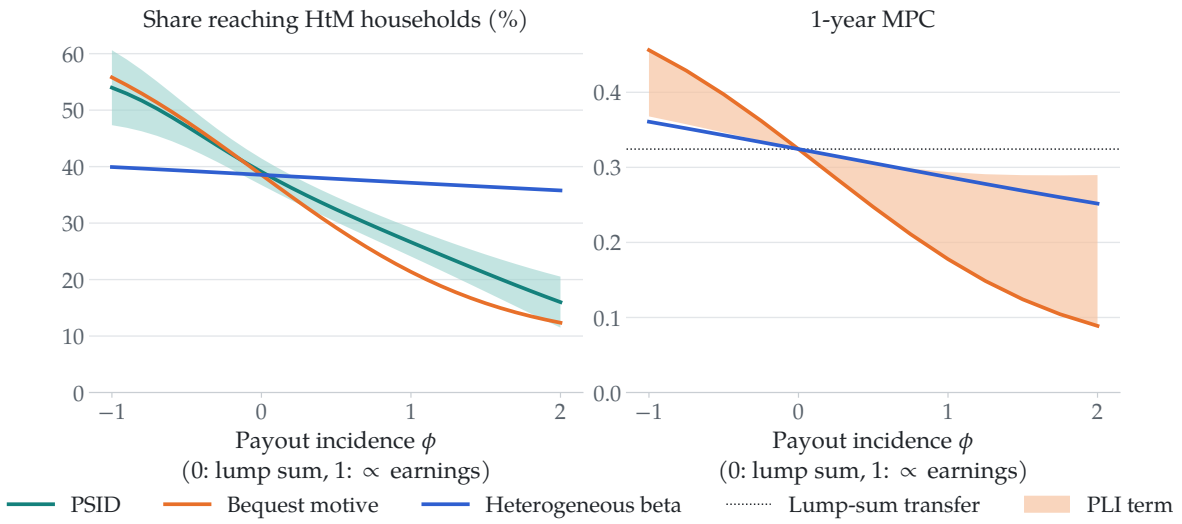


Figure 6: Who receives a transfer and how fast it is spent, by incidence

Note: Transfer paid today in proportion to $(\alpha_i e)^\phi$, against its incidence ϕ. Left: share of the transfer reaching hand-to-mouth households. PSID: 2019 wave, pre-tax labor income of head and spouse, with hand-to-mouth thresholds on that income; in the transfer weights, labor income is floored at 2.7% of its mean, the lowest earnings state of the model, and below $\phi = -1$ a few households would receive most of the transfer; shaded area: 95% interval from 200 bootstrap draws of households. Right: 1-year MPC of the transfer; shaded area: the PLI term of Proposition 1 in the bequest-motive model, zero in the heterogeneous beta model. 2019 calibration.

5.3 Cumulative multipliers and incidence

Matching the HtM–PLI gradient changes fiscal multipliers in two ways. First, how long the debt lasts matters less, and deficits finance themselves less. Second, who receives a package and who finances it matter much more.

To build intuition, consider the case in which households do not anticipate future payments. The cumulative multiplier is then a static Keynesian cross in which each MPC is replaced by its cumulative MPC, summed over horizons (Proposition 3 in

[16] In the bequest-motive model, hand-to-mouth status accounts for about two thirds of the gap in 1-year MPCs between a lump-sum transfer and a transfer proportional to earnings.

Appendix D.3),

$$\mathcal{M} = \underbrace{\frac{D}{\sum_{h\geq 1}\left(1-\mathrm{MPC}_h^Y\right)}}_{\text{Debt-financing effect}} + \underbrace{\frac{\sum_{h\geq 1}\mathrm{Cov}\left(\iota_{\phi_p}-\iota_{\phi_f},\, c_h\right)}{\sum_{h\geq 1}\left(1-\mathrm{MPC}_h^Y\right)}}_{\text{Redistribution effect}}. \tag{21}$$

Here c_h is the cumulative MPC of a household, the share of an income flow it has spent within h quarters, in present value, Cov is the covariance across all households, and MPC_h^Y is the average cumulative MPC out of the new income an expansion creates, paid as wages, dividends and tax receipts returned to payers. D is the debt duration, and all sums are in quarters, like D.

As in a static Keynesian cross, the numerators are the initial impulse to demand, and the common denominator captures the general-equilibrium amplification through further rounds of spending. The slower households spend the new income an expansion creates, the larger the denominator and the weaker this amplification. The numerator of the redistribution effect is the covariance between what a household receives net of what it pays and how fast it spends, and by Proposition 1 it splits into a PLI term and an idiosyncratic term. With anticipation, payers save part of the financing before it is due, and deficits raise output less than the formula implies.

Debt duration matters less, and deficits finance themselves less. In the bequest-motive model, a deficit raises output less than in the heterogeneous beta model, and the gap grows with the debt duration (Figure 7). Doubling the duration from 10 to 20 years raises the multiplier of a lump-sum deficit by 1.0, against 1.5. The reason is that the new income an expansion creates goes largely to high-PLI households, since wages are paid in proportion to earnings and dividends in proportion to illiquid wealth. In the bequest-motive model, these households have low MPCs, so each round of spending is smaller and the amplification in the denominator of Equation (21) is weaker. In the heterogeneous beta model, MPCs do not depend on PLI, so new income is spent at the same speed whoever receives it. Since tax receipts are proportional to output, deficits also finance themselves less, in the sense of Angeletos et al. (2024). At 8.4 years, the extra tax receipts cover 9% of the payout, against 12%.[17]

[17] In the one-asset version of the model (Appendix F), the gap is larger: at 8.4 years, the lump-sum deficit has a multiplier of 0.20, against 0.44.

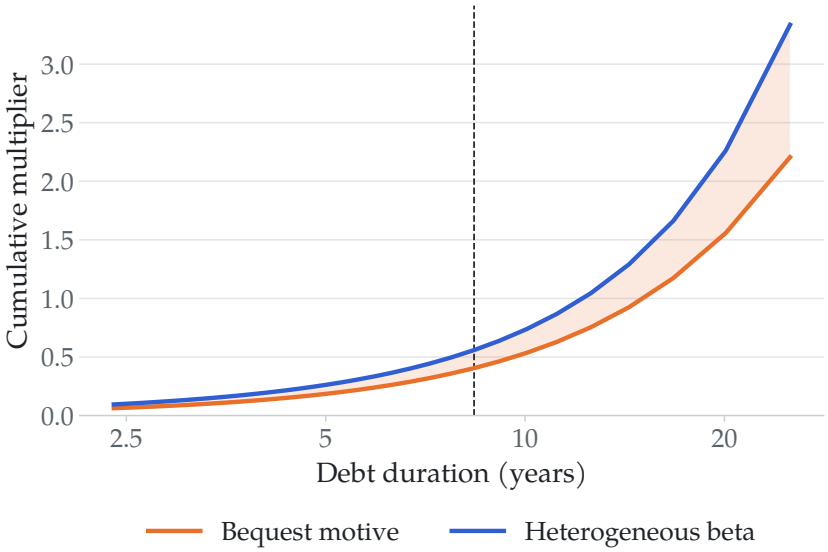


Figure 7: Cumulative multiplier of a lump-sum deficit by debt duration

Note: Cumulative multiplier of a lump-sum payout of 1% of quarterly output with persistence 0.9, financed lump sum, against the debt duration D in years, on a log scale. The multiplier of a pure deficit barely depends on its incidence. Dashed line: baseline, 8.4 years. 2019 calibration.

Fiscal incidence becomes first order. In the bequest-motive model, who receives a package and who finances it strongly affect the cumulative multiplier. A pure redistribution from high to low earners ($\phi_p = -1$, $\phi_f = 2$) has a cumulative multiplier of 1.2, three times that of a lump-sum deficit at the baseline debt duration. In the heterogeneous beta model, the same redistribution has a multiplier of 0.1 (Figure 8).

The covariance in Equation (21) explains the difference. In the bequest-motive model, low-PLI households spend extra income quickly, because many of them are hand to mouth, while high-PLI households spend it slowly and pass much of it to their heirs. Taking income from high earners and giving it to low earners therefore raises spending, and the covariance is large and positive. In the heterogeneous beta model, how fast a household spends is only weakly related to its labor income, so the covariance is close to zero at every incidence.

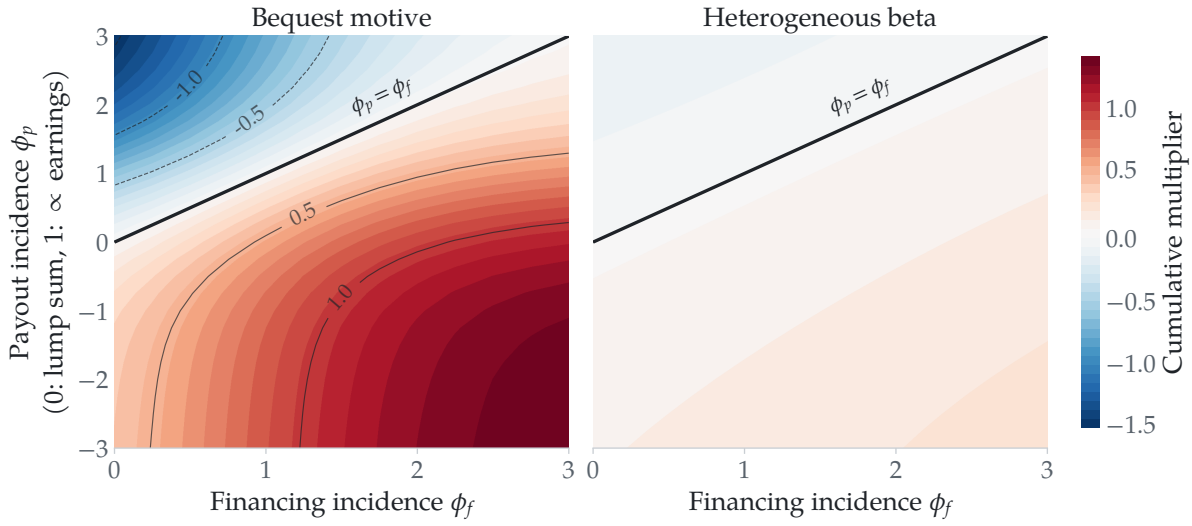


Figure 8: Cumulative multipliers of pure redistributions

Note: Cumulative multiplier of a payout of 1% of quarterly output with persistence 0.9 at incidence ϕ_p (rows), financed in the same quarter at incidence $\phi_f \geq 0$ (columns), with no debt. On the dashed line, $\phi_p = \phi_f$, and the package is zero. 2019 calibration.

Debt can stay low for years without lowering output. Since a redistribution from high to low earners raises output, it leaves room to lower debt. Suppose the government finances a package in proportion to earnings ($\phi_f = 1$), pays it out lump sum ($\phi_p = 0$), and delays the payout just enough to keep output at its steady state in every quarter. In the bequest-motive model, each unit of financing then keeps debt one unit lower for 8 years, about our baseline debt duration. In the heterogeneous beta model, it does so for less than a year (Figure 19 in Appendix D.4).

The reason is the one behind fiscal incidence. Payers cut their spending slowly, and beneficiaries spend quickly, so a small payout is enough to replace the spending that payers give up in the first years. The government keeps the rest of the financing as a primary surplus, and it pays this out as payers' spending cuts continue. In the heterogeneous beta model, payers and beneficiaries spend at similar speeds, so the payout must closely follow the financing.

This room is temporary. Debt returns to its steady state and beneficiaries receive the whole financing in present value, so the primary surplus has a present value of zero. High earners' labor supply and tax avoidance also do not respond in the model, so we read these durations as upper bounds.

5.4 Application: the One Big Beautiful Bill Act

With the HtM–PLI gradient, a package that redistributes toward high earners can lower output, even when deficits finance part of it. How much do these results matter in practice? We apply the model to the One Big Beautiful Bill Act, passed by Congress in July 2025. The law cuts taxes by about 1.5% of GDP a year over 2025–2034 and finances 45% of the cost with tax increases and transfer cuts, mostly to Medicaid, ACA premium tax credits, SNAP and student loans (Gale & Page 2026). Through the lens of our model, the tax cuts are a payout, and the transfer cuts and tax increases are negative transfers, the financing of the package. We set their incidences to match CBO's change in household resources by income decile for the House-passed bill. This gives $\phi_p = 1$, so the top 10% receive about 45% of the payout, and $\phi_f = 0.25$, a financing that rises slightly with earnings. Debt finances the rest, \$2.9 trillion over 2025–2034.[18] We assume none of this debt is repaid before 2035. From then on, payers finance it at the same incidence, with our baseline debt duration of 8.4 years (Appendix D.5).

The law lowers output in the bequest-motive model and raises it in the heterogeneous beta model, which has the same aggregate MPC but no HtM–PLI gradient. Its cumulative multiplier is -0.06 in the first and 0.57 in the second (Table 3). The redistribution effect drives the gap. The law pays in proportion to earnings and charges households a financing that rises much less with earnings, so it moves resources from low-PLI households, who spend most of their extra income at once, to high-PLI households, who save most of it.

Table 3: Cumulative multiplier of the One Big Beautiful Bill Act

	Bequest-motive model	Heterogeneous beta model
Debt-financing effect	0.45	0.60
Redistribution effect	-0.51	-0.04
Cumulative multiplier	-0.06	0.57

Note: Present value of output per present value of the payout, split as in Equation (19). The payout, the tax cuts of the law, about 1.5% of GDP a year spread evenly over 2025–2034, has incidence $\phi_p = 1$. The financing has incidence $\phi_f = 0.25$: the transfer cuts and tax increases of the law, 45% of the payout, over 2025–2034, then the debt, on which interest accrues through 2034, at persistence $\rho_B = 0.975$. Real rate fixed. 2019 calibration.

[18] CBO's estimate of the law's deficits, \$3.4 trillion (Congressional Budget Office 2025*b*), also counts the spending on defense and border security that we leave out.

5.5 Monetary policy

Matching the HtM–PLI gradient also changes the transmission of monetary policy, because a rate cut directly affects the government budget constraint. Indeed, an interest rate cut lowers the interest payment on public debt and, as output rises, raises tax receipts. The fiscal authority must then decide what to do with this windfall. Under the fiscal rule (16), debt falls first, and the savings return to households as a rebate at incidence ϕ_p, with an average delay equal to the debt duration D. Goods market clearing then gives the intertemporal Keynesian cross of a rate cut (Appendix D.1),

$$\big[\mathbf{I} - \mathbf{K}_R(\phi_p)\big]\, d\mathbf{Y} = \underbrace{\overline{\mathbf{M}}^r d\mathbf{r}}_{\text{Direct effect}} \quad \underbrace{-\,B\,\mathbf{M}^{\phi_p}\mathbf{R}\, d\mathbf{r}}_{\text{Rebate}}, \qquad \mathbf{K}_R(\phi) \equiv (1-\tau)\overline{\mathbf{M}}^Z + \tau\,\mathbf{M}^{\phi}\mathbf{R}, \quad (22)$$

where $\overline{\mathbf{M}}^r$ is the response of spending to the rate, including the capital gain on equity, and $\mathbf{R}$ is the repayment schedule of the fiscal rule (Appendix B.1).

Who receives the interest savings matters more. In the bequest-motive model, the response of output to a rate cut depends on how the fiscal authority returns the interest savings (Figure 9).[19] It is largest when the rebate goes quickly to low earners, in the bottom left of the left panel, and smallest when it goes slowly to high earners, in the top right, where the rate cut even lowers output. At the baseline duration, moving the rebate from low to high earners lowers the response by 72%.[20] In the heterogeneous beta model, the response depends on how fast the rebate is paid but barely on who receives it, and the same move lowers it by 13%. The reason is the one behind fiscal incidence. Changing who receives the rebate is a pure redistribution of it, since the direct effect of the rate cut is the same at every incidence (Equation (41) in Appendix D.3). How fast the rebate is paid matters in both models. A rate cut lowers the interest that bondholders receive today. If the rebate comes years later, households lose this income now and get it back only in the future, as if the government ran a temporary surplus. Households cut their spending more in response to income lost today than they raise it for income returned later, so a slower rebate weakens the response of output. This is the debt-financing

[19] The debt duration sets the timing of the rebate. With a short duration, the fiscal rule returns the interest savings in large transfers over a few years. With a long duration, it returns them in small transfers spread over decades. The present value of the rebate is the same in both cases.

[20] In the one-asset version of the model (Appendix F), moving the rebate from low to high earners at a delay of 8.4 years lowers the semi-elasticity from 12.2 to 2.9, against 10.8 to 9.4.

effect of Section 5.1 in reverse.

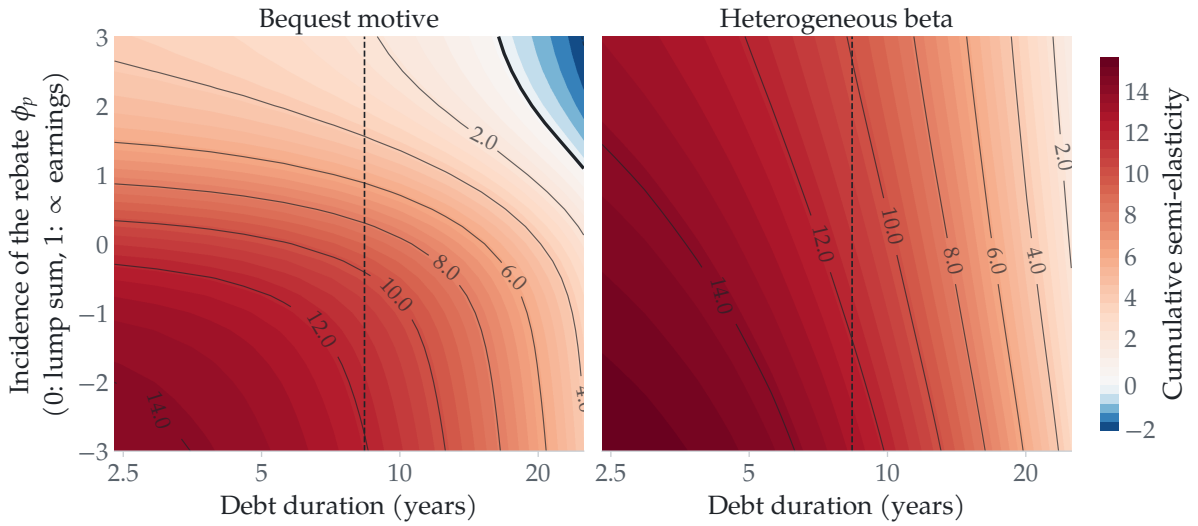


Figure 9: Response to a rate cut by the incidence and the delay of the rebate

Note: Cumulative output semi-elasticity, the present value of dY_t/Y per present value of the rate cut, for a cut of 1 percentage point in the annual real rate with quarterly persistence 0.9. Rows: incidence ϕ_p of the rebate paid by the fiscal rule (16). Columns: average delay of the rebate, the debt duration D, in years, on a log scale. Dashed line: baseline, 8.4 years. 2019 calibration.

6 The Effects of Rising Permanent Labor Income Inequality

Section 5 showed that, once a model matches the HtM–PLI gradient, the incidence of a fiscal package is first order. This section asks how the rise in PLI inequality since 1970 changed this, and with it the transmission of fiscal and monetary policy. We re-solve the bequest-motive model with the 1970 PLI distribution and keep every other parameter at its 2019 value. The real rate clears the asset market, and the tax rate balances the government budget. Under homothetic preferences, this experiment would change no iMPC matrix and no multiplier (Proposition 2), so every change below comes from the non-homotheticities that generate the HtM–PLI gradient.

We find three results. First, households below the top spend extra income faster, while labor income shifts toward high-PLI households, who save it. The gap between how fast households spend a transfer and how fast they spend labor income therefore widens. Second, redistributions between PLI groups move output more. At a given debt duration, deficits move output less, but at a given fiscal rule the lower rate lengthens the

debt and offsets this loss. Third, the response to a rate cut barely changes on average, but it depends more on who receives the rebate.

6.1 Rising inequality and cumulative MPCs

Rising PLI inequality lowers the real rate and moves households below the top toward the borrowing constraint (Table 4). High-PLI households receive a larger share of labor income and, because bequests are a luxury good, save most of it. For the asset market to clear, the annual real rate falls from 2.5% to 2.0%. The hand-to-mouth share rises from 36% to 39%.

Table 4: Real rate, HtM share, 1-year MPCs and EIS, 1970 and 2019

	r (%)	HtM	MPC, lump sum	MPC, labor income	EIS
1970	2.53	0.36	0.30	0.18	0.95
Partial equilibrium	2.53	0.37	0.28	0.15	0.95
2019	2.00	0.39	0.32	0.18	0.95

The changes are largest in the middle of the PLI distribution (Table 5). The bottom four quintiles lose 15 to 19% of their PLI relative to the mean. The hand-to-mouth share and the 1-year MPC out of a lump-sum transfer rise most in the second to fourth quintiles. They rise less in the bottom quintile, because almost two thirds of these households were already hand to mouth in 1970, and they barely move in the top quintile.

Table 5: Rising PLI inequality by PLI quintile, 1970 to 2019

	Q1	Q2	Q3	Q4	Q5
PLI relative to the mean (% change)	-17.6	-17.8	-18.9	-14.9	19.1
HtM share (change, pp)	2.4	4.0	5.6	3.9	-1.1
1-year MPC, lump sum (change, pp)	1.4	3.2	5.2	4.5	0.1
1-year MPC, labor income (change, pp)	1.6	3.5	4.8	3.6	-1.3

These changes reach policy through the iMPC matrices. Rising inequality changes the iMPC matrix of a transfer in two ways. First, the same incidence rule hands each PLI group a different share of the transfer, because the PLI distribution has changed. Second, each group spends its share differently, because its PLI and the prices it faces have changed. Proposition 2 separates the two, extending the split of Proposition 1 to a comparison of two economies.

Proposition 2 (Comparing economies)**.** *Consider two economies with the same population shares ω_i and different PLI distributions. With the notation of Proposition 1, let $\mathbf{M}_i^\phi \equiv \mathbb{E}_i[\tilde{e}^\phi m]$ be the iMPC matrix of PLI group i per unit of income it receives at incidence ϕ, so that $\mathbf{M}^\phi = \sum_i \omega_i \tilde{\alpha}_i^\phi \mathbf{M}_i^\phi$. Let Δ denote the change from the first economy to the second, and a prime the second economy. For every incidence ϕ,*

$$\Delta \mathbf{M}^\phi = \underbrace{\mathrm{Cov}_\omega\left(\Delta\tilde{\alpha}_i^\phi,\, \mathbf{M}_i^{\phi\prime}\right)}_{\textit{composition}} + \underbrace{\sum_i \omega_i \tilde{\alpha}_i^\phi\, \Delta\mathbf{M}_i^\phi}_{\textit{behavior}}, \tag{23}$$

and the same split holds for cumulative MPCs. The composition term is zero for a lump-sum transfer. Under homothetic preferences, with discount factors independent of PLI and adjustment costs proportional to PLI, both terms are zero at every incidence, and every iMPC matrix and every multiplier is the same in the two economies.

Proof. Appendix E.2. □

Figure 10 shows the two terms for the 1-year MPC. Behavior raises spending for most incidences, because households below the top now spend faster. Composition lowers spending for rules tilted toward high earners. As the top's share of earnings rises, a rule proportional to earnings hands more of each transfer to high-PLI households, who save it. The 1-year MPC of a lump-sum transfer thus rises by 2.9 points, all through behavior, while that of labor income barely changes, as composition offsets behavior. The change peaks near a lump-sum transfer. Transfers targeted to the very bottom gain less, because most of their beneficiaries were already hand to mouth in 1970. As a result, the gap between the 1-year MPCs of a lump-sum transfer and of labor income widens by 3.1 points, about two thirds of it through composition. Summed over all horizons, which is what the multipliers of the next subsection depend on, almost all of the widening comes from composition. In the heterogeneous beta model, both terms are zero.[21]

[21] Two features break neutrality in the bequest-motive model: the bequest motive itself, and an adjustment cost that does not scale with PLI. With discount factors correlated with PLI instead of a bequest motive, the rate falls by 16 basis points, against 53, composition drives the change in MPCs, deficits lose more at a given debt duration, and the rebate gap widens (Appendix G.4).

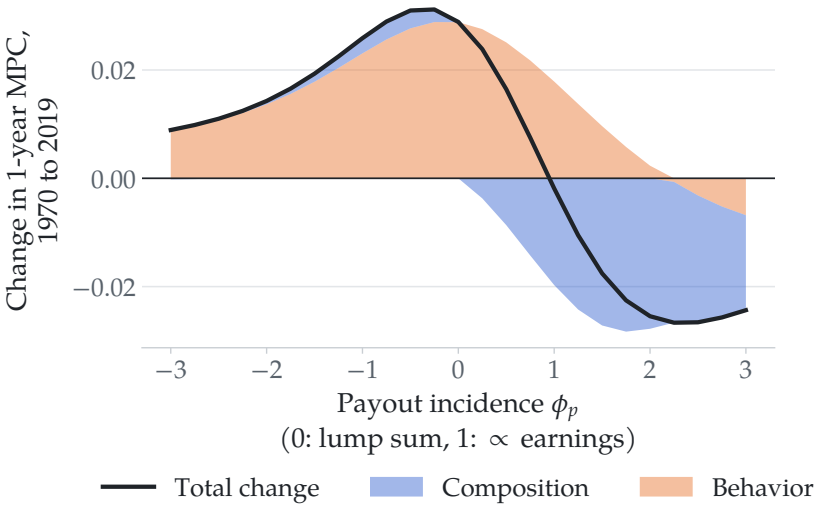


Figure 10: Change in the 1-year MPC of a transfer from 1970 to 2019

Note: Change from 1970 to 2019 in the 1-year MPC of a transfer paid in proportion to $(\alpha_i e)^\phi$, against its incidence ϕ, split as in Proposition 2. Composition: the same rule hands each PLI group a different share of the transfer. Behavior: each PLI group spends its share differently. Labor income is the transfer at $\phi = 1$. Bequest-motive model.

6.2 Fiscal policy depends more on incidence

Rising inequality changes fiscal multipliers in two ways. First, redistributions between PLI groups move output more. Second, deficits move output less at a given debt duration, although at a given fiscal rule the lower rate lengthens the debt and offsets this loss. The no-anticipation formula of Section 5.3 shows where each change comes from.

Corollary 1 (Rising inequality and the cumulative multiplier)**.** *Under the assumptions of Proposition 3, consider the same package in two economies, with debt durations D and D', and let $\mathcal{M}$ be its cumulative multiplier in the first. The change in its cumulative multiplier is*

$$\Delta\mathcal{M} = \frac{1}{\sum_{h\geq 1}\left(1-\mathrm{MPC}_h^{Y\prime}\right)}\left[\underbrace{\Delta D}_{\textit{debt duration}} + \underbrace{\Delta\sum_{h\geq 1}\mathrm{Cov}\left(\iota_{\phi_p}-\iota_{\phi_f},\, c_h\right)}_{\textit{redistribution effect}} - \underbrace{\mathcal{M}\,\Delta\sum_{h\geq 1}\left(1-\mathrm{MPC}_h^{Y}\right)}_{\textit{amplification}}\right]. \tag{24}$$

Proof. Appendix E.2. □

The first term is the debt duration. At the same fiscal rule ρ_B, the lower rate makes the debt last longer, from 8.0 to 8.4 years at the baseline rule (Appendix E.1), which raises the multiplier of any deficit. The second term is the change in the redistribution effect. It is zero for a pure deficit, and for a redistribution it moves with the gap between how fast beneficiaries and payers spend, which Proposition 2 splits into composition and behavior. The third term is the change in amplification. When the new income an expansion creates is spent more slowly, every multiplier shrinks toward zero.

Redistributions between PLI groups move output more. Figure 11 shows the change in the multipliers of the pure redistributions of Figure 8. Payouts that redistribute toward the top lower output more than in 1970, and payouts financed by high earners raise it more. The multiplier of a payout targeted to low earners and financed by high earners rises by 0.14, and that of a payout proportional to earnings financed lump sum falls by 0.12. Both changes come from composition, since a payout or a financing tilted toward earnings now reaches more high-PLI households, who spend slowly. Redistributions toward low earners that are financed lump sum or close to it move output slightly less, by at most 0.06. Behavior speeds up the spending of their payers more than that of their beneficiaries, most of whom were already hand to mouth in 1970. In the heterogeneous beta model, no multiplier changes by more than 0.01.

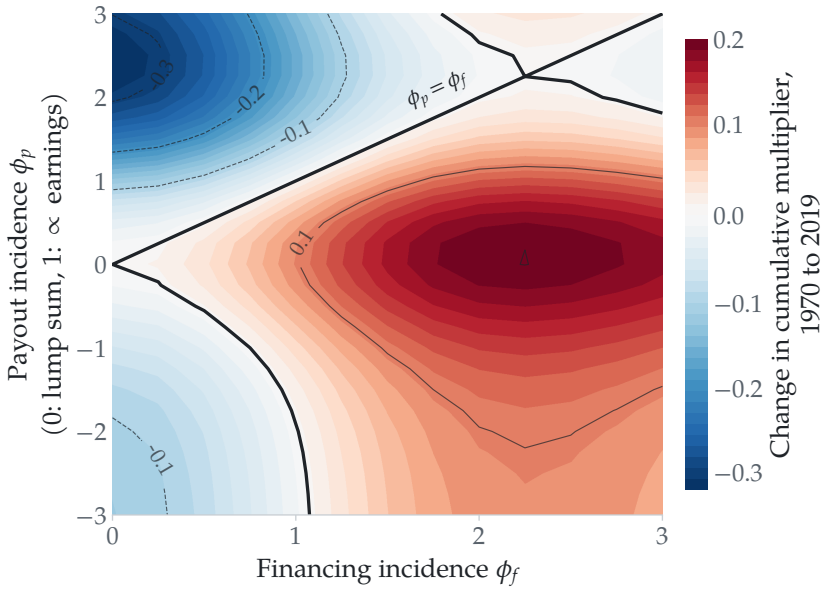


Figure 11: Change in the cumulative multipliers of pure redistributions from 1970 to 2019

Note: Change from 1970 to 2019 in the cumulative multiplier of a payout of 1% of quarterly output with persistence 0.9 at incidence ϕ_p (rows), financed in the same quarter at incidence ϕ_f (columns), with no debt. On the dashed line, $\phi_p = \phi_f$, and the package is zero. Bequest-motive model; the change is below 0.01 in the heterogeneous beta model.

At a given debt duration, deficits move output less. The multiplier of a pure deficit barely depends on its incidence, so Figure 12 shows the multiplier of a lump-sum deficit in 1970 and 2019, against the debt duration. At every duration, the 2019 multiplier is lower, and the gap grows with the duration. The deficit loses 7% of its 1970 multiplier at the baseline duration and 11% at 25 years. The loss comes from amplification, the third term of the corollary. As the top's share of earnings rises, more of the new income an expansion creates goes to high-PLI households, so more of it remains unspent at long horizons. Since the extra tax receipts of the expansion finance a share $\tau\mathcal{M}$ of the payout, deficits also finance themselves less.

At a given fiscal rule, the lower rate offsets this loss. At the same ρ_B, the lower 2019 rate lengthens the debt, the first term of Equation (24). At the baseline rule, the two effects cancel, and the multiplier of a lump-sum deficit is unchanged. At a more persistent rule, $\rho_B = 0.995$, the debt duration rises from 22 to 25 years, and the multiplier rises by 0.16.

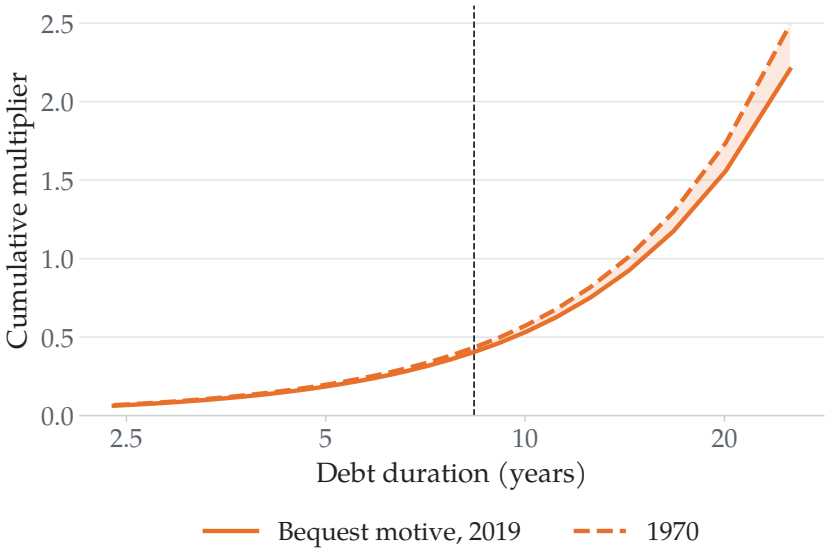


Figure 12: Cumulative multiplier of a lump-sum deficit by debt duration, 1970 and 2019

Note: Cumulative multiplier of a lump-sum payout of 1% of quarterly output with persistence 0.9, financed lump sum, against the debt duration D in years, on a log scale, the same in both years. Dashed line: baseline, 8.4 years. Bequest-motive model; the heterogeneous beta model is the same in both years.

6.3 Monetary policy depends more on the fiscal response

The response to a rate cut barely changes on average, but it depends more on who receives the rebate (Figure 13). With a lump-sum rebate paid with the baseline delay, the cumulative output semi-elasticity of the rate cut of Section 5.5 is almost the same in 1970 and 2019. Moving the rebate from low to high earners lowers it by 72% in 2019, against 62% in 1970. The reason is the one behind the fiscal results. Changing who receives the rebate is a pure redistribution of it (Equation (41) in Appendix D.3), so it inherits the widening gap between how fast low- and high-PLI households spend. In the heterogeneous beta model, the response is the same in both years, and moving the rebate lowers it by only 13%.[22]

[22] In the one-asset version of the model, the response with a lump-sum rebate rises from 1970 to 2019 instead (Appendix F).

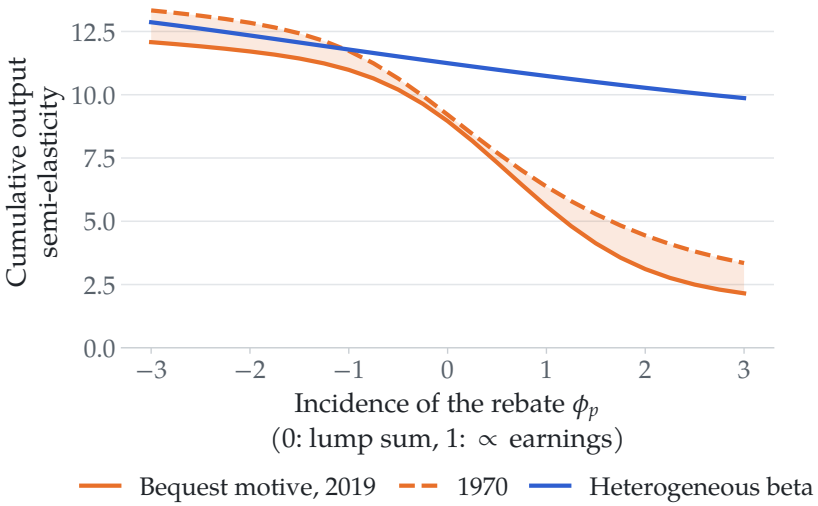


Figure 13: Response to a rate cut by the incidence of the rebate, 1970 and 2019

Note: Cumulative output semi-elasticity, the present value of dY_t/Y per present value of the rate cut, for a cut of 1 percentage point in the annual real rate with quarterly persistence 0.9, against the incidence ϕ_p of the rebate paid by the fiscal rule (16), with an average delay of 8.4 years in both years. The heterogeneous beta model is the same in both years.

7 Conclusion

In standard heterogeneous-agent models, the distribution of permanent labor income is neutral for the transmission of monetary and fiscal policy. In the PSID, however, hand-to-mouth status is concentrated among households with low permanent labor income, and this concentration has grown as inequality rose. A two-asset HANK model with a non-homothetic bequest motive reproduces this gradient and breaks the neutrality. High-PLI households save to leave bequests, which they value as a luxury good, the equilibrium interest rate falls, and low-PLI households dissave toward the borrowing constraint.

Once the hand-to-mouth households are the low-PLI ones, the effect of a policy depends on who receives it and who pays for it. The same payout can stimulate or contract output depending on who finances it, and the effect of a rate cut depends on who receives the interest the government saves on its debt. Rising PLI inequality since 1970 makes incidence matter more. The shift of earnings toward the top widens the gap between spending out of a transfer and out of labor income, so redistributions between PLI groups move output more and the response to a rate cut depends more on the fiscal rule. Persistent deficits create less output per year of debt, but the lower interest

rate lets a given fiscal rule repay it later. The composition of a fiscal package, and the coordination between fiscal and monetary policy, matter more today than fifty years ago.

References

Aguiar, M. A., Bils, M. & Boar, C. (2025), 'Who are the hand-to-mouth?', *Review of Economic Studies* . Forthcoming.

Altonji, J. G. & Segal, L. M. (1996), 'Small-sample bias in GMM estimation of covariance structures', *Journal of Business & Economic Statistics* **14**(3), 353–366.

Alves, F., Kaplan, G., Moll, B. & Violante, G. L. (2020), 'A Further Look at the Propagation of Monetary Policy Shocks in HANK', *Journal of Money, Credit and Banking* **52**(S2), 521–559.

Angeletos, G.-M., Lian, C. & Wolf, C. K. (2024), 'Can deficits finance themselves?', *Econometrica* **92**(5), 1351–1390.

Auclert, A. (2019), 'Monetary policy and the redistribution channel', *American Economic Review* **109**(6), 2333–2367.

Auclert, A., Bardóczy, B., Rognlie, M. & Straub, L. (2021), 'Using the sequence-space jacobian to solve and estimate heterogeneous-agent models', *Econometrica* **89**(5), 2375–2408.

Auclert, A. & Rognlie, M. (2020), 'Inequality and Aggregate Demand', p. 82.

Auclert, A., Rognlie, M. & Straub, L. (2024), 'The intertemporal Keynesian cross', *Journal of Political Economy* **132**(12), 4068–4121.

Auclert, A., Rognlie, M. & Straub, L. (2025), 'Fiscal and monetary policy with heterogeneous agents', *Annual Review of Economics* **17**, 539–562.

Bloom, N., Guvenen, F., Pistaferri, L., Sabelhaus, J., Salgado, S. & Song, J. (2017), The great micro moderation. Working paper.

Boehm, J., Fize, E. & Jaravel, X. (2025), 'Five facts about MPCs: Evidence from a randomized experiment', *American Economic Review* **115**(1), 1–42.

Braxton, J. C., Herkenhoff, K. F., Rothbaum, J. L. & Schmidt, L. (2021), Changing

income risk across the us skill distribution: Evidence from a generalized kalman filter, Technical report, National Bureau of Economic Research.

Brinca, P., Ferreira, M. H., Franco, F., Holter, H. A. & Malafry, L. (2021), 'Fiscal consolidation programs and income inequality', *International Economic Review* **62**(1), 405–460.

Brinca, P., Holter, H. A., Krusell, P. & Malafry, L. (2016), 'Fiscal multipliers in the 21st century', *Journal of Monetary Economics* **77**, 53–69.

Carroll, C. D. (1998), 'Why do the rich save so much?'.

Carroll, C., Slacalek, J., Tokuoka, K. & White, M. N. (2017), 'The distribution of wealth and the marginal propensity to consume', *Quantitative Economics* **8**(3), 977–1020.

Coibion, O., Gorodnichenko, Y., Kueng, L. & Silvia, J. (2017), 'Innocent bystanders? monetary policy and inequality', *Journal of Monetary Economics* **88**, 70–89.

Commault, J. (2025), Heterogeneity in MPC beyond liquid wealth: The role of permanent earnings, Economics Discussion Paper 2022-11, Sciences Po.

Congressional Budget Office (2025*a*), Distributional effects of public law 119-21, Technical report, Congressional Budget Office.

Congressional Budget Office (2025*b*), Estimated budgetary effects of public law 119-21, Technical report, Congressional Budget Office.

Congressional Budget Office (2025*c*), Preliminary analysis of the distributional effects of the one big beautiful bill act, Technical report, Congressional Budget Office.

Crawley, E. & Kuchler, A. (2023), 'Consumption heterogeneity: Micro drivers and macro implications', *American Economic Journal: Macroeconomics* **15**(1), 314–341.

De Nardi, M. (2004), 'Wealth inequality and intergenerational links', *Review of Economic Studies* **71**(3), 743–768.

De Nardi, M., French, E. & Jones, J. B. (2016), 'Savings after retirement: A survey', *Annual Review of Economics* **8**, 177–204.

DeBacker, J. M., Heim, B. T., Vidangos, I. & Panousi, V. (2011), 'Rising inequality:

Transitory or permanent? new evidence from a US panel of household income 1987-2006'.

Dynan, K. E., Skinner, J. & Zeldes, S. P. (2004), 'Do the rich save more?', *Journal of political economy* **112**(2), 397–444.

Fagereng, A., Holm, M. B. & Natvik, G. J. (2021), 'MPC heterogeneity and household balance sheets', *American Economic Journal: Macroeconomics* **13**(4), 1–54.

Ferriere, A. & Navarro, G. (2025), 'The heterogeneous effects of government spending: It's all about taxes', *Review of Economic Studies* **92**(2), 1061–1125.

Gaillard, A., Wangner, P., Hellwig, C. & Werquin, N. (2023), 'Consumption, wealth, and income inequality: A tale of tails', *Available at SSRN 4636704* .

Gale, W. & Page, B. (2026), One big beautiful bill? a preliminary assessment, Technical report, Urban-Brookings Tax Policy Center.

Guvenen, F., Kaplan, G., Song, J. & Weidner, J. (2022), 'Lifetime earnings in the United States over six decades', *American Economic Journal: Applied Economics* **14**(4), 446–479.

Heathcote, J., Perri, F. & Violante, G. L. (2010), 'Unequal we stand: An empirical analysis of economic inequality in the United States, 1967–2006', *Review of Economic Dynamics* **13**(1), 15–51.

Holm, M. B., Paul, P. & Tischbirek, A. (2021), 'The transmission of monetary policy under the microscope', *Journal of Political Economy* **129**(10), 2861–2904.

Kaplan, G., Moll, B. & Violante, G. L. (2018), 'Monetary Policy According to HANK', *American Economic Review* **108**(3), 697–743.

Kaplan, G. & Violante, G. L. (2014), 'A model of the consumption response to fiscal stimulus payments', *Econometrica* **82**(4), 1199–1239.

Kaplan, G., Violante, G. L. & Weidner, J. (2014), 'The wealthy hand-to-mouth', *Brookings Papers on Economic Activity* **45**(1), 77–138.

Krusell, P. & Smith, Jr., A. A. (1998), 'Income and Wealth Heterogeneity in the Macroeconomy', *Journal of Political Economy* **106**(5), 867–896.

Kueng, L. (2018), 'Excess sensitivity of high-income consumers', *Quarterly Journal of Economics* **133**(4), 1693–1751.

Kumhof, M., Rancière, R. & Winant, P. (2015), 'Inequality, leverage, and crises', *American Economic Review* **105**(3), 1217–1245.

Lewis, D., Melcangi, D. & Pilossoph, L. (2026), 'Latent heterogeneity in the marginal propensity to consume', *Review of Economic Studies* **93**(5), 3274–3302.

Melcangi, D. & Sterk, V. (2020), Stock market participation, inequality, and monetary policy, Staff Report 932, Federal Reserve Bank of New York.

Mian, A., Straub, L. & Sufi, A. (2021), 'Indebted Demand*', *The Quarterly Journal of Economics* **136**(4), 2243–2307.

Michau, J.-B. (2018), 'Secular stagnation: Theory and remedies', *Journal of Economic Theory* **176**, 552–618.

Parker, J. A. (2017), 'Why don't households smooth consumption? Evidence from a $25 million experiment', *American Economic Journal: Macroeconomics* **9**(4), 153–183.

Patterson, C. (2023), 'The matching multiplier and the amplification of recessions', *American Economic Review* **113**(4), 982–1012.

Piketty, T., Saez, E. & Zucman, G. (2018), 'Distributional national accounts: methods and estimates for the united states', *The Quarterly Journal of Economics* **133**(2), 553–609.

Saez, E. & Zucman, G. (2020), 'The rise of income and wealth inequality in america: Evidence from distributional macroeconomic accounts', *Journal of Economic Perspectives* **34**(4), 3–26.

Straub, L. (2019), 'Consumption, Savings, and the Distribution of Permanent Income', p. 94.

The Budget Lab at Yale (2025), Combined distributional effects of the one big beautiful bill act and of tariffs, Technical report, Yale University.

Wolf, C. K. (2025), 'Interest rate cuts versus stimulus payments: An equivalence result', *Journal of Political Economy* **133**(4), 1235–1275.

A Additional Empirical Results

A.1 Summary statistics

Table 6 reports weighted summary statistics by PLI quintile for the analysis sample of Section 2. Mean labor earnings rise from \$25,000 in the bottom quintile to \$157,000 in the top, median net worth from \$10,000 to \$272,000, and the median household in the bottom quintile holds no liquid assets, against about \$100 in the second.

Table 6: Summary statistics by PLI quintile (PSID, 1999–2019)

	Q1	Q2	Q3	Q4	Q5	All
Labor earnings (mean)	24.7	45.1	63.9	85.1	157.1	75.2
Nondurable consumption (mean)	23.1	30.6	35.9	43.7	59.5	38.5
Net worth (median)	10.2	36.9	75.5	123.8	272.2	73.6
Liquid assets (median)	0.00	0.10	0.97	4.80	22.98	0.95
Age (mean)	45.2	45.1	44.4	44.8	45.1	44.9
Family size (mean)	2.3	2.5	2.5	2.6	2.4	2.5
HtM_{NW} share (%)	37.9	27.5	21.9	15.8	12.2	23.1
HtM_{LIQ} share (%)	26.1	19.2	17.0	14.8	9.9	17.5
Observations	6,378	6,795	6,862	6,421	5,922	32,378
Households	1,089	1,053	1,029	929	845	4,945

Notes: Weighted by PSID family weights. Dollar amounts in thousands of 2009 dollars (PCE deflator). Nondurable consumption follows the 21-category aggregate of Aguiar et al. (2025). PLI quintiles based on the averaged residuals of Straub (2019). HtM shares are computed among households with non-missing hand-to-mouth status.

A.2 Robustness of the HtM–PLI gradient

Figure 14 recomputes the gradient under three alternatives to the baseline Straub (2019) averaged residuals: the original Aguiar et al. (2025) residuals (age cubic and year effects only), a household fixed-effects estimator with the Straub (2019) first-stage controls, and the empirical Bayes shrinkage estimator of Appendix A.5. All three yield a steep, near-monotonic decline from decile 1 to decile 10. Figure 15 halves the net-worth cutoff to one month of earnings or raises it to three months. The threshold shifts the level of the hand-to-mouth share in every quintile but leaves the gradient intact: the bottom-quintile share is about three times the top-quintile share at all three cutoffs.

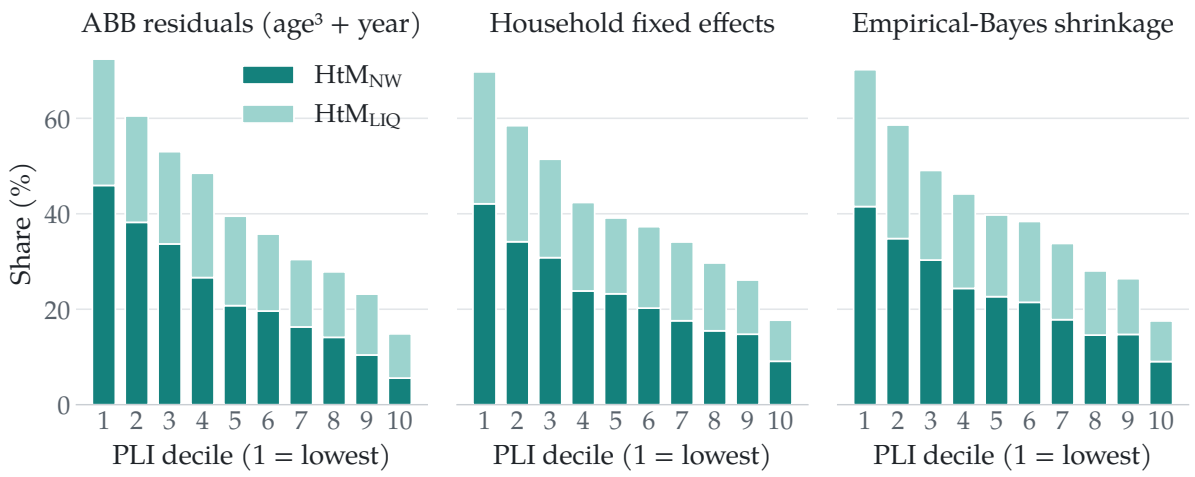


Figure 14: HtM shares by PLI decile: comparison across PLI measures

Notes: Left: Aguiar et al. (2025) residuals (age cubic + year FE). Center: household fixed-effects estimator with Straub (2019) controls. Right: empirical Bayes shrinkage estimator using the minimum-distance income process parameters. All panels use PSID 1999–2019, weighted.

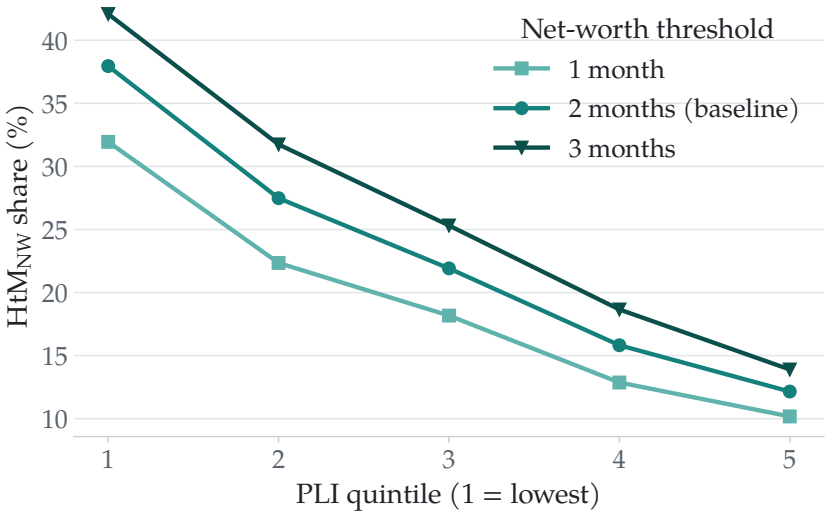


Figure 15: HtM_{NW} shares by PLI quintile: alternative net-worth thresholds

Notes: Weighted HtM_{NW} shares by PLI quintile when the hand-to-mouth cutoff is net worth below one, two (baseline), or three months of household earnings. Shares are computed among households with non-missing net worth and earnings. PSID 1999–2019.

A.3 What explains hand-to-mouth status

Table 7 splits the variance of hand-to-mouth status among five groups of household characteristics. We regress total hand-to-mouth status on year fixed effects and dummies for each group. The groups are correlated, so we report the R^2 of each group alone, its gain when added last, and its Shapley value, the average gain in R^2 over all orders in which the groups can be added. The Shapley values sum to the total R^2. PLI contributes the most by all three measures. Deviations of current earnings from PLI add almost nothing. With the PLI from pre-tax labor income used in Section 4, PLI contributes 7.0 points and these deviations 0.5.

The five groups explain 17.7% of the variance, and much of the rest is specific to households. Net of year effects, household fixed effects explain 54% of the variance, or 44% after adjusting for the number of households, since with about six waves per household the fixed effects also absorb transitory variation. A check that avoids this bias compares the same household's status in two waves far apart. If hand-to-mouth status is a fixed household component plus shocks that fade within a decade, the correlation between two waves at least ten years apart equals the share of the fixed component in the variance. This correlation falls as the gap between waves widens and is stable, near one third, between ten and twenty years. PLI alone explains 9.3% of the variance, between 17% and 27% of the household component across these measures.

A.4 The gradient and the level of inequality

The weighted standard deviation of residualized log household earnings rises from 0.72 in the 1999 wave to 0.78 in 2019, and the P90–P10 gap from 1.69 to 1.84 log points, most of it before 2008. Table 8 relates the steepening of Fact 3 to this dispersion directly. We regress HtM_{NW} status on continuous PLI, its interaction with the wave-level dispersion (demeaned), and year fixed effects. The interaction is negative and significant under both dispersion measures: the HtM–PLI gradient is steeper in waves in which residual earnings dispersion is higher.

Table 7: Contributions to the variance of hand-to-mouth status (PSID, 1999–2019)

	Alone	Shapley	Added last	Share of explained
PLI	9.3	7.8	6.6	44%
		(0.7)		
Income shocks	0.2	0.3	0.3	2%
		(0.1)		
Age	4.4	3.8	3.3	21%
		(0.5)		
Family structure	6.3	4.1	2.4	23%
		(0.4)		
Race	2.9	1.7	0.9	10%
		(0.3)		
All five groups		17.7		100%
		(0.9)		

Notes: Contribution of each group of characteristics to the R^2 of total hand-to-mouth status (HtM_{NW} or HtM_{LIQ}), in percentage points of its variance, net of year fixed effects. Alone: R^2 gain of the group on its own. Shapley: average R^2 gain over all orders in which the groups are added. Added last: R^2 gain over the four other groups. PLI: deciles of the averaged residuals of Straub (2019). Income shocks: deciles of the deviation of current residual log earnings from PLI. Age: five-year groups. Family structure: marital status, family size and a dual-earner indicator. Race: white, black, other. Weighted by PSID family weights; 29,595 household-years. Bootstrap standard errors in parentheses (50 draws of households).

Table 8: The HtM–PLI gradient and the level of inequality

	(1)	(2)
PLI	-0.148	-0.148
	(0.009)	(0.008)
PLI × Ineq (std)	-0.759	
	(0.168)	
PLI × Ineq (P90–P10)		-0.267
		(0.045)
R^2	0.06	0.06
N	29,705	29,705

Notes: Linear probability regressions of HtM_{NW} status on continuous PLI, its interaction with the wave-level dispersion of residual log earnings (demeaned; standard deviation in column 1, P90–P10 gap in column 2), and year fixed effects. The PLI coefficient is the gradient at average inequality. Weighted by PSID family weights; standard errors two-way clustered by household and year.

A.5 Estimation of the income process

Let $\hat{y}_{it}$ denote residual log labor earnings of household i in income year t, from the first-stage regression of Section 2. Following Straub (2019), the residual is the sum of a permanent component α_i with variance σ_α^2, a persistent component $z_{it} = \rho\, z_{i,t-1} + \varepsilon_{it}$ with innovation variance σ_ε^2 at annual frequency, drawn from its stationary distribution, a transitory shock ν_{it} with variance σ_ν^2, and measurement error m_{it} with variance fixed at $\sigma_m^2 = 0.02$ (Heathcote et al. 2010). The model implies, for any wave t and lag $s \geq 0$ in biennial waves,

$$\operatorname{Cov}(\hat{y}_{it},\, \hat{y}_{i,t+s}) = \sigma_\alpha^2 + \rho^{2s}\, \frac{\sigma_\varepsilon^2}{1-\rho^2} + \mathbf{1}\{s=0\}\left(\sigma_\nu^2 + \sigma_m^2\right). \tag{25}$$

The estimate of $(\sigma_\alpha^2, \rho, \sigma_\varepsilon^2, \sigma_\nu^2)$ minimizes the equally weighted sum of squared deviations between these moments and the 66 empirical autocovariances of the eleven waves with income years 1998–2018, for households aged 30–55 with at least five earnings observations (2,397 households). Equal weights avoid the bias of an estimated optimal weighting matrix in samples of this size (Altonji & Segal 1996).

Table 9: Minimum-distance estimates of the income process (annual frequency)

		This paper	Straub (2019)
Persistence	ρ	0.872	0.904
Variance, permanent component	σ_α^2	0.284	0.131
Variance, persistent innovation	σ_ε^2	0.032	0.026
Variance, transitory shock	σ_ν^2	0.088	0.052
Variance, measurement error	σ_m^2	0.020 (fixed)	—

Notes: Equally weighted minimum distance on the 66 autocovariances of residual log household labor earnings, PSID income years 1998–2018, households aged 30–55 with at least five observations (2,397 households). The last column reproduces the corresponding estimates in Straub (2019).

Table 9 reports the estimates. Persistence is close to Straub (2019), and the permanent variance is about twice as large, consistent with our later sample and different first-stage controls. The estimates imply an unconditional standard deviation of the persistent component of $\sqrt{\sigma_\varepsilon^2/(1-\rho^2)} = 0.36$ and a cross-sectional variance of residual earnings of 0.52, of which the permanent component accounts for 54 percent. The transitory variance exceeds Straub (2019) because the nearest observed covariance in a biennial panel is two years away, so a shock that mean-reverts within two years is attributed to σ_ν^2. For the model, we estimate the same process on pre-tax labor income with the

estimator of Straub (2019, Appendix D.4), which matches autocovariances between ages instead of waves and lets the variance of the persistent component build up from age 30. On 2,086 households and 171 moments, it gives $\rho = 0.938$, $\sigma_\varepsilon^2 = 0.017$, $\sigma_\nu^2 = 0.042$ and $\sigma_\alpha^2 = 0.197$, and the model carries both the persistent and the transitory component.

The empirical Bayes estimator of Figure 14 shrinks each household's average residual $\bar{y}_i$ toward the population mean, $\hat{\alpha}_i = \lambda_i \bar{y}_i$ with $\lambda_i = \sigma_\alpha^2/(\sigma_\alpha^2 + \sigma_{u,i}^2)$. The noise variance $\sigma_{u,i}^2$ is the variance of the average of the other components over the household's T_i waves, $\sigma_{u,i}^2 = \bar{r}_i\,\sigma_\varepsilon^2/(1-\rho^2) + (\sigma_\nu^2 + \sigma_m^2)/T_i$, where $\bar{r}_i$ is the mean of $\rho^{|t-t'|}$ over all pairs (t, t') of the household's income years, including $t = t'$. Because the persistent component is correlated across waves, it averages out more slowly than the transitory shock and the measurement error. The average shrinkage factor is 0.74.

B Model Details

This appendix derives the debt path implied by the fiscal rule, describes the homothetic benchmark, and proves the scaling property of the household problem under homothetic preferences on which the propositions of Sections 5 and 6 rest.

B.1 The fiscal rule and debt duration

The fiscal and the monetary experiments return public debt to steady state at the same geometric rate ρ_B. They differ only in what enters debt. Linearizing the budget constraint (10) around the steady state, with receipts $dT_t = \tau\, dY_t$,

$$dB_t = (1+r)\, dB_{t-1} + B\, dr_t + dX_t^p - dX_t^f - \tau\, dY_t.$$

Monetary experiments. There is no financing, and the payout follows the fiscal rule (16). Substituting,

$$dB_t = \rho_B\, dB_{t-1} + B\, dr_t - \tau\, dY_t, \qquad \text{so that} \qquad dB_t = \sum_{k=0}^{t} \rho_B^{t-k}\big(B\, dr_k - \tau\, dY_k\big).$$

Each quarter, the payout returns the interest on the debt gap, $-r\, dB_{t-1}$, plus a share

$1-\rho_B$ of this gap, and debt returns to steady state if and only if $\rho_B < 1$. In operator form, $d\mathbf{X}^p = \mathbf{R}\big(-B\,d\mathbf{r} + \tau\,d\mathbf{Y}\big)$ with $\mathbf{R} \equiv (1+r-\rho_B)\,\mathbf{L}\big(\mathbf{I}-\rho_B\mathbf{L}\big)^{-1}$. A dollar of budget impulse at date s is returned as $(1+r-\rho_B)\rho_B^{t-s-1}$ at each date $t > s$, which sums to one in present value, so $\mathbf{q}'\mathbf{R} = \mathbf{q}'$.

Fiscal experiments. The debt path is set directly by Equation (12), $dB_t = \rho_B\,dB_{t-1} + dX_t^p$, and the financing clears the budget. At $d\mathbf{r} = 0$,

$$dX_t^f = (1+r-\rho_B)\,dB_{t-1} - \tau\,dY_t.$$

The financing repays debt on the same schedule as the rule (16), and it also returns the extra tax receipts $\tau\,dY_t$ in the quarter they accrue. Only the payout enters debt, whereas in the monetary experiments the whole budget impulse enters debt and is repaid at the rate ρ_B. *Debt duration.* A dollar of payout at date s adds ρ_B^{t-s} to debt at each date $t \geq s$, so for any payout path the debt duration (13) is

$$D \equiv \frac{\mathbf{q}'d\mathbf{B}}{\mathbf{q}'d\mathbf{X}^p} = \sum_{k\geq 0}\Big(\frac{\rho_B}{1+r}\Big)^k = \frac{1+r}{1+r-\rho_B}.$$

The same number is the average delay, weighted by present value, with which a dollar of debt is repaid: it is repaid as $(1+r-\rho_B)\rho_B^{k-1}$ after $k \geq 1$ quarters, and $\sum_{k\geq 1} k\,(1+r-\rho_B)\rho_B^{k-1}(1+r)^{-k} = (1+r)/(1+r-\rho_B)$. At the 2019 rate, D is 8.4 years at $\rho_B = 0.975$ and 16.8 years at $\rho_B = 0.99$.

B.2 Homothetic benchmark

The heterogeneous beta model sets $\gamma = 0$ and generates wealth concentration and hand-to-mouth households through heterogeneous discount factors (calibration in Appendix C.3). The discount factor takes two values, $\beta_j \in \{\beta_L, \beta_H\}$, and follows a two-state Markov chain independent of permanent income, with a share s_H of patient households and a quarterly switching probability q_β (Krusell & Smith 1998, Auclert et al. 2025). The adjustment-cost scale is proportional to PLI, $\psi_{0,i} = \alpha_i\psi_0$, so preferences and frictions are homothetic. By Lemma 1, the PLI distribution is neutral for its iMPC matrices.

B.3 Scaling under homothetic preferences

The two neutrality results of the paper, the absence of a PLI term in the iMPC matrices (Lemma 3) and the invariance of every iMPC matrix to the PLI distribution (Proposition 4), follow from one property of the household problem.

Lemma 1 (Scaling)**.** *Suppose preferences are homothetic, $\gamma = 0$ (or $\gamma > 0$ with $\eta = 1$ and $\zeta = 0$), the adjustment-cost scale is proportional to PLI, $\psi_{0,i} = \alpha_i \psi_0$, and the discount-factor process is independent of the PLI group. Let starred objects belong to the problem of a household with $\alpha = 1$ facing the same prices, and write $x = (b, a)$ for the two accounts. Then in steady state*

$$c_i(x, e) = \alpha_i \, c^*(x/\alpha_i, e), \qquad x_i'(x, e) = \alpha_i \, x^{*\prime}(x/\alpha_i, e), \qquad \Phi_i(x, e) = \Phi^*(x/\alpha_i, e),$$

the steady-state real rate and tax rate do not depend on the PLI distribution $\boldsymbol{\alpha}$, and the response of a type-i household's consumption at date t to a unit of income received at date s is $m_{t,s}(x, e, i) = m^*_{t,s}(x/\alpha_i, e)$.

Proof. Divide the budget constraint (6) by α_i. Post-tax labor income is proportional to α_i, financial income scales with (b, a) on both accounts, and the adjustment cost (4) is homogeneous of degree one in (a_t, a_{t-1}, ψ_0), so with $\psi_{0,i} = \alpha_i \psi_0$ it scales with α_i. The bequest received by a newborn is proportional to α_i, steady-state transfers and capital gains are zero, and the borrowing limits $b \geq 0$ and $a \geq 0$ are invariant to scale. The rescaled budget set is therefore that of the $\alpha = 1$ household. With $\gamma = 0$ the objective is logarithmic in consumption, so scaling consumption by α_i adds the constant $\log \alpha_i$ to period utility without affecting choices.[23] The scaled policies verify the Bellman equation (5), and since the idiosyncratic process, persistent and transitory, the discount-factor process and the rebirth rule are independent of type, the stationary distribution satisfies $\Phi_i(x, e) = \Phi^*(x/\alpha_i, e)$.

Aggregate wealth is then $A(r) = \sum_i \omega_i \alpha_i \, A^*(r) = A^*(r)$ by the normalization $\sum_i \omega_i \alpha_i = 1$, so the rate that clears the asset market does not depend on $\boldsymbol{\alpha}$. Output, debt, spending and dividends do not depend on $\boldsymbol{\alpha}$ either, so neither does the tax rate that balances the

[23] With $\gamma > 0$, $\eta = 1$ and $\zeta = 0$, the wealth term is logarithmic too, $v(\alpha(a + b)) = v(a + b) + \gamma \log \alpha$, and the same argument applies.

budget.

For the income kernel, a unit of income raises the deposits of a type-i household by one unit, and differentiating $c_i(x,e) = \alpha_i\, c^*(x/\alpha_i, e)$ with respect to deposits cancels the factor α_i, so the first-order response of consumption at every later date is $m_{t,s}(x,e,i) = m^*_{t,s}(x/\alpha_i, e)$. The date-0 capital gain is paid in proportion to illiquid wealth, which scales with α_i, so the response to it scales in the same way. □

The bequest-motive model violates the lemma in two ways: its wealth term is non-homothetic, and its scale ψ_0 is fixed in units of output. We solve each group's problem in units of its own PLI, so the scaling is exact numerically: in the heterogeneous beta model, the 1970 and 2019 PLI distributions give the same rate, MPC and hand-to-mouth shares to five digits.

Averaging the kernel over the stationary distribution of group i gives

$$\bar{m}_{i,t,s} \equiv \mathbb{E}_i[m_{t,s}] = \int m^*_{t,s}(x/\alpha_i, e)\, d\Phi^*(x/\alpha_i, e) = \bar{m}^*_{t,s}, \tag{26}$$

the same in every PLI group, and likewise $\mathbb{E}_i[\tilde{e}^{\phi} m_{t,s}] = \mathbb{E}^*[\tilde{e}^{\phi} m^*_{t,s}]$ for every incidence ϕ: a PLI group spends a dollar it receives in the same way whatever its permanent income.

C Calibration Details

This appendix derives the Pareto tails that pin down η and ξ, shows hand-to-mouth status by current labor income, reports the calibration of the heterogeneous beta model, and compares MPCs by income in the two models with the empirical evidence.

C.1 Pareto tails and the bequest parameters

Consider a household of permanent type i in steady state, with $r_t = r$, no transfers and no bequest endowment; the derivation extends when these are positive, as long as they do not grow with individual wealth. At the top of the distribution, households hold almost all their wealth in the illiquid account, which earns r, and finance consumption

by withdrawals. Withdrawing c_t costs $\psi_1 c_t^2/\big(2((1+r)a_{t-1}+\psi_0)\big)$, which vanishes relative to c_t as a_{t-1} grows when c_t grows less than proportionally with wealth, and so do the derivatives of the cost in the first-order conditions. The Euler equation for illiquid wealth then tends to

$$c_t^{-1} = (1-\xi)\beta(1+r)\,\mathbb{E}_t\big[c_{t+1}^{-1}\big] + \xi\gamma\,(a_t+\zeta)^{-\eta}\,.$$

As $a_{t-1}\to\infty$, labor income and idiosyncratic shocks become negligible next to financial income, the budget constraint tends to $c_t + a_t = (1+r)a_{t-1}$, and the shifter ζ becomes negligible next to a_t. We guess a power-law consumption function, $c_t = B\,a_{t-1}^{\eta}$, which grows less than proportionally since $\eta<1$. Then $a_t = (1+r)a_{t-1} - B\,a_{t-1}^{\eta} \sim (1+r)a_{t-1}$ and $c_{t+1}\sim B(1+r)^{\eta}a_{t-1}^{\eta}$. Substituting into the Euler equation and dividing by $a_{t-1}^{-\eta}$,

$$B^{-1} = (1-\xi)\beta(1+r)^{1-\eta}B^{-1} + \xi\gamma(1+r)^{-\eta}, \qquad \text{so} \qquad B = \frac{1-(1-\xi)\beta(1+r)^{1-\eta}}{\xi\gamma(1+r)^{-\eta}}.$$

Wealthy households consume a vanishing share of their wealth and save at the gross rate $1+r$.

At the top of the distribution, wealth therefore grows by the factor $1+r$ with probability $1-\xi$ and falls to zero with probability ξ, when the household dies. A random growth process with a constant growth factor and a constant exit probability has a Pareto tail whose exponent θ^a solves $(1-\xi)(1+r)^{\theta^a}=1$, the condition that the expected θ^a-th power of the growth factor equals one:

$$\theta^a = \frac{-\log(1-\xi)}{\log(1+r)}.$$

Since $c_t = B\,a_{t-1}^{\eta}$ at the top, $\mathbb{P}(c>y) = \mathbb{P}\big(a>(y/B)^{1/\eta}\big)\sim C\,y^{-\theta^a/\eta}$, so consumption has a Pareto tail with exponent

$$\theta^c = \frac{\theta^a}{\eta}.$$

Given r, the two targets θ^a and θ^c pin down ξ and η, as in Section 4. Both exponents are limits. Where consumption is still a sizable share of wealth, wealth grows more slowly and the tail is thinner (Figure 16). Between \$10 million and \$1 billion, the model tail has a slope of 2.3, because households in this range still consume 0.5% to 3.1% of their

wealth each year, so their wealth grows by about 1.3% a year instead of the 2% the limit assumes. Hence the top-0.1% share is 8%, against 18% in the data.

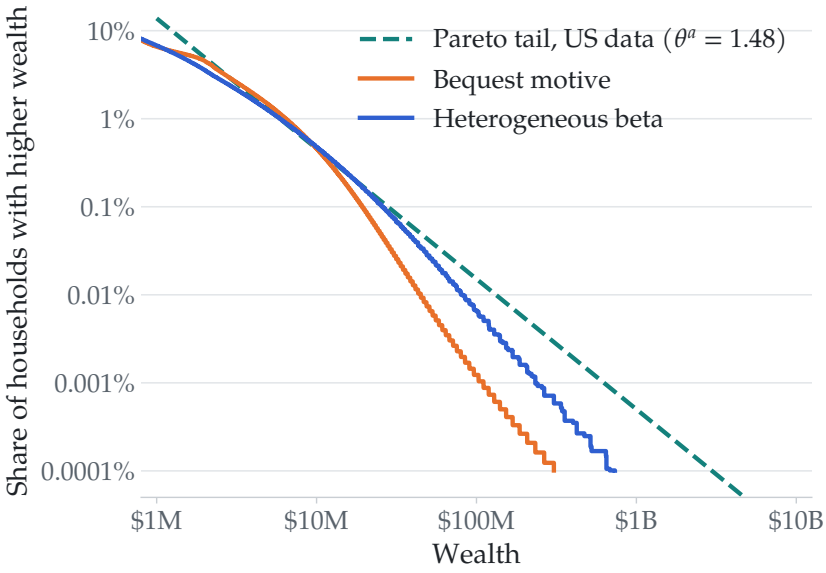


Figure 16: Wealth Pareto tail

Note: Share of households with total wealth above each level, in the bequest-motive model and the heterogeneous beta model of Section 4.4. The dashed line is a Pareto tail with the empirical coefficient $\theta^a = 1.48$, drawn through the bequest-motive model at \$10 million. The model curves stop at one household in a million, below which the mass sits on the top points of the wealth grid.

C.2 Hand-to-mouth status by current labor income

Hand-to-mouth status falls with current labor income in the bequest-motive model as in the PSID (Figure 17). Across quintiles of current labor income, the hand-to-mouth share falls from 66% to 12% in the model, against 59% and 18% in the PSID. In the heterogeneous beta model it is between 35% and 41% in every quintile. The model match follows largely from the PLI targets, since PLI accounts for 84% of the variance of log current labor income in the model. In the PSID it accounts for about half, yet hand-to-mouth status falls almost as steeply with current labor income as with PLI.

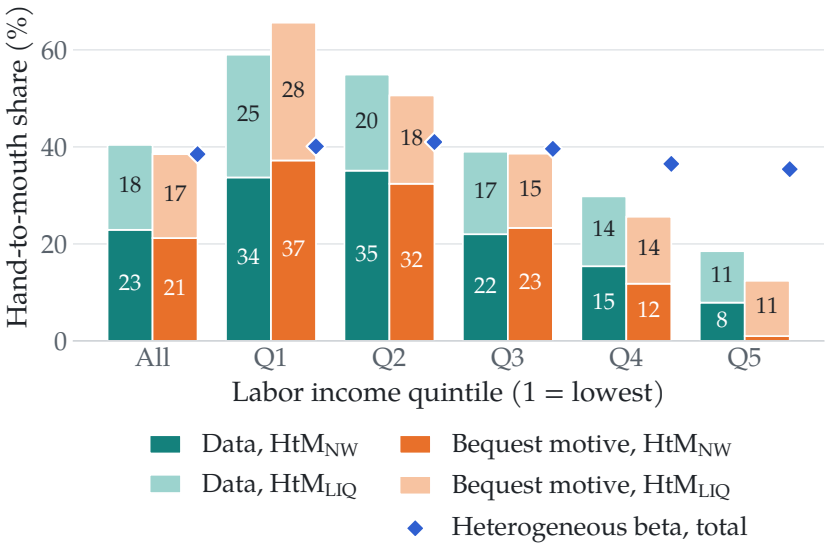


Figure 17: Hand-to-mouth shares by current labor income quintile

Note: As in Figure 5, by quintile of the current pre-tax labor income of the head and spouse, ranked within each wave in the data.

C.3 Heterogeneous beta model

Two parameters of the discount-factor chain of Appendix B.2, the patient share $s_H = 0.14$ and the spread $\beta_H - \beta_L = 0.024$, and the adjustment-cost curvature $\psi_1 = 15.8$ reproduce three moments of the bequest-motive model exactly: the 1-year MPC out of a lump-sum transfer (0.32) and the net-worth and liquid hand-to-mouth shares (21% and 17%). The switching probability is fixed at $q_\beta = 0.005$, and the scale ψ_0 is the bequest-motive value at mean PLI, proportional to PLI. The mean discount factor is 0.990. The wealth distribution is untargeted: the top-10% and top-1% shares are 82% and 40%, against 80% and 33% in the bequest-motive model and 69% and 34% in the data. The liquid share of wealth is 9%, against 21.5% in the bequest-motive model and 21.8% in the PSID, because patient households hold their wealth in the illiquid account. The hand-to-mouth share is 38.5% in every PLI quintile (diamonds in Figure 5).

C.4 MPCs fall with income because wealth rises with income

The empirical literature has found that MPCs out of transfers fall with income when nothing else is held fixed, and that this relation weakens or reverses once wealth is held fixed (Table 10). The bequest-motive model reproduces both patterns, while the heterogeneous beta model reproduces only the second. In most studies, low-income

households spend a larger share of a transfer. The fall is steeper when households are sorted by proxies for PLI, such as past income or spending over the previous year (Parker 2017, Boehm et al. 2025). Once liquid and illiquid wealth are held fixed, the MPC does not fall with income (Fagereng et al. 2021), and it rises with PLI (Commault 2025). Three studies do not find this fall when nothing is held fixed: in Fagereng et al. (2021) the MPC does not vary with income, and in Kueng (2018) and Lewis et al. (2026) it rises. Neither model reproduces the rise in Kueng (2018), for a dividend paid every year and known in advance, and in Lewis et al. (2026), for the 2008 stimulus payments.

Two forces determine how the MPC varies with PLI. First, a given amount of wealth covers fewer months of income for a high-PLI household. That household is thus closer to its borrowing constraint, which raises its MPC. Second, the bequest motive leads high-PLI households to hold more wealth relative to their income, which lowers their MPC. Because the model matches the HtM–PLI gradient, the second force dominates. Doubling PLI lowers the 1-year MPC by 8 percentage points, against a mean of 0.32. Holding the level of wealth fixed shuts down the second force, so the MPC then rises with PLI. In the heterogeneous beta model, preferences and frictions are homothetic, so the MPC depends only on wealth relative to PLI. This ratio has the same distribution in every PLI group. The MPC therefore does not vary with PLI unless the level of wealth is held fixed, in which case the first force acts alone. The MPC evidence and the HtM–PLI gradient of Section 2 thus test the same prediction of the bequest motive: low-PLI households hold little wealth relative to their income.

D Policy at the 2019 Distribution: Derivations and Proofs

Throughout, all derivatives are evaluated at the steady state, $\mathbf{q}_t = (1+r)^{-t}$, and every iMPC matrix counts the spending of the newborns who inherit the estates of dying households, as defined below.

Table 10: MPCs by income in the data and in the models

		Bequest motive		Heterogeneous beta	
Held fixed	Data	PLI	Labor income	PLI	Labor income
Nothing	Falls (Patterson 2023, Parker 2017, Crawley & Kuchler 2023, Boehm et al. 2025); no relation (Fagereng et al. 2021); rises (Kueng 2018, Lewis et al. 2026)	−8	−9	0	−2
Liquid wealth	Still falls with past income (Parker 2017); little relation (Crawley & Kuchler 2023); rises (Lewis et al. 2026)	−2	−4	4	1
Liquid and illiquid wealth	No relation with income (Fagereng et al. 2021); rises with PLI (Commault 2025)	6	0	7	5

Note: Model columns: change in the 1-year MPC, in percentage points, when PLI or annual labor income doubles. The mean 1-year MPC is 0.32 in both models. Each household's MPC out of a small transfer is computed from the policy functions of the 2019 steady state, then regressed on log PLI or log labor income, weighted by the stationary distribution. The regression has no controls, or fixed effects for bins of liquid wealth, or of liquid and illiquid wealth, in dollars, each bin 10% wide. The studies differ in the transfer, the spending measure and the horizon, so the data column reports only the sign of the relation. Fagereng et al. (2021) use after-tax household income.

D.1 From the consumption function to the IKC

With $w = 1/\mu$ and $N_t = Y_t$, post-tax labor income satisfies

$$d\mathbf{Z} = \frac{1-\tau}{\mu}\, d\mathbf{Y}. \tag{27}$$

Since dividends are taxed at the same rate as labor, after-tax dividends are $(1-\tau)\frac{\mu-1}{\mu}Y_t$: post-tax income $(1-\tau)Y_t$ splits between wages and dividends in the constant shares $\frac{1}{\mu}$ and $\frac{\mu-1}{\mu}$. Differentiating the no-arbitrage condition (8) and the mutual fund balance sheet (9) yields the following lemma.

Lemma 2 (Capital gains)**.** *To first order, the aggregate date-0 capital gain $dcap \equiv \bar{S}\, dcap_0$ is*

$$dcap = (1-\tau)\,\frac{\mu-1}{\mu}\,\mathbf{q}'d\mathbf{Y} - p\,\mathbf{q}'d\mathbf{r}, \tag{28}$$

with $dr_0 = 0$, since the date-0 return is predetermined.

Proof. Write the no-arbitrage condition (8) as $(1+r_{t+1})\,p_t = p_{t+1} + (1-\tau)d_{t+1}$ and

linearize around the steady state, using $(1-\tau)\,dd_t = (1-\tau)\frac{\mu-1}{\mu}\,dY_t$:

$$dp_t = (1+r)^{-1}\Big[dp_{t+1} + (1-\tau)\tfrac{\mu-1}{\mu}\,dY_{t+1} - p\,dr_{t+1}\Big].$$

Iterating forward, with $dp_t \to 0$, gives $dp_0 = \sum_{k\geq 1}(1+r)^{-k}\big[(1-\tau)\frac{\mu-1}{\mu}\,dY_k - p\,dr_k\big]$. The mutual fund balance sheet (9) defines the aggregate capital gain as the unexpected excess value of equity, $\bar{S}\,dcap_0 = dp_0 + (1-\tau)\,dd_0$. Adding the date-0 dividend extends the sum to $k \geq 0$ and gives Equation (28), with $dr_0 = 0$ since the date-0 return was set at $t = -1$. □

Substituting Equation (27) and Lemma 2 into Equation (17) and collecting the $d\mathbf{Y}$ and $d\mathbf{r}$ terms gives

$$d\mathbf{C} = \underbrace{\Big[\mathbf{M}^r - p\,\mathbf{m}^{cap}\mathbf{q}'\Big]}_{=\overline{\mathbf{M}}^r}\,d\mathbf{r} + \underbrace{\Big[\frac{1}{\mu}\,\mathbf{M}^Z + \frac{\mu-1}{\mu}\,\mathbf{m}^{cap}\mathbf{q}'\Big]}_{=\overline{\mathbf{M}}^Z}(1-\tau)\,d\mathbf{Y} + \mathbf{M}^{\phi_p}d\mathbf{X}^p - \mathbf{M}^{\phi_f}d\mathbf{X}^f, \tag{29}$$

the analogue of equation 9 in Auclert et al. (2025).

For the government budget, differentiating Equation (10) with G and τ fixed,

$$d\mathbf{X}^f = d\mathbf{X}^p + B\,d\mathbf{r} - \tau\,d\mathbf{Y} - \big[\mathbf{I} - (1+r)\mathbf{L}\big]\,d\mathbf{B}. \tag{30}$$

The financing covers the payout and the change in the interest bill, net of the extra tax receipts the expansion generates and of new borrowing. After a fiscal shock, the government sets the paths of debt and of the payout, and the financing adjusts so that Equation (30) holds each period (Auclert et al. 2025); Appendix B.1 gives the debt paths. The debt term is present-value neutral by telescoping,

$$\mathbf{q}'\big[\mathbf{I}-(1+r)\mathbf{L}\big]d\mathbf{B} = \sum_{t\geq 0}(1+r)^{-t}\big[dB_t - (1+r)dB_{t-1}\big] = \lim_{T\to\infty}(1+r)^{-T}dB_T - (1+r)\,dB_{-1} = 0,$$

since $dB_{-1} = 0$ and debt returns to its steady state.

Combining Equation (29) with goods market clearing $d\mathbf{Y} = d\mathbf{C}$ gives the intertemporal

Keynesian cross,

$$d\mathbf{Y} = (1-\tau)\,\overline{\mathbf{M}}^{Z}\,d\mathbf{Y} + \mathbf{M}^{\phi_p} d\mathbf{X}^{p} - \mathbf{M}^{\phi_f} d\mathbf{X}^{f} + \overline{\mathbf{M}}^{r}\,d\mathbf{r}, \tag{31}$$

with the transfers tied down by Equation (30).

Fiscal shocks. At $d\mathbf{r} = 0$, substituting the financing from Equation (30) into Equation (31),

$$d\mathbf{Y} = (1-\tau)\overline{\mathbf{M}}^{Z}\,d\mathbf{Y} + \mathbf{M}^{\phi_p} d\mathbf{X}^{p} - \mathbf{M}^{\phi_f}\Big(d\mathbf{X}^{p} - \tau\,d\mathbf{Y} - \big[\mathbf{I} - (1+r)\mathbf{L}\big]d\mathbf{B}\Big).$$

The endogenous receipts $\tau\,d\mathbf{Y}$ reduce post-tax income spent through $\overline{\mathbf{M}}^{Z}$ and reduce the financing, which payers spend through $\mathbf{M}^{\phi_f}$. Collecting the $d\mathbf{Y}$ terms into $\mathbf{K} = (1-\tau)\overline{\mathbf{M}}^{Z} + \tau\mathbf{M}^{\phi_f}$ gives the cross for any debt path,

$$(\mathbf{I} - \mathbf{K})\,d\mathbf{Y} = \mathbf{M}^{\phi_f}\big[\mathbf{I} - (1+r)\mathbf{L}\big]\,d\mathbf{B} + \big(\mathbf{M}^{\phi_p} - \mathbf{M}^{\phi_f}\big)\,d\mathbf{X}^{p}. \tag{32}$$

Under the debt rule (12), $d\mathbf{B} = (\mathbf{I} - \rho_B\mathbf{L})^{-1} d\mathbf{X}^{p}$, and Equation (14) gives $1 + r - \rho_B = (1+r)/D$, so that $\rho_B = (1+r)(1-1/D)$. Hence

$$\big[\mathbf{I} - (1+r)\mathbf{L}\big]\,d\mathbf{B} = \Big[\mathbf{I} - \frac{1+r}{D}\,\mathbf{L}\,(\mathbf{I} - \rho_B\mathbf{L})^{-1}\Big]\,d\mathbf{X}^{p} = \Big[\mathbf{I} - \sum_{k\geq 1} \pi_k\,(1+r)^{k}\,\mathbf{L}^{k}\Big]\,d\mathbf{X}^{p},$$

with $\pi_k = \frac{1}{D}\big(1 - \frac{1}{D}\big)^{k-1}$, which gives Equation (18).

Monetary shocks. With no financing, the payout follows the fiscal rule, $d\mathbf{X}^{p} = \mathbf{R}(-B\,d\mathbf{r} + \tau\,d\mathbf{Y})$ with $\mathbf{R} = (1 + r - \rho_B)\mathbf{L}(\mathbf{I} - \rho_B\mathbf{L})^{-1}$ (Appendix B.1). Substituting into Equation (31) and collecting the $d\mathbf{Y}$ terms gives Equation (22).

Present value of a rate cut. Counting heirs, $\mathbf{q}'\overline{\mathbf{M}}^{r} = B\,\mathbf{q}'$, so the direct effect lowers the present value of spending by the interest savings $P \equiv -B\,\mathbf{q}'d\mathbf{r}$, and the rebate raises it by P, since $\mathbf{q}'\mathbf{M}^{\phi} = \mathbf{q}'$ and $\mathbf{q}'\mathbf{R} = \mathbf{q}'$. Equity holders lose nothing in present value, since higher share prices offset the lower return. For a cut of 1 percentage point in the annual real rate with quarterly persistence 0.9, P is 9.5% of quarterly output.

Estates. The household block takes the bequest flow as an input. Dying households leave their wealth and the interest on it, $d\mathcal{E}_t = \xi(1+r)\,dA_{t-1} + \xi A\,dr_t$, plus their share

ξ of the date-0 capital gain, and newborns receive it as endowments. Let $\mathbf{J}^{C,x}$ and $\mathbf{J}^{A,x}$ be the consumption and wealth Jacobians of the household block to an input x, holding the bequest flow fixed, and $\mathbf{J}^{C,\mathcal{E}}$ and $\mathbf{J}^{A,\mathcal{E}}$ the Jacobians to the bequest flow. Closing the loop, the Jacobians used throughout are

$$\mathbf{M}^x = \mathbf{J}^{C,x} + \mathbf{J}^{C,\mathcal{E}} \left(\mathbf{I} - \xi(1+r)\mathbf{L}\,\mathbf{J}^{A,\mathcal{E}}\right)^{-1} \xi(1+r)\mathbf{L}\,\mathbf{J}^{A,x},$$

with the estates' share of the capital gain added for $\mathbf{m}^{cap}$ and the interest on estates, $\xi A\,\mathbf{J}^{C,\mathcal{E}} \left(\mathbf{I} - \xi(1+r)\mathbf{L}\,\mathbf{J}^{A,\mathcal{E}}\right)^{-1}$, added for $\mathbf{M}^r$. They count the spending of the household and of the newborns who inherit its estate, and spending includes the adjustment cost and the intermediation cost on deposits. With them, Equation (18) holds with no separate bequest term: against the general-equilibrium paths of the quantitative model, the residual is below 0.001% of the peak output response, in both models.

D.2 Incidence and iMPCs

A transfer dX_s at incidence ϕ raises the date-s income of a household with earnings $\alpha_i e$ by $\iota_\phi\, dX_s$, and the household-level response $m_{t,s}$ of date-t consumption per unit of date-s income paid into the liquid account does not depend on the label of the income. By linearity of aggregation, and using $\mathbb{E}[\iota_\phi] = 1$,

$$\mathbf{M}^{\phi}_{t,s} = \int \iota_\phi\, m_{t,s}\, d\Phi = \mathbb{E}\left[m_{t,s}\right] + \mathrm{Cov}\left(\iota_\phi, m_{t,s}\right),$$

where the expectation is over the cross-section of households at date s along the steady state, and the first term is the lump-sum iMPC matrix $\mathbf{M}^0_{t,s}$, since $\iota_0 = 1$. In particular, labor income reaches households at incidence one, so $\mathbf{M}^Z = \mathbf{M}^1$, and the capital gain reaches them in proportion to their illiquid wealth, which is held by wealthy households with low iMPCs and is costly to spend, so $\mathbf{m}^{cap}$ falls short of $\mathbf{M}^0_{\cdot,0}$ in the bequest-motive model.

Proof of Proposition 1. The stationary distribution of e is the same in every PLI group, so $\mathbb{E}[(\alpha e)^\phi] = \mathbb{E}[\alpha^\phi]\,\mathbb{E}[e^\phi]$ and $\iota_\phi = \tilde{\alpha}_i^\phi\, \tilde{e}^\phi$, with $\mathbb{E}_i[\tilde{e}^\phi] = 1$. Splitting the expectation by

PLI group,

$$\mathbb{E}\left[\iota_\phi\, m_{t,s}\right] - \mathbb{E}\left[m_{t,s}\right] = \sum_i \omega_i \tilde{\alpha}_i^\phi \left(\bar{m}_{i,t,s} + \mathrm{Cov}_i(\tilde{e}^\phi, m_{t,s})\right) - \sum_i \omega_i\, \bar{m}_{i,t,s},$$

and collecting the mean terms gives

$$\mathbf{M}_{t,s}^\phi = \mathbf{M}_{t,s}^0 + \underbrace{\sum_i \omega_i \left(\tilde{\alpha}_i^\phi - 1\right) \bar{m}_{i,t,s}}_{\text{PLI term}} + \underbrace{\sum_i \omega_i \tilde{\alpha}_i^\phi\, \mathrm{Cov}_i\left(\tilde{e}^\phi,\, m_{t,s}\right)}_{\text{idiosyncratic term}}. \tag{33}$$

Since $\sum_i \omega_i \tilde{\alpha}_i^\phi = 1$, the PLI term is the covariance $\mathrm{Cov}_\omega(\tilde{\alpha}_i^\phi, \bar{m}_{i,t,s})$ of Equation (20). The same steps apply to any linear function of the column, such as the cumulative MPC. □

Lemma 3 (Irrelevance of permanent income under homothetic preferences)**.** *Under the assumptions of Lemma 1, at every incidence ϕ, $\bar{m}_{i,t,s}$ and $\mathrm{Cov}_i(\tilde{e}^\phi, m_{t,s})$ are independent of i: the PLI term in Equation* (33) *is zero, and the wedge $\mathbf{M}^\phi - \mathbf{M}^0$ reduces to the idiosyncratic term, which is invariant to the PLI distribution $\boldsymbol{\alpha}$.*

Proof. By Equation (26), $\bar{m}_{i,t,s} = \bar{m}^*_{t,s}$ and $\mathrm{Cov}_i(\tilde{e}^\phi, m_{t,s}) = \mathrm{Cov}^*(\tilde{e}^\phi, m^*_{t,s})$ for every i. The PLI term is then $\bar{m}^*_{t,s} \sum_i \omega_i(\tilde{\alpha}_i^\phi - 1) = 0$, and the idiosyncratic term is $\mathrm{Cov}^*(\tilde{e}^\phi, m^*_{t,s}) \sum_i \omega_i \tilde{\alpha}_i^\phi = \mathrm{Cov}^*(\tilde{e}^\phi, m^*_{t,s})$, which does not depend on $\boldsymbol{\alpha}$. □

Figure 18 shows the split of Proposition 1 for the 1-year MPC in both models, over a wider range of incidences. In the heterogeneous beta model, only the idiosyncratic term is present, as Lemma 3 implies.

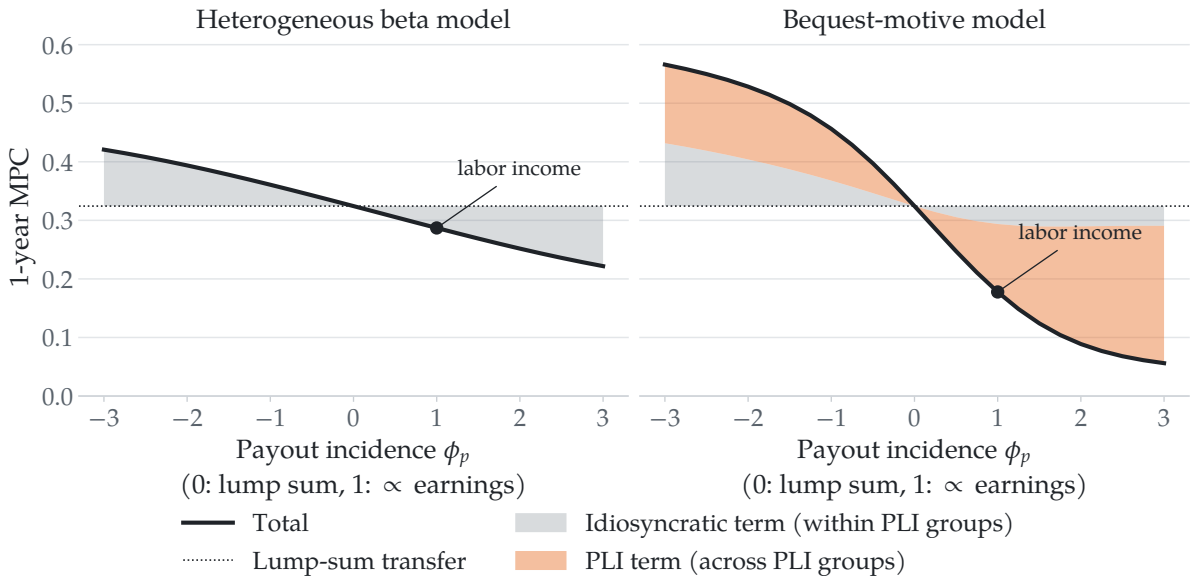


Figure 18: 1-year MPC of a transfer by incidence

Note: 1-year MPC of a transfer paid today in proportion to $(\alpha_i e)^\phi$, against its incidence ϕ, in the heterogeneous beta model (left) and the bequest-motive model (right), split as in Proposition 1. Labor income is the transfer at $\phi = 1$. 2019 calibration.

D.3 Cumulative MPCs and the cumulative multiplier

The cumulative MPC of an iMPC matrix $\mathbf{M}$ at horizon h is the share of a dollar received today that households have spent within h quarters, in present value,

$$\mathrm{MPC}_h(\mathbf{M}) \equiv \sum_{t=0}^{h-1} (1+r)^{-t} \mathbf{M}_{t,0}, \tag{34}$$

so that the 1-year MPC is MPC_4. We write $\mathrm{MPC}_h^p \equiv \mathrm{MPC}_h(\mathbf{M}^{\phi_p})$ and $\mathrm{MPC}_h^f \equiv \mathrm{MPC}_h(\mathbf{M}^{\phi_f})$, and c_h for the cumulative MPC of a household.

Proposition 3 (The cumulative multiplier without anticipation)**.** *Consider a fiscal shock made of a payout $d\mathbf{X}^p$ at incidence ϕ_p, a debt path $d\mathbf{B}$ with $dB_{-1} = 0$ that returns to zero, and the financing that balances the budget, at incidence ϕ_f, with the real rate fixed. Suppose that households do not anticipate: they respond to income and to financing only when these are paid, and receive dividends as firms pay them out. Then the two terms of Equation* (19) *are*

$$\mathcal{M}(\phi_f, \phi_f, D) = \frac{D}{\sum_{h\geq 1}\left(1 - \mathrm{MPC}_h^Y\right)}, \qquad \mathcal{M}(\phi_p, \phi_f, 0) = \frac{\sum_{h\geq 1}\left(\mathrm{MPC}_h^p - \mathrm{MPC}_h^f\right)}{\sum_{h\geq 1}\left(1 - \mathrm{MPC}_h^Y\right)}, \tag{35}$$

where $D \equiv \mathbf{q}' d\mathbf{B} / \mathbf{q}' d\mathbf{X}^p$ *is the debt duration and* $\mathrm{MPC}_h^Y \equiv (1-\tau)\,\mathrm{MPC}_h(\overline{\mathbf{M}}^Z) + \tau\,\mathrm{MPC}_h^f$ *is the cumulative MPC of a dollar of output. The gap between the cumulative MPCs of the payout and of the financing splits into a PLI term and an idiosyncratic term,*

$$\mathrm{MPC}_h^p - \mathrm{MPC}_h^f = \mathrm{Cov}\big(\iota_{\phi_p} - \iota_{\phi_f},\, c_h\big) = \underbrace{\sum_i \omega_i \big(\tilde{\alpha}_i^{\phi_p} - \tilde{\alpha}_i^{\phi_f}\big)\,\bar{c}_{i,h}}_{\textit{PLI term}} + \underbrace{\sum_i \omega_i\, \mathrm{Cov}_i\big(\iota_{\phi_p} - \iota_{\phi_f},\, c_h\big)}_{\textit{idiosyncratic term}}, \tag{36}$$

where $\bar{c}_{i,h} \equiv \mathbb{E}_i[c_h]$ *is the average cumulative MPC of PLI group* i*.*

Mean spending horizons. The mean spending horizon of a dollar received at date s is the average delay before it is spent, weighted by the discounted iMPCs,

$$\lambda_s(\mathbf{M}) \equiv \sum_{t \geq 0} (t-s)\,(1+r)^{-(t-s)}\,\mathbf{M}_{t,s}, \tag{37}$$

where spending done in anticipation enters with a negative delay. Throughout, iMPC matrices count the spending of heirs, so their columns have present value one, $\sum_t (1+r)^{-(t-s)}\mathbf{M}_{t,s} = 1$, and paths decay geometrically, so all sums converge.

First, for a dollar received today, the horizon is the shortfall of the cumulative MPCs (34) from one, summed over horizons. By the unit present value of the column, $1 - \mathrm{MPC}_h(\mathbf{M}) = \sum_{t \geq h} (1+r)^{-t}\mathbf{M}_{t,0}$, and swapping the sums,

$$\sum_{h \geq 1} \big(1 - \mathrm{MPC}_h(\mathbf{M})\big) = \sum_{t \geq 0} \sum_{h=1}^{t} (1+r)^{-t}\mathbf{M}_{t,0} = \sum_{t \geq 0} t\,(1+r)^{-t}\mathbf{M}_{t,0} = \lambda_0(\mathbf{M}). \tag{38}$$

Second, let $\mathbf{n}_t \equiv t\,\mathbf{q}_t$ be the first-moment vector, so that $\mathbf{n}'\mathbf{y}$ is the present value of a path weighted by its dates. For any $\mathbf{M}$ with unit present-value columns and any path $\mathbf{y}$, using $\mathbf{q}_t = \mathbf{q}_s (1+r)^{-(t-s)}$,

$$\mathbf{n}'\mathbf{M}\mathbf{y} = \sum_s \mathbf{q}_s\, y_s \sum_t \big[(t-s) + s\big](1+r)^{-(t-s)}\mathbf{M}_{t,s} = \mathbf{n}'\mathbf{y} + \sum_s \mathbf{q}_s\, \lambda_s(\mathbf{M})\, y_s : \tag{39}$$

spending happens, on average, one mean spending horizon after the income.

Cumulative MPCs by incidence. The cumulative MPC is linear in the column, $\mathrm{MPC}_h(\mathbf{M}^\phi) = \mathbb{E}[\iota_\phi\, c_h]$, so the split of Appendix D.2 applies to it, and its PLI term is zero under the

assumptions of Lemma 3. Since $\mathbb{E}[\iota_\phi] = 1$ at every incidence, $\text{MPC}_h^p - \text{MPC}_h^f = \text{Cov}\big(\iota_{\phi_p} - \iota_{\phi_f}, c_h\big)$, and the law of total covariance splits it into the covariance of the group means, the PLI term of Equation (36), and the average within-group covariance.

Proof of Proposition 3. Premultiply Equation (32) by $\mathbf{n}'$ and apply Equation (39) to each of its three matrices, writing λ_s^K, λ_s^p and λ_s^f for the horizons of $\mathbf{K}$, $\mathbf{M}^{\phi_p}$ and $\mathbf{M}^{\phi_f}$. On the left, $\mathbf{n}'(\mathbf{I} - \mathbf{K})\, d\mathbf{Y} = -\sum_s \mathbf{q}_s \lambda_s^K\, dY_s$. On the right, the redistribution term gives $\sum_s \mathbf{q}_s(\lambda_s^p - \lambda_s^f)\, dX_s^p$, since the $\mathbf{n}'d\mathbf{X}^p$ terms cancel. For the debt-financing term, let $\mathbf{x} \equiv [\mathbf{I} - (1+r)\mathbf{L}]\, d\mathbf{B}$; two sums telescope, using $\mathbf{q}_{t+1}(1+r) = \mathbf{q}_t$ and $dB_{-1} = 0$:

$$\mathbf{n}'\mathbf{x} = \sum_t \big[t - (t+1)\big]\, \mathbf{q}_t\, dB_t = -\mathbf{q}' d\mathbf{B}, \qquad \sum_s \mathbf{q}_s \lambda_s^f\, x_s = \sum_s \mathbf{q}_s \big(\lambda_s^f - \lambda_{s+1}^f\big)\, dB_s.$$

Collecting terms,

$$\sum_s \mathbf{q}_s\, \lambda_s^K\, dY_s = \sum_s \mathbf{q}_s \big(1 + \lambda_{s+1}^f - \lambda_s^f\big)\, dB_s + \sum_s \mathbf{q}_s \big(\lambda_s^f - \lambda_s^p\big)\, dX_s^p, \tag{40}$$

an identity that holds whether or not households anticipate. Without anticipation, households respond to a dollar only from the date it is paid, and, since the steady state is stationary, in the same way whatever that date: $\mathbf{M}_{t,s} = m_{t-s}$ for $t \geq s$ and zero otherwise. With dividends paid as they accrue, the same holds for $\overline{\mathbf{M}}^Z$, hence for $\mathbf{K}$. Every column of each matrix then has the horizon of its first column, so Equation (40) divided by $\mathbf{q}'d\mathbf{X}^p$ reads $\lambda_0(\mathbf{K})\, \mathcal{M} = D + \lambda_0(\mathbf{M}^{\phi_f}) - \lambda_0(\mathbf{M}^{\phi_p})$. Horizons are linear in the matrix, so $\lambda_0(\mathbf{K}) = (1-\tau)\lambda_0(\overline{\mathbf{M}}^Z) + \tau\lambda_0(\mathbf{M}^{\phi_f})$, and Equation (38) turns each horizon into a sum of cumulative MPCs: $\lambda_0(\mathbf{K}) = \sum_{h\geq 1}(1 - \text{MPC}_h^Y)$ and $\lambda_0(\mathbf{M}^{\phi_f}) - \lambda_0(\mathbf{M}^{\phi_p}) = \sum_{h\geq 1}(\text{MPC}_h^p - \text{MPC}_h^f)$. For the pure deficit the second difference is zero, and for the pure redistribution $D = 0$, which gives Equation (35); Equation (36) is shown above. □

Asset market clearing. If households receive a path $\mathbf{y}$ and spend $\mathbf{My}$, their saving $z_t \equiv \mathbf{q}_t\big(y_t - (\mathbf{My})_t\big)$ sums to zero and their assets satisfy $\mathbf{q}_t dA_t = \mathbf{q}_{t-1} dA_{t-1} + z_t$, so $\sum_t \mathbf{q}_t\, dA_t = \sum_t \sum_{u\leq t} z_u = -\sum_u u\, z_u = \sum_s \mathbf{q}_s \lambda_s(\mathbf{M})\, y_s$ by Equation (39): the horizon of a dollar is the present value of the dollar-quarters of assets it generates. The left side of Equation (40) is therefore the dollar-quarters of assets households accumulate out of the income the expansion creates, net of the extra equity $\sum_t \mathbf{q}_t dp_t$ they hold because

dividends arrive up front as a capital gain, and the right side is the dollar-quarters of public debt plus those the payout and the financing generate. Equation (40) is thus $\sum_t \mathbf{q}_t \, dA_t^{hh} = \sum_t \mathbf{q}_t \, (dB_t + dp_t)$, asset market clearing in present value.

Self-financing. Premultiplying Equation (30) by $\mathbf{q}'$ at $d\mathbf{r} = 0$, the debt term vanishes by telescoping, so $\mathbf{q}' d\mathbf{X}^f = \mathbf{q}' d\mathbf{X}^p - \tau \, \mathbf{q}' d\mathbf{Y}$: the receipts of the expansion finance a share $\tau\mathcal{M}$ of the payout, the self-financing share of Angeletos et al. (2024).

Persistence and payout incidence add up. The right side of Equation (32) is the sum of the term of the package without debt, $(\mathbf{M}^{\phi_p} - \mathbf{M}^{\phi_f}) \, d\mathbf{X}^p$, and the term of a pure deficit at incidence ϕ_f, $\mathbf{M}^{\phi_f}[\mathbf{I} - (1+r)\mathbf{L}] \, d\mathbf{B}$. The operator $\mathbf{I} - \mathbf{K}$ depends on ϕ_f only, and asset market clearing selects the solution linearly, so $\mathcal{M}(\phi_p, \phi_f, D) = \mathcal{M}(\phi_p, \phi_f, 0) + \mathcal{M}(\phi_f, \phi_f, D)$, which is Equation (19).

The fiscal rule as a redistribution. Write Equation (22) at ϕ_2 and at ϕ_1 and subtract. The direct term cancels, and adding and subtracting $\mathbf{K}_R(\phi_2) \, d\mathbf{Y}(\phi_1)$,

$$\big[\mathbf{I} - \mathbf{K}_R(\phi_2)\big] \, \Delta d\mathbf{Y} = \big(\mathbf{M}^{\phi_2} - \mathbf{M}^{\phi_1}\big) \, d\mathbf{X}^p(\phi_1), \qquad d\mathbf{X}^p(\phi_1) \equiv \mathbf{R}\big(- B \, d\mathbf{r} + \tau \, d\mathbf{Y}(\phi_1)\big), \tag{41}$$

where $\Delta d\mathbf{Y} \equiv d\mathbf{Y}(\phi_2) - d\mathbf{Y}(\phi_1)$ and $d\mathbf{X}^p(\phi_1)$ is the rebate the rule pays at incidence ϕ_1.

Anticipation in the model. In the model households anticipate in two ways, and Equation (40) shows how each enters. First, payers save part of a future financing before it is due. The weight on dB_s, $1 + \lambda^f_{s+1} - \lambda^f_s$, is the delay in the payers' spending cut when a dollar of financing due at s is postponed by one quarter: one quarter if payers ignore the news, zero if they save the whole amount in advance. For a lump-sum deficit at $\rho_B = 0.975$, its average over the debt path is 0.74 in the bequest-motive model and 0.75 in the heterogeneous beta model. Second, income expected far ahead is partly spent before it arrives, so horizons fall with the date of payment: in the bequest-motive model, a lump-sum dollar paid in ten years has a horizon of 2.9 years, against 6.8 years for one paid today, and the horizon of post-tax income falls from 14.6 years on impact to 10.3 years at ten years.

D.4 Lowering debt at zero output

This appendix computes how long debt can stay below its steady state with output at its steady state in every quarter (Section 5.3). We fix a financing path $d\mathbf{X}^f$ at incidence ϕ_f, of 1% of quarterly output with persistence 0.9, and solve for the payout $d\mathbf{X}^p$ at incidence ϕ_p and for the debt path, with no fiscal rule. With $d\mathbf{Y} = 0$ and $d\mathbf{r} = 0$, wages, dividends and tax receipts do not change, so Equations (32) and (30) reduce to

$$\mathbf{M}^{\phi_p} d\mathbf{X}^p = \mathbf{M}^{\phi_f} d\mathbf{X}^f, \qquad \left[\mathbf{I} - (1+r)\mathbf{L}\right] d\mathbf{B} = d\mathbf{X}^p - d\mathbf{X}^f. \tag{42}$$

The spending of beneficiaries replaces the spending that payers give up, date by date. Since the columns of iMPC matrices have present value one, premultiplying the first equation by $\mathbf{q}'$ gives $\mathbf{q}' d\mathbf{X}^p = \mathbf{q}' d\mathbf{X}^f$. The whole financing thus reaches beneficiaries, and the primary surplus has a present value of zero.

The debt duration of this path is close to the gap between cumulative MPCs, summed over horizons. At $d\mathbf{Y} = 0$, the left side of Equation (40) is zero. Without anticipation, the weights on debt are one and the horizons do not depend on the date, so

$$-\frac{\mathbf{q}' d\mathbf{B}}{\mathbf{q}' d\mathbf{X}^p} = \sum_{h \geq 1} \left(\mathrm{MPC}_h^p - \mathrm{MPC}_h^f\right). \tag{43}$$

In the model, payers save part of the financing before it is due, so the weights are below one and the debt duration exceeds the gap. The gap is about 0.8 of the duration in the bequest-motive model, and about half of it in the heterogeneous beta model.

Figure 19 shows the debt duration for each pair of incidences. A redistribution from high to low earners ($\phi_p = -1$, $\phi_f = 2$) keeps debt lower for 19 years in the bequest-motive model, against 2.5 years in the heterogeneous beta model. The duration barely depends on the path of the financing: with a financing paid over one year or with persistence 0.99, it moves by less than 10% at this pair.

Three limits apply. First, the map stops at $\phi_p = 1.5$, because the solution is numerically unstable for a payout concentrated further on high earners. Second, the payout is slightly negative on impact in part of the map, because beneficiaries spend ahead of later payouts. At the pair above, this is 0.5% of the financing, and at most 5% in the

map. Third, the longest durations need a large financing from the highest earners. A financing of 1% of quarterly output takes at most 1.4% of the income of any income state at $\phi_f = 1$, and 9% at $\phi_f = 2$.

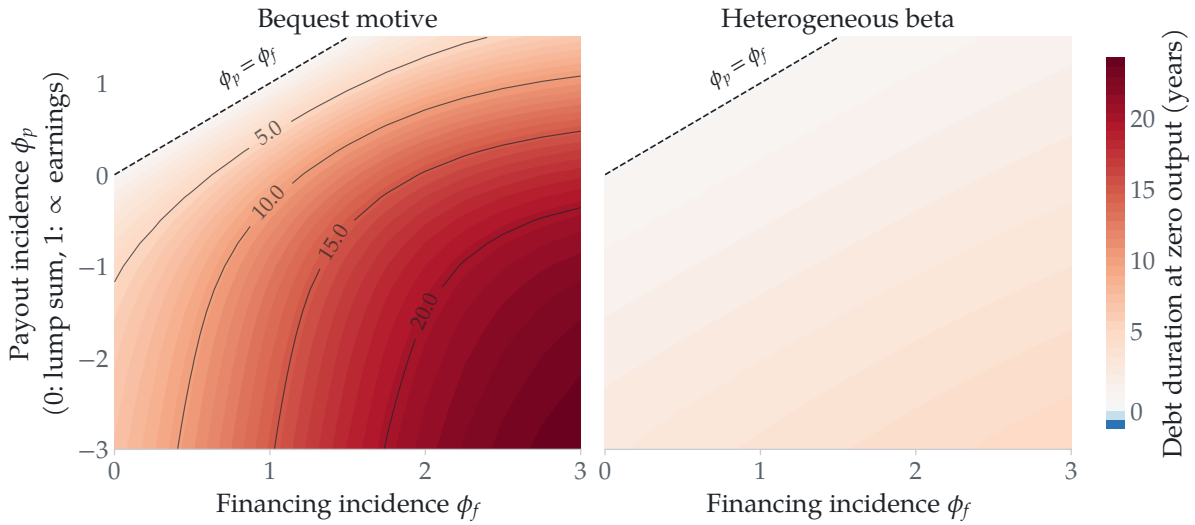


Figure 19: Debt duration at zero output, by incidence

Note: Debt duration $-\mathbf{q}'d\mathbf{B}/\mathbf{q}'d\mathbf{X}^f$, in years, of the path that keeps output at its steady state in every quarter (Equation (42)), with a financing of 1% of quarterly output with persistence 0.9 at incidence ϕ_f (columns) and a payout at incidence ϕ_p (rows). Above the dashed line, $\phi_p > \phi_f$, debt would rise. 2019 calibration.

D.5 The One Big Beautiful Bill Act

This appendix explains how we set the three parameters of Section 5.4: the payout incidence ϕ_p, the financing incidence ϕ_f and the debt duration D.

Payout and financing. We count every tax cut in the law as a payout and every transfer cut or tax increase as financing. Over 2025–2034, the tax cuts cost $5.4 trillion, about 1.5% of GDP a year: $4.3 trillion for individuals and $1.1 trillion for businesses (Gale & Page 2026). The transfer cuts save $1.7 trillion, half of it from Medicaid. The tax increases raise $0.8 trillion, mostly by repealing clean-energy credits. The financing is therefore 45% of the payout. We leave out spending on defense and border security, and tariffs.

Payout and financing incidences, $\phi_p = 1$ and $\phi_f = 0.25$. CBO's change in resources by income decile nets the payout and the financing, so we fit their incidences jointly. With households ranked by post-tax labor and capital income, let $s_d(\phi)$ be the share of a payout at incidence ϕ that decile d receives. Per household and per unit of the mean

payout, the law changes the resources of decile d by

$$\Delta_d(\phi_p, \phi_f) = 10\left[s_d(\phi_p) - 0.45\, s_d(\phi_f)\right].$$

We choose (ϕ_p, ϕ_f) on the grid of Section 5.3, in steps of 0.25, to minimize the squared distance between Δ_d and CBO's ten numbers divided by the mean payout, CBO's mean net change over 0.55. We use the profile of the House-passed bill (The Budget Lab at Yale 2025). The enacted law moves its end points from −\$1,550 and \$12,000 a year to −\$1,200 and \$13,600 (Congressional Budget Office 2025*a*). Both models give $\phi_p = 1$, a payout proportional to earnings, and $\phi_f = 0.25$, a financing that rises slightly with earnings (Figure 20). The top deciles pin down ϕ_p: the top 10% receive about 45% of the payout. The bottom deciles pin down ϕ_f: they lose on net, as do 41% of households in the model.

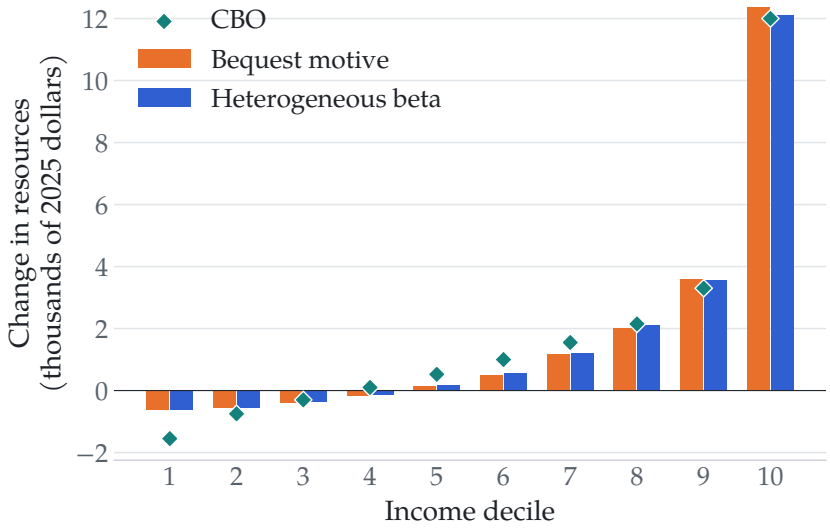


Figure 20: Change in household resources by income decile

Note: Average change in resources per household and year over 2026–2034, in thousands of 2025 dollars. Diamonds: CBO, House-passed bill, read from Figure 2 of The Budget Lab at Yale (2025) to within about \$100. Bars: payout at incidence $\phi_p = 1$ minus financing of 45% of the payout at incidence $\phi_f = 0.25$, scaled to CBO's mean. 2019 calibration.

Timing. We spread the payout and the financing evenly over 2025–2034. The law front-loads some tax cuts, which expire after 2028, and back-loads its transfer cuts. Letting the payout fall by half and the financing double between 2027 and 2033, as the changes of the top and bottom deciles do in Congressional Budget Office (2025*c*), raises the multiplier by 0.06 in the bequest-motive model and by 0.08 in the heterogeneous beta

model.

Debt duration. The payout net of the financing, $2.9 trillion over 2025–2034, adds to debt, and we assume that none of this debt is repaid before 2035. CBO puts the law's deficits at $3.4 trillion (Congressional Budget Office 2025*b*), a figure that also counts the spending we leave out. The debt follows Equation (12) with $\rho_B = 1 + r$ through 2034, so it accumulates with interest and reaches 9% of annual GDP at the end of 2034, and with our baseline $\rho_B = 0.975$ from 2035, a debt duration of 8.4 years. In every quarter, the financing clears the budget at the incidence of the law's own financing, $\phi_f = 0.25$.

Robustness. Moving ϕ_p by 0.25 moves the multiplier by about 0.18 in the bequest-motive model and by at most 0.01 in the heterogeneous beta model. Raising ϕ_f by 0.5 raises it by 0.35, against 0.04.

Limits. The split into payout and financing depends on how scorers define provisions: the extension of the 2017 individual provisions nets a $1.9 trillion tax increase from repealing personal exemptions against larger rate cuts. Some tax increases fall first on firms. Medicaid cuts are a cash loss valued at their cost to the government. CBO ranks households by income after taxes and transfers, the model by labor and capital income. Finally, the payout stops after 2034, although the law makes the 2017 individual tax rates permanent.

E Rising PLI Inequality: Proofs and Robustness

This appendix explains how the 1970 and 2019 economies are compared, proves Proposition 2, Corollary 1 and the neutrality of the PLI distribution under homothetic preferences. Appendix G.4 repeats the results of Section 6 for the model in which discount factors rise with the PLI rank.

E.1 Comparing the two economies

We compare the 2019 and 1970 economies in two ways: at the same debt persistence ρ_B, the same fiscal rule, and at the same debt duration $D = (1 + r)/(1 + r - \rho_B)$. The steady-state real rate is 0.50% per quarter in 2019 against 0.63% in 1970, because the

extra saving of high-PLI households lowers the rate. At a common ρ_B, the lower rate lengthens the debt in present value, from 8.0 to 8.4 years at $\rho_B = 0.975$ and from 22.2 to 25.1 years at $\rho_B = 0.995$. At the same duration, we set the 1970 debt persistence to $1 + r - (1 + r)/D$, with D its 2019 value, which isolates the response of households. In the heterogeneous beta model, the steady-state rate is the same in the two economies (Lemma 1), and the two comparisons coincide.

E.2 Proofs of Proposition 2 and Corollary 1

As in Appendix D.2, $\mathbf{M}^{\phi} = \mathbb{E}[\iota_{\phi} m] = \sum_i \omega_i \tilde{\alpha}_i^{\phi} \mathbf{M}_i^{\phi}$ with $\mathbf{M}_i^{\phi} \equiv \mathbb{E}_i[\tilde{e}^{\phi} m]$. With population shares held fixed,

$$\Delta \mathbf{M}^{\phi} = \sum_i \omega_i \left(\tilde{\alpha}_i^{\phi\prime} \mathbf{M}_i^{\phi\prime} - \tilde{\alpha}_i^{\phi} \mathbf{M}_i^{\phi} \right) = \sum_i \omega_i \, \Delta\tilde{\alpha}_i^{\phi} \, \mathbf{M}_i^{\phi\prime} + \sum_i \omega_i \tilde{\alpha}_i^{\phi} \, \Delta \mathbf{M}_i^{\phi}.$$

Since $\sum_i \omega_i \tilde{\alpha}_i^{\phi} = 1$ in both economies, $\sum_i \omega_i \Delta\tilde{\alpha}_i^{\phi} = 0$, and the first sum is the covariance $\mathrm{Cov}_{\omega}\left(\Delta\tilde{\alpha}_i^{\phi}, \mathbf{M}_i^{\phi\prime}\right)$. At $\phi = 0$, $\tilde{\alpha}_i^{0} = 1$ in both economies, so the composition term is zero. Cumulative MPCs are linear in the columns of $\mathbf{M}^{\phi}$, so the same split holds for them. Under the assumptions of Lemma 1, $\mathbf{M}_i^{\phi} = \mathbf{M}^{\phi *}$ for every group in both economies by Equation (26), so the composition term is $\mathbf{M}^{\phi *} \sum_i \omega_i \Delta\tilde{\alpha}_i^{\phi} = 0$ and the behavior term is zero. Proposition 4 extends the invariance to $\mathbf{M}^{r}$, $\mathbf{m}^{cap}$ and every multiplier. □

Proof of Corollary 1. By Equation (19), Proposition 3 and Equation (36), in each economy

$$\mathcal{M} = \frac{D + \sum_{h \geq 1} \mathrm{Cov}\left(\iota_{\phi_p} - \iota_{\phi_f}, \, c_h\right)}{\sum_{h \geq 1} \left(1 - \mathrm{MPC}_h^{Y}\right)}.$$

Writing this equation in both economies and subtracting gives Equation (24).

E.3 Neutrality of the PLI distribution

The neutrality of aggregate consumption with respect to the permanent income distribution is due to Straub (2019); the proposition extends it to every sequence-space Jacobian of the household block, and hence to the full transmission of fiscal and monetary shocks.

Proposition 4 (Neutrality)**.** *Under the assumptions of Lemma 1,* $\mathbf{M}^{\phi}$ *at every incidence, including* $\mathbf{M}^{Z} = \mathbf{M}^{1}$*,* $\mathbf{M}^{r}$ *and* $\mathbf{m}^{cap}$*, and hence* $\overline{\mathbf{M}}^{Z}$*,* $\overline{\mathbf{M}}^{r}$ *and the output responses of Equations* (18) *and* (22)*, are invariant to the permanent income distribution* $\boldsymbol{\alpha}$*.*

Proof. By Lemma 1, the steady-state rate and tax rate do not depend on $\boldsymbol{\alpha}$, and neither do output, debt, dividends and the firm value, so the scalars (τ, μ, B, p) in Equations (18) and (22) are invariant. For transfers, Equation (26) gives $\mathbf{M}^{\phi} = \sum_i \omega_i \tilde{\alpha}_i^{\phi} \, \mathbb{E}^*[\tilde{e}^{\phi} m^*] = \mathbf{M}^{\phi *}$, since $\sum_i \omega_i \tilde{\alpha}_i^{\phi} = 1$; labor income is the case $\phi = 1$. A rate change dr_s moves the return on both assets, so it raises the income of a type-i household in proportion to its liquid and illiquid holdings, which scale with α_i under the scaling of Lemma 1, and it changes intertemporal prices identically for all types. The consumption response of group i is therefore $\alpha_i \, \mathbf{M}^{r*}$. Capital gains accrue to illiquid holdings, so the response of group i to them is $\alpha_i \, \mathbf{m}^{cap*}$. Aggregating with $\sum_i \omega_i \alpha_i = 1$ gives $\mathbf{M}^{r} = \mathbf{M}^{r*}$ and $\mathbf{m}^{cap} = \mathbf{m}^{cap*}$, none of which depend on $\boldsymbol{\alpha}$. The combined matrices $\overline{\mathbf{M}}^{Z}$ and $\overline{\mathbf{M}}^{r}$, the operators and the impulses of Equations (18) and (22), and the output responses they select inherit the invariance. □

F One-Asset Model

This appendix describes a one-asset version of the model and compares it with the two-asset model. Two results hold only with one asset: the rise in the response to a rate cut from 1970 to 2019, and the fall in the EIS.

Model. Households save in a single account a_t at the return r_t, without adjustment or intermediation costs, the bequest motive applies to a_t, and labor productivity has no transitory component. The date-0 capital gain reaches each household in proportion to its equity exposure $\vartheta(a_{t-1}/A_{t-1})\, a_{t-1}$, with

$$\vartheta(a_{t-1}/A_{t-1}) = \frac{\vartheta_0}{1 + \exp\big(-\vartheta_1\big(\log(a_{t-1}/A_{t-1}) - \vartheta_2\big)\big)}, \tag{44}$$

a reduced-form rule for the rising equity share of wealthier households (Auclert & Rognlie 2020), and $\bar{S}$ is aggregate equity exposure.

Calibration. η, ξ, the fiscal block and the markup are those of Section 4. Idiosyncratic

productivity follows the AR(1) estimated on household earnings in Appendix A.5, with annual persistence 0.87 and unconditional standard deviation 0.36. $(\gamma, \zeta) = (7.7, 89.9)$ match the total hand-to-mouth share in each PLI quintile and the top-1% wealth share, and $\beta = 0.990$ clears the asset market. The hand-to-mouth share falls from 54% in the bottom PLI quintile to 12% in the top one, against 65% and 19% in the data, the top-1% share is 34%, and the untargeted 1-year MPC is 0.35. Figure 21 shows the policy functions at the top of the wealth distribution, which converge to the asymptote of Appendix C.1.

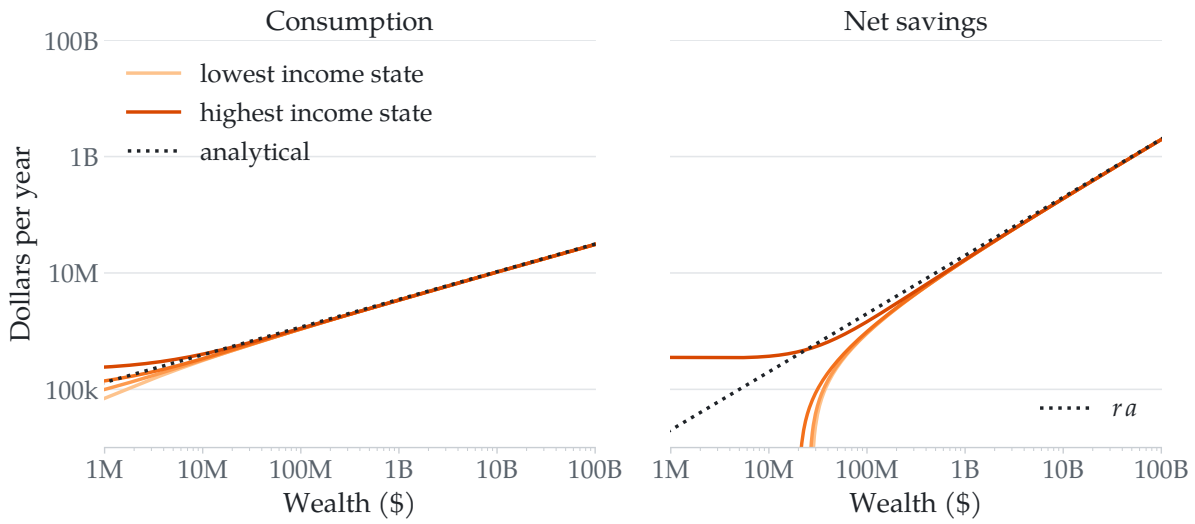


Figure 21: Savings and consumption of wealthy households, one-asset model

Note: Numerical policy functions for net savings and consumption as functions of wealth in dollars (x-axis), for different idiosyncratic and permanent types (darker lines for higher income states). Dotted lines: the analytical approximation as wealth $\to \infty$ of Appendix C.1.

Portfolio shares. The rule (44) is fitted to the share of equity and business wealth in net worth in the Surveys of Consumer Finances, pooling the waves from 1989 to 2019 (Auclert & Rognlie 2020). Equity includes stocks held through funds and retirement accounts, and housing is excluded. The fit yields $\vartheta_0 = 1.00$, $\vartheta_1 = 0.37$, and $\vartheta_2 = 2.60$ (Figure 22).

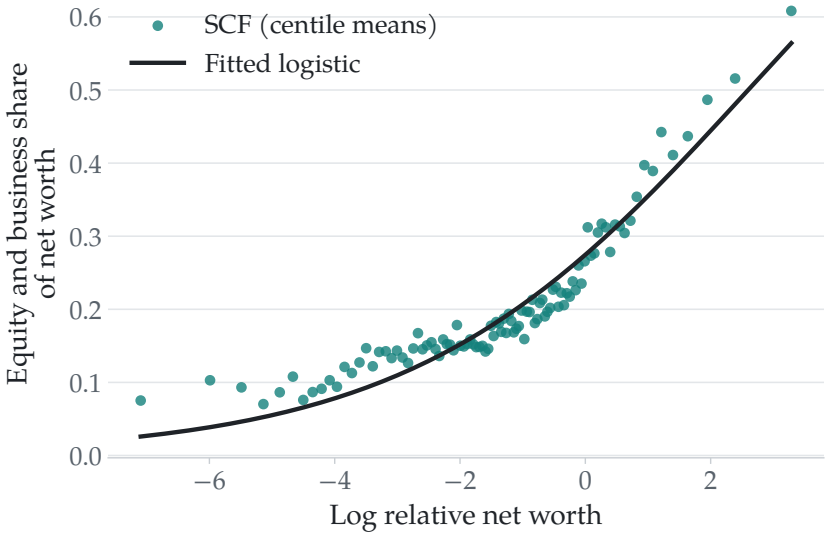


Figure 22: Portfolio shares in the SCF

Note: The dots show the share of equity and business wealth in net worth by centile of relative net worth in the Survey of Consumer Finances, waves 1989 to 2019 pooled. The line shows the fitted rule of Equation (44).

Results. The incidence results are smaller with two assets (Table 11): the gap between the cumulative MPCs of a lump-sum transfer and of labor income, summed over horizons, is 6.4 years, against 8.4. With two assets, the transitory shock and the adjustment cost give every PLI group a liquid hand-to-mouth margin, so less of the variation in MPCs lines up with PLI. Rising PLI inequality raises the aggregate MPC in both models, by half as much with two assets.

Table 11: One-asset and two-asset models

	One asset	Two assets
2019, bequest-motive model (heterogeneous beta model in parentheses)		
Cumulative MPC gap, lump sum vs labor income, summed over horizons (years)	8.4 (0.6)	6.4 (0.4)
$\mathcal{M}$, pure redistribution, payout $\propto$ earnings	-0.69	-0.62
$\mathcal{M}$, lump-sum deficit, $D = 8.4$ years	0.20 (0.44)	0.41 (0.56)
$\mathcal{M}$, OBBBA	-0.49 (0.43)	-0.06 (0.57)
1970 to 2019, bequest-motive model		
Real rate (%, annual)	2.73 to 2.00	2.53 to 2.00
Annual MPC, lump-sum transfer	0.29 to 0.35	0.30 to 0.32
EIS	0.81 to 0.76	0.95 to 0.95
Semi-elasticity, lump-sum rebate	9.1 to 10.5	9.2 to 9.0

Note: $\mathcal{M}$: cumulative multiplier of a payout of 1% of quarterly output with persistence 0.9; the pure redistribution is financed lump sum within the period. OBBBA: incidences fitted to the law, $(\phi_p, \phi_f) = (1, 0)$ with one asset and $(1, 0.25)$ with two. Semi-elasticity: cumulative output semi-elasticity of a 1 point cut in the annual real rate with persistence 0.9, lump-sum rebate repaid at the 2019 debt duration.

G Discount Factors Correlated with PLI

This appendix shows that our results do not rest on the bequest motive. We replace it with discount factors that rise with PLI, fit this model to the same moments, and repeat the experiments of Sections 5 and 6. First, the three results of Section 5 hold, mostly more strongly. Second, the results of Section 6 hold too, but rising inequality changes spending mostly through composition, since behavior barely moves. The two models agree because the bequest motive itself works like a discount factor that rises with PLI (Appendix G.1). Appendix G.2 describes the model and its calibration, and Appendices G.3 and G.4 report the results.

G.1 The bequest motive as a discount factor that rises with PLI

The bequest-motive model can be read as a model in which households with a higher PLI are more patient. To see this, we write the problem of a household of PLI group i in units of its own PLI, $\hat{c} = c/\alpha_i$, $\hat{b} = b/\alpha_i$ and $\hat{a} = a/\alpha_i$, as in Lemma 1.

Lemma 4 (The bequest motive as a discount factor)**.** *(i) In units of its own PLI, a household of PLI group i in the bequest-motive model solves the problem of a household with PLI one, bequest intensity $\gamma\alpha_i^{1-\eta}$, Stone-Geary shifter ζ/α_i and adjustment-cost scale ψ_0/α_i.*

(ii) Its choice of deposits satisfies the first-order condition of a household without a bequest motive whose discount factor is

$$\tilde{\beta}_t = \beta + \frac{\xi\, v'(a_t + b_t)}{(1-\xi)\,\mathbb{E}_t\left[V_{b,t+1}\right]}, \tag{45}$$

where $V_{b,t+1}$ is the derivative of the value function (5) *at $t+1$ with respect to deposits b_t. When the household holds deposits, $b_t > 0$, this effective discount factor is*

$$\tilde{\beta}_t = \frac{\beta}{1 - \xi\gamma\,\alpha_i^{1-\eta}\,\hat{c}_t\left(\hat{a}_t + \hat{b}_t + \zeta/\alpha_i\right)^{-\eta}}. \tag{46}$$

Proof. (i) Log utility gives $u(\alpha_i\hat{c}) = \log\alpha_i + \log\hat{c}$, and the bequest motive is

$$\xi\, v\big(\alpha_i(\hat{a}+\hat{b})\big) = \xi\,\gamma\,\alpha_i^{1-\eta}\,\frac{\left(\hat{a}+\hat{b}+\zeta/\alpha_i\right)^{1-\eta}}{1-\eta}.$$

The adjustment cost (4) is homogeneous of degree one in (a_t, a_{t-1}, ψ_0), so $\Psi(\alpha_i \hat{a}_t, \alpha_i \hat{a}_{t-1}; \psi_0) = \alpha_i \Psi(\hat{a}_t, \hat{a}_{t-1}; \psi_0/\alpha_i)$, and the rest of the budget constraint scales with α_i as in the proof of Lemma 1. The constant $\log \alpha_i$ does not affect choices.

(ii) The first-order condition for deposits in problem (5) is

$$u'(c_t) \geq \xi\, v'(a_t + b_t) + (1 - \xi)\beta\, \mathbb{E}_t\big[V_{b,t+1}\big],$$

with equality when $b_t > 0$, and that of a household without a bequest motive and with discount factor $\tilde{\beta}_t$ is $u'(c_t) \geq (1 - \xi)\tilde{\beta}_t\, \mathbb{E}_t[V_{b,t+1}]$. The two right-hand sides are equal at $\tilde{\beta}_t$ in (45). When $b_t > 0$, the first condition holds with equality, so $(1 - \xi)\beta\, \mathbb{E}_t[V_{b,t+1}] = u'(c_t) - \xi v'(a_t + b_t)$ and $\tilde{\beta}_t = \beta / \big(1 - \xi v'(a_t + b_t)/u'(c_t)\big)$. With $u'(c) = 1/c$, $\xi v'(a_t + b_t)/u'(c_t) = \xi\gamma\, c_t\, (a_t + b_t + \zeta)^{-\eta}$, which gives (46) in units of PLI. □

A household with a higher PLI thus behaves as if it were more patient. Take two households in the same position relative to their own PLI, with the same consumption $\hat{c}_t$ and the same wealth $\hat{a}_t + \hat{b}_t$. By (46), the one with the higher PLI has the higher effective discount factor, for two reasons. Its bequest intensity $\gamma \alpha_i^{1-\eta}$ is larger, because the curvature on wealth is below that of log consumption, $\eta < 1$: at $\eta = 0.48$, doubling PLI raises it by more than 40%. Its shifter ζ/α_i is also smaller, so its bequest motive is already strong at low wealth. Under homothetic preferences, $\eta = 1$ and $\zeta = 0$, α_i drops out of (46), and households in the same position have the same effective discount factor whatever their PLI.

The model of Appendix G.2 builds the same feature in directly: each PLI group has a fixed discount factor that rises with its PLI rank. In both models, high-PLI households therefore save a larger share of their income, hold wealth far above the hand-to-mouth threshold, and have low MPCs. This correlation between PLI and MPCs is the PLI term of Proposition 1, which drives the results of Section 5. Since both models are fitted to the same HtM–PLI gradient, they produce PLI terms of similar size, and the results carry over.

The lemma also shows how the two models differ. First, in the bequest-motive model the effective discount factor varies with consumption and wealth within a PLI group,

while it is fixed in the other model. Second, it depends on the level of PLI, through $\alpha_i^{1-\eta}$ and ζ/α_i, while the schedule (47) depends on the PLI rank. When PLI inequality rises, high-PLI households thus become more patient in the bequest-motive model and low-PLI households less patient, while no discount factor moves in the model with discount factors correlated with PLI. This is why, from 1970 to 2019, behavior barely moves spending in that model, and its real rate falls by 16 basis points, against 53 in the bequest-motive model (Appendix G.4).

G.2 Model and calibration

This model sets $\gamma = 0$ and generates the HtM–PLI gradient through patience that rises with PLI. Each PLI group i has its own fixed discount factor,

$$\beta_i = \beta + \delta\,\Lambda\big(\lambda\,(F_i - F_0)\big) - \bar{\delta}, \tag{47}$$

where $F_i \in (0,1)$ is the mid-rank of group i in the PLI distribution, Λ the logistic function, and $\bar{\delta}$ the population mean of the logistic term, so that β is the mean discount factor. The adjustment-cost scale ψ_0 is fixed in units of output, as in the bequest-motive model, so the two models differ only in preferences. The PLI distribution is not neutral in this model, both because discount factors differ across PLI groups and because ψ_0 does not scale with PLI.

The three parameters of the schedule, $\delta = 0.100$, $\lambda = 0.89$ and $F_0 = 0.24$, and the adjustment-cost curvature $\psi_1 = 18.8$ minimize the distance to the same twelve moments as the bequest-motive model, with ψ_0, the deposit wedge and the income process at their bequest-motive values, and $\beta = 0.994$ clears the asset market. Only the shape of the schedule is identified. The spread δ sits at its upper bound of 0.1, and a wider spread with a flatter logistic gives similar moments. Discount factors rise from 0.985 in the bottom quintile to 1.004 in the top 5%. The model fits the HtM–PLI gradient as well as the bequest-motive model (Figure 23), its top-1% wealth share is 34%, as in the data, and its untargeted 1-year MPC is 0.32. It misses the liquid share of wealth, 13% against 22% in the data.

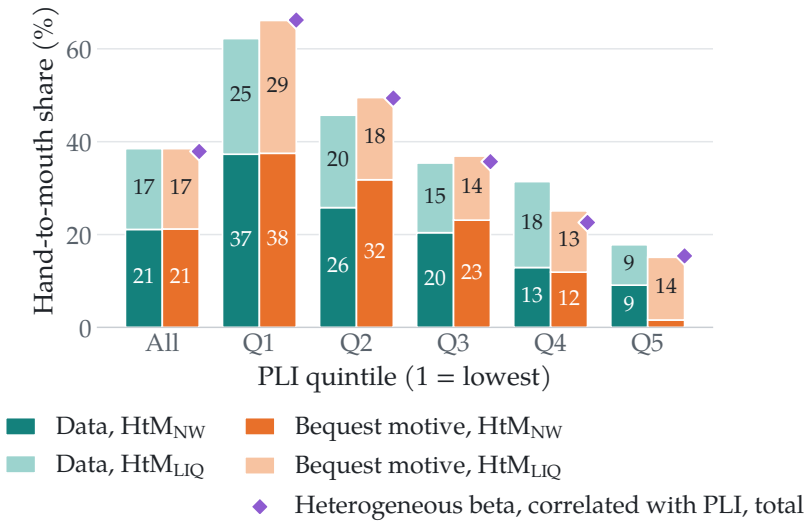


Figure 23: Hand-to-mouth shares by PLI quintile, discount factors correlated with PLI

Note: As in Figure 5, with diamonds for the model with discount factors correlated with PLI.

G.3 Policy at the 2019 distribution

The three results of Section 5 hold without a bequest motive, mostly more strongly. Figures 24 to 26 repeat the policy figures of Section 5 with this model as a third panel or curve. First, the multiplier depends less on the debt duration: doubling the duration from 10 to 20 years raises the multiplier of a lump-sum deficit by 0.7, against 1.5 in the heterogeneous beta model. Second, incidence matters more: a payout proportional to earnings, financed lump sum with debt repaid at the baseline fiscal rule, has a cumulative multiplier of -0.48, against 0.52 in the heterogeneous beta model. Third, moving the rebate of a rate cut from low to high earners lowers the cumulative output semi-elasticity by two thirds, against an eighth.

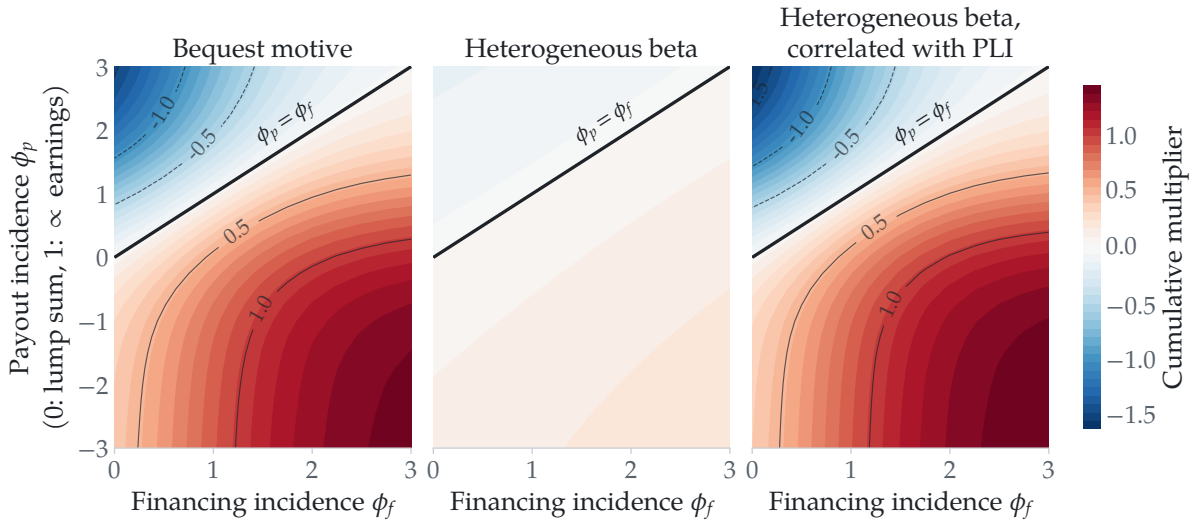


Figure 24: Cumulative multipliers of pure redistributions, three models

Note: Cumulative multiplier of a payout of 1% of quarterly output with persistence 0.9 at incidence ϕ_p (rows), financed in the same quarter at incidence ϕ_f (columns), with no debt, in the bequest-motive model (left), the heterogeneous beta model (center) and the model with discount factors correlated with PLI (right). On the dashed line, $\phi_p = \phi_f$, and the package is zero. 2019 calibration.

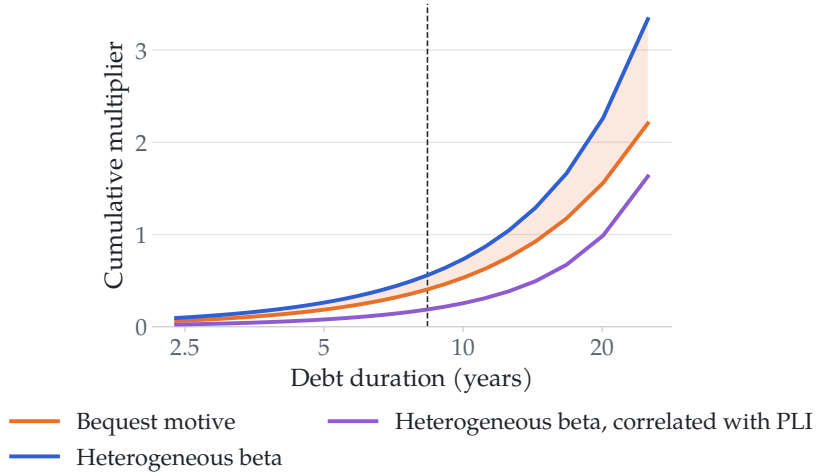


Figure 25: Cumulative multiplier of a lump-sum deficit by debt duration, three models

Note: Cumulative multiplier of a lump-sum payout of 1% of quarterly output with persistence 0.9, financed lump sum, against the debt duration D in years, on a log scale, in the bequest-motive model, the heterogeneous beta model and the model with discount factors correlated with PLI. Dashed line: baseline, 8.4 years. 2019 calibration.

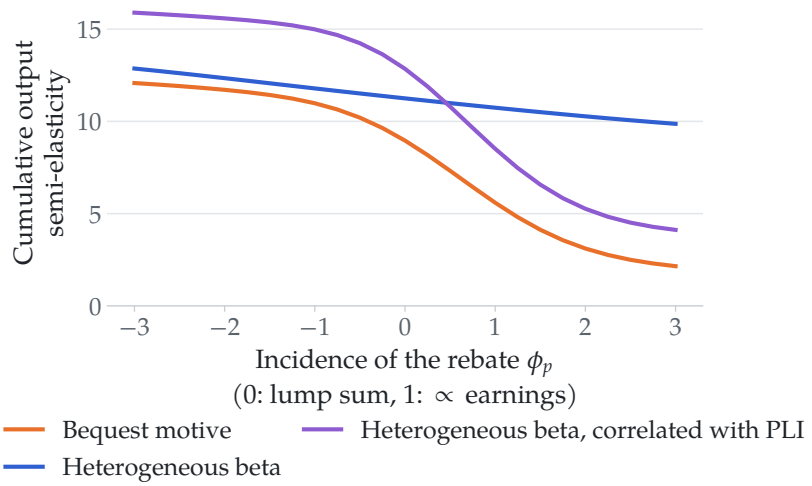


Figure 26: Response to a rate cut by the incidence of the rebate, three models

Note: Cumulative output semi-elasticity, the present value of dY_t/Y per present value of the rate cut, for a cut of 1 percentage point in the annual real rate with quarterly persistence 0.9, against the incidence ϕ_p of the rebate paid by the fiscal rule (16) with an average delay of 8.4 years, in the bequest-motive model, the heterogeneous beta model and the model with discount factors correlated with PLI. 2019 calibration.

G.4 Rising PLI inequality

The results of Section 6 also hold in this model, but rising inequality changes spending mostly through composition. The model is not neutral to the PLI distribution, since its patient top groups hold most of the wealth and their share of earnings rises. Behavior, however, barely moves the 1-year MPC: that of a lump-sum transfer, which composition leaves unchanged, rises by only 0.2 points, against 2.9 in the bequest-motive model (Figure 27). As Appendix G.1 explains, discount factors depend on PLI ranks here, so they do not move when inequality rises. The changes in fiscal and monetary policy have the same sign and a similar size as in the bequest-motive model. At a debt duration of 25 years, the lump-sum deficit loses 0.33, against 0.29, and moving the rebate of a rate cut from low to high earners lowers the semi-elasticity by 9.7 in 2019, against 9.3 in 1970.

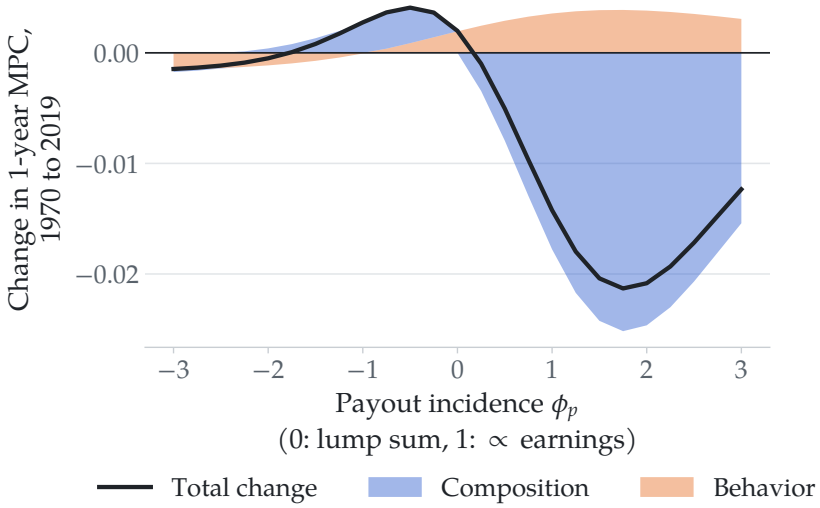


Figure 27: Change in the 1-year MPC of a transfer from 1970 to 2019, discount factors correlated with PLI

Note: As in Figure 10, in the model with discount factors correlated with PLI.